In [ ]:

```
#Advanced EDA for FIFA Sport Dataset
```

In [1]:

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

import warnings
warnings.filterwarnings('ignore')
```

In [2]:

```
fifa = pd.read_csv(r'C:\Users\SID Balurkar\Desktop\Full Stack Data Science\Classwork\Nov 2025\26th, 27th- Advanced Ed
fifa
```

Out[2]:

	Unnamed: 0	ID	Name	Age	Photo	Nationality	
0	0	158023	L. Messi	31	https://cdn.sofifa.org/players/4/19/158023.png	Argentina	https://cdn.sofifa.org/flags
1	1	20801	Cristiano Ronaldo	33	https://cdn.sofifa.org/players/4/19/20801.png	Portugal	https://cdn.sofifa.org/flags
2	2	190871	Neymar Jr	26	https://cdn.sofifa.org/players/4/19/190871.png	Brazil	https://cdn.sofifa.org/flags
3	3	193080	De Gea	27	https://cdn.sofifa.org/players/4/19/193080.png	Spain	https://cdn.sofifa.org/flags
4	4	192985	K. De Bruyne	27	https://cdn.sofifa.org/players/4/19/192985.png	Belgium	https://cdn.sofifa.org/flag
...	...	...	...	...	...	...	
18202	18202	238813	J. Lundstram	19	https://cdn.sofifa.org/players/4/19/238813.png	England	https://cdn.sofifa.org/flags
18203	18203	243165	N. Christoffersson	19	https://cdn.sofifa.org/players/4/19/243165.png	Sweden	https://cdn.sofifa.org/flags
18204	18204	241638	B. Worman	16	https://cdn.sofifa.org/players/4/19/241638.png	England	https://cdn.sofifa.org/flags
18205	18205	246268	D. Walker-Rice	17	https://cdn.sofifa.org/players/4/19/246268.png	England	https://cdn.sofifa.org/flags
18206	18206	246269	G. Nugent	16	https://cdn.sofifa.org/players/4/19/246269.png	England	https://cdn.sofifa.org/flags

18207 rows × 89 columns

In [3]:

```
fifa.head()
```

Out[3]:

	Unnamed: 0	ID	Name	Age	Photo	Nationality	Flag	C
0	0	158023	L. Messi	31	https://cdn.sofifa.org/players/4/19/158023.png	Argentina	https://cdn.sofifa.org/flags/52.png	
1	1	20801	Cristiano Ronaldo	33	https://cdn.sofifa.org/players/4/19/20801.png	Portugal	https://cdn.sofifa.org/flags/38.png	
2	2	190871	Neymar Jr	26	https://cdn.sofifa.org/players/4/19/190871.png	Brazil	https://cdn.sofifa.org/flags/54.png	
3	3	193080	De Gea	27	https://cdn.sofifa.org/players/4/19/193080.png	Spain	https://cdn.sofifa.org/flags/45.png	
4	4	192985	K. De Bruyne	27	https://cdn.sofifa.org/players/4/19/192985.png	Belgium	https://cdn.sofifa.org/flags/7.png	

5 rows × 89 columns

In [7]:

```
fifa.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 18207 entries, 0 to 18206
Data columns (total 89 columns):
 #   Column                    Non-Null Count  Dtype  
---  ------                    --------------  -----  
 0   Unnamed: 0                18207 non-null  int64  
 1   ID                        18207 non-null  int64  
 2   Name                      18207 non-null  object 
 3   Age                       18207 non-null  int64  
 4   Photo                     18207 non-null  object 
 5   Nationality               18207 non-null  object 
 6   Flag                      18207 non-null  object 
 7   Overall                   18207 non-null  int64  
 8   Potential                 18207 non-null  int64  
 9   Club                      17966 non-null  object 
 10  Club Logo                 18207 non-null  object 
 11  Value                     18207 non-null  object 
 12  Wage                      18207 non-null  object 
 13  Special                   18207 non-null  int64  
 14  Preferred Foot            18159 non-null  object 
 15  International Reputation  18159 non-null  float64
 16  Weak Foot                 18159 non-null  float64
 17  Skill Moves               18159 non-null  float64
 18  Work Rate                 18159 non-null  object 
 19  Body Type                 18159 non-null  object 
 20  Real Face                 18159 non-null  object 
 21  Position                  18147 non-null  object 
 22  Jersey Number             18147 non-null  float64
 23  Joined                    16654 non-null  object 
 24  Loaned From               1264 non-null   object 
 25  Contract Valid Until      17918 non-null  object 
 26  Height                    18159 non-null  object 
 27  Weight                    18159 non-null  object 
 28  LS                        16122 non-null  object 
 29  ST                        16122 non-null  object 
 30  RS                        16122 non-null  object 
 31  LW                        16122 non-null  object 
 32  LF                        16122 non-null  object 
 33  CF                        16122 non-null  object 
 34  RF                        16122 non-null  object 
 35  RW                        16122 non-null  object 
 36  LAM                       16122 non-null  object 
```

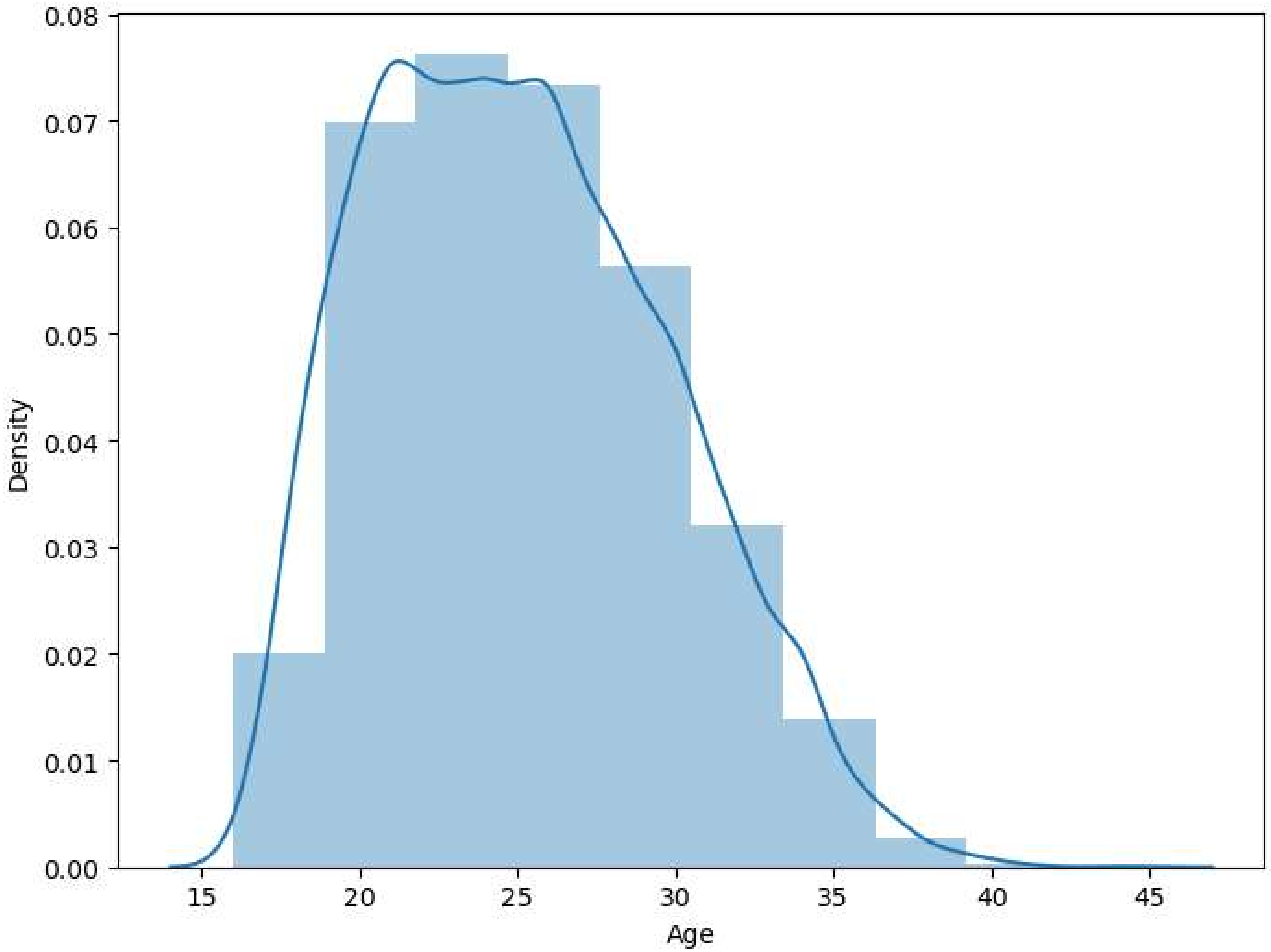
```
 37  CAM                       16122 non-null  object 
 38  RAM                       16122 non-null  object 
 39  LM                        16122 non-null  object 
 40  LCM                       16122 non-null  object 
 41  CM                        16122 non-null  object 
 42  RCM                       16122 non-null  object 
 43  RM                        16122 non-null  object 
 44  LWB                       16122 non-null  object 
 45  LDM                       16122 non-null  object 
 46  CDM                       16122 non-null  object 
 47  RDM                       16122 non-null  object 
 48  RWB                       16122 non-null  object 
 49  LB                        16122 non-null  object 
 50  LCB                       16122 non-null  object 
 51  CB                        16122 non-null  object 
 52  RCB                       16122 non-null  object 
 53  RB                        16122 non-null  object 
 54  Crossing                  18159 non-null  float64
 55  Finishing                 18159 non-null  float64
 56  HeadingAccuracy           18159 non-null  float64
 57  ShortPassing              18159 non-null  float64
 58  Volleys                   18159 non-null  float64
 59  Dribbling                 18159 non-null  float64
 60  Curve                     18159 non-null  float64
 61  FKAccuracy                18159 non-null  float64
 62  LongPassing               18159 non-null  float64
 63  BallControl               18159 non-null  float64
 64  Acceleration              18159 non-null  float64
 65  SprintSpeed               18159 non-null  float64
 66  Agility                   18159 non-null  float64
 67  Reactions                 18159 non-null  float64
 68  Balance                   18159 non-null  float64
 69  ShotPower                 18159 non-null  float64
 70  Jumping                   18159 non-null  float64
 71  Stamina                   18159 non-null  float64
 72  Strength                  18159 non-null  float64
 73  LongShots                 18159 non-null  float64
 74  Aggression                18159 non-null  float64
 75  Interceptions             18159 non-null  float64
 76  Positioning               18159 non-null  float64
 77  Vision                    18159 non-null  float64
 78  Penalties                 18159 non-null  float64
```

```
 79  Composure                 18159 non-null  float64
 80  Marking                   18159 non-null  float64
 81  StandingTackle            18159 non-null  float64
 82  SlidingTackle             18159 non-null  float64
 83  GKDiving                  18159 non-null  float64
 84  GKHandling                18159 non-null  float64
 85  GKKicking                 18159 non-null  float64
 86  GKPositioning             18159 non-null  float64
 87  GKReflexes                18159 non-null  float64
 88  Release Clause            16643 non-null  object
dtypes: float64(38), int64(6), object(45)
memory usage: 12.4+ MB
```

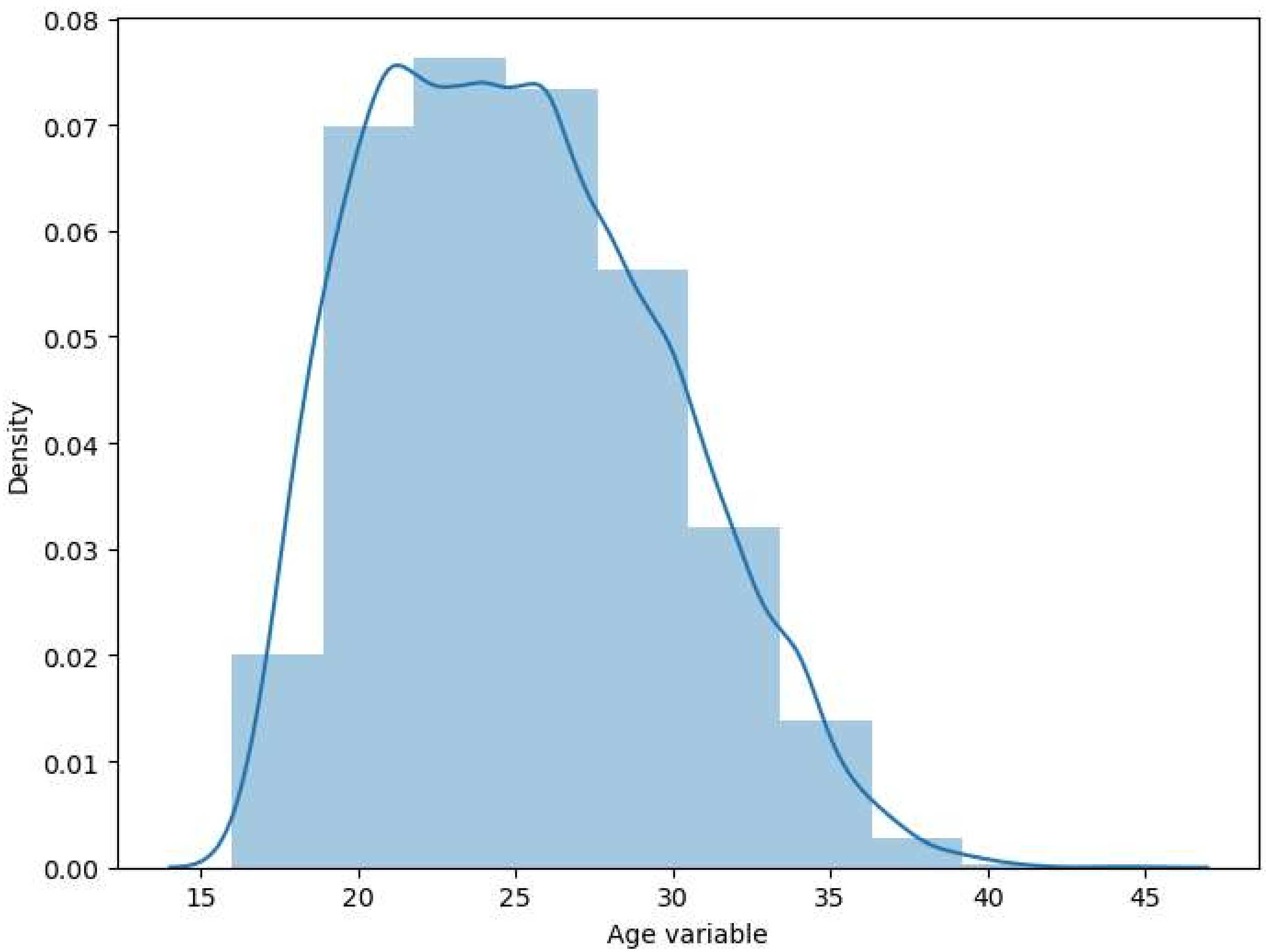
In [8]:
```python
fifa['Body Type'].value_counts()
```

Out[8]:
```
Body Type
Normal                 10595
Lean                    6417
Stocky                  1140
Messi                      1
C. Ronaldo                 1
Neymar                     1
Courtois                   1
PLAYER_BODY_TYPE_25        1
Shaqiri                    1
Akinfenwa                  1
Name: count, dtype: int64
```

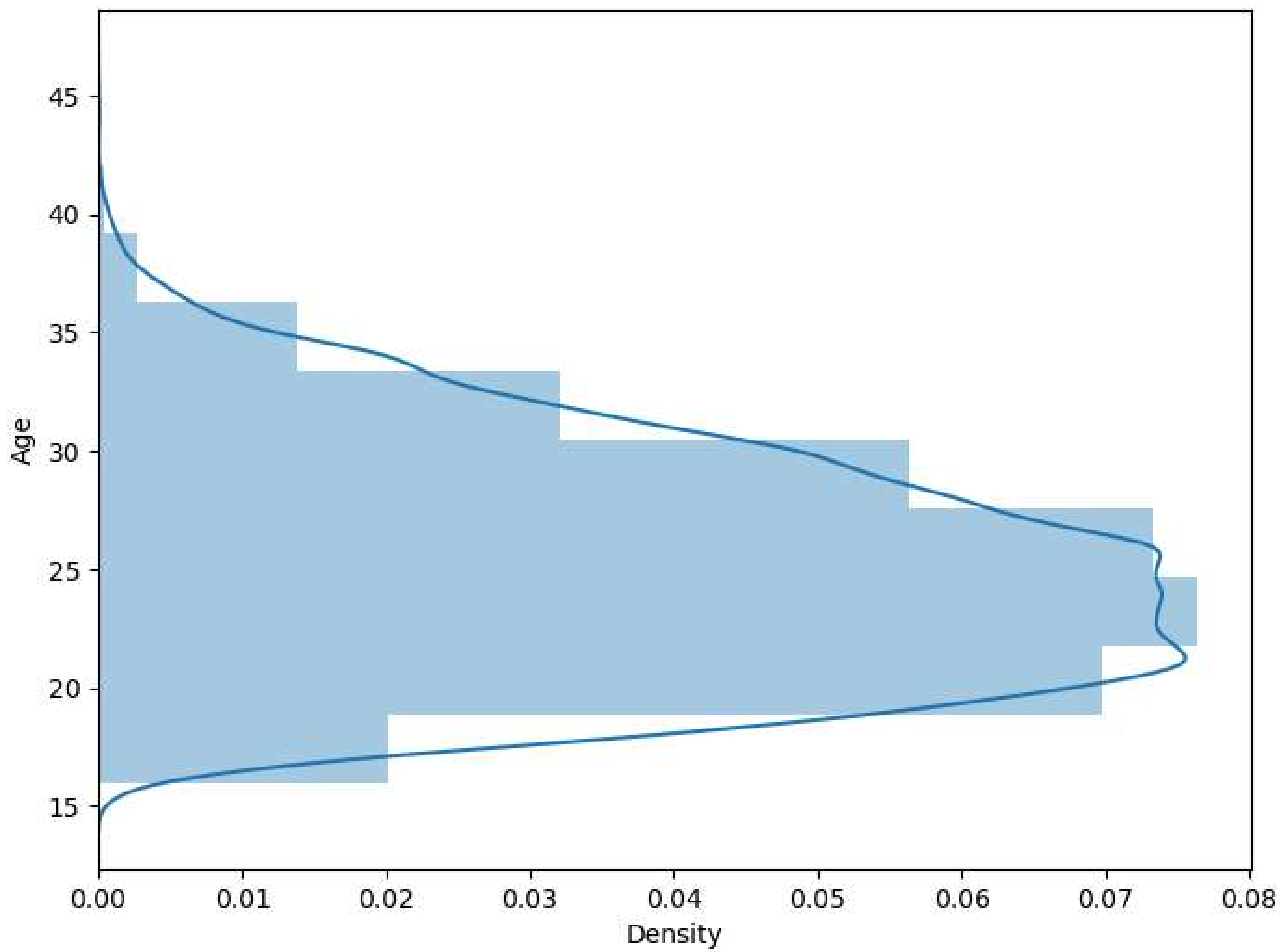
In [9]:
```python
f, ax = plt.subplots(figsize = (8,6))
x = fifa['Age']
ax = sns.distplot(x, bins = 10)
plt.show()
```



In [10]:
```
f, ax = plt.subplots(figsize = (8,6))
x = fifa['Age']
x = pd.Series(x, name = "Age variable")
ax = sns.distplot(x, bins = 10)
plt.show()
```

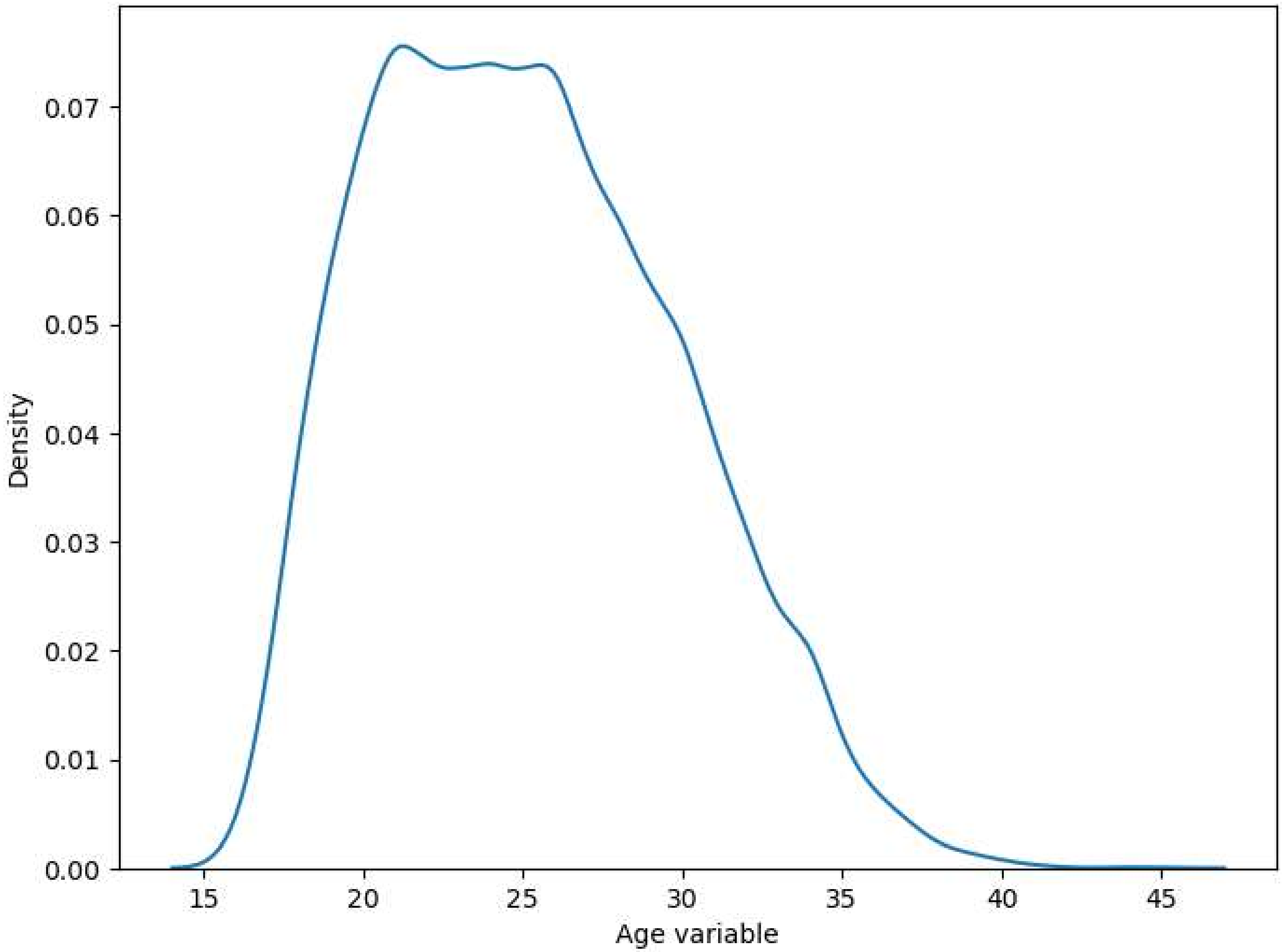


In [11]:
```
f, ax = plt.subplots(figsize = (8,6))
x = fifa['Age']
ax = sns.distplot(x, bins = 10, vertical = True)
plt.show()
```



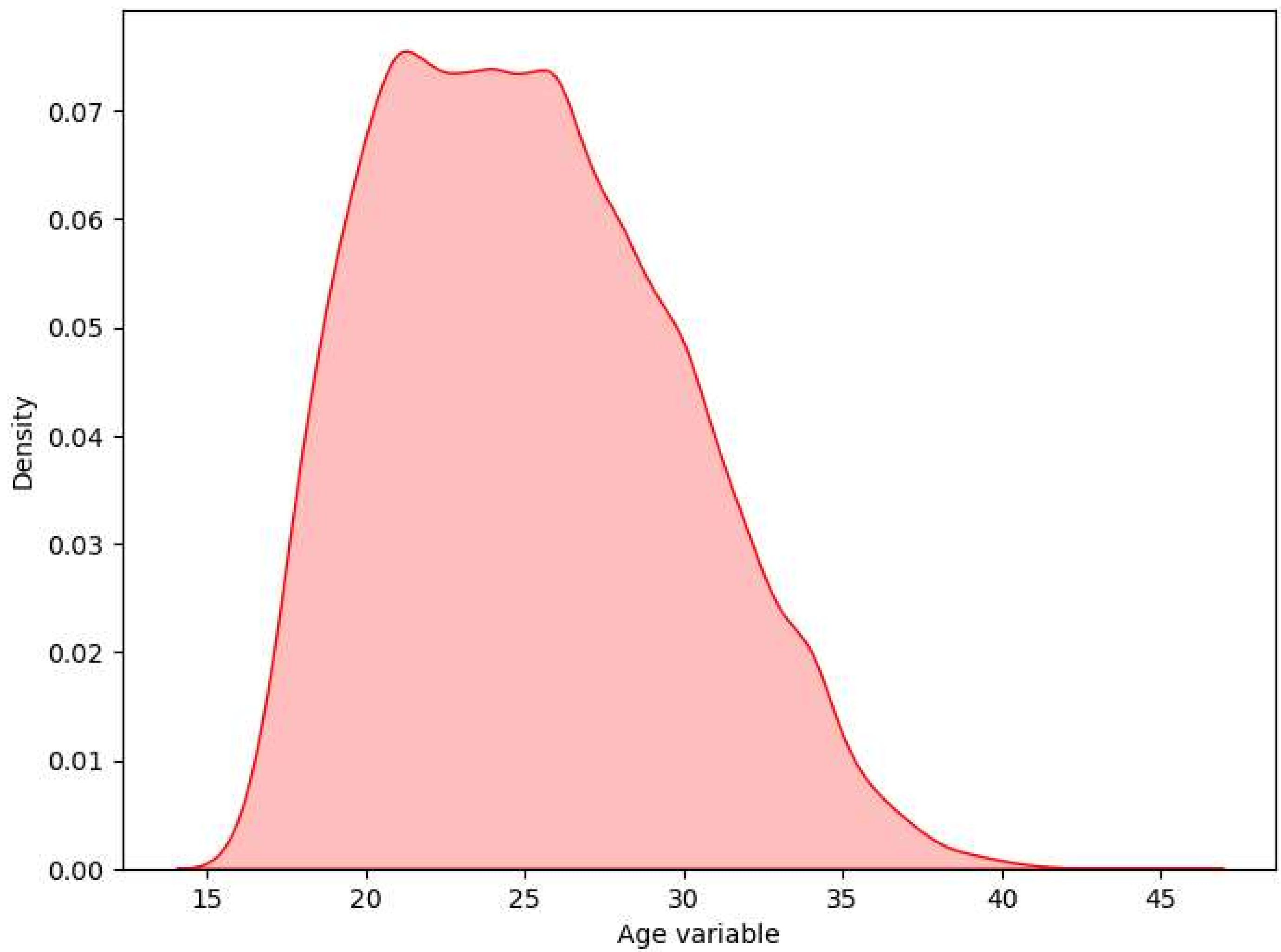
In [12]:

```
f, ax = plt.subplots(figsize = (8,6))
x = fifa['Age']
x = pd.Series(x, name = "Age variable")
ax = sns.kdeplot(x)
plt.show()
```

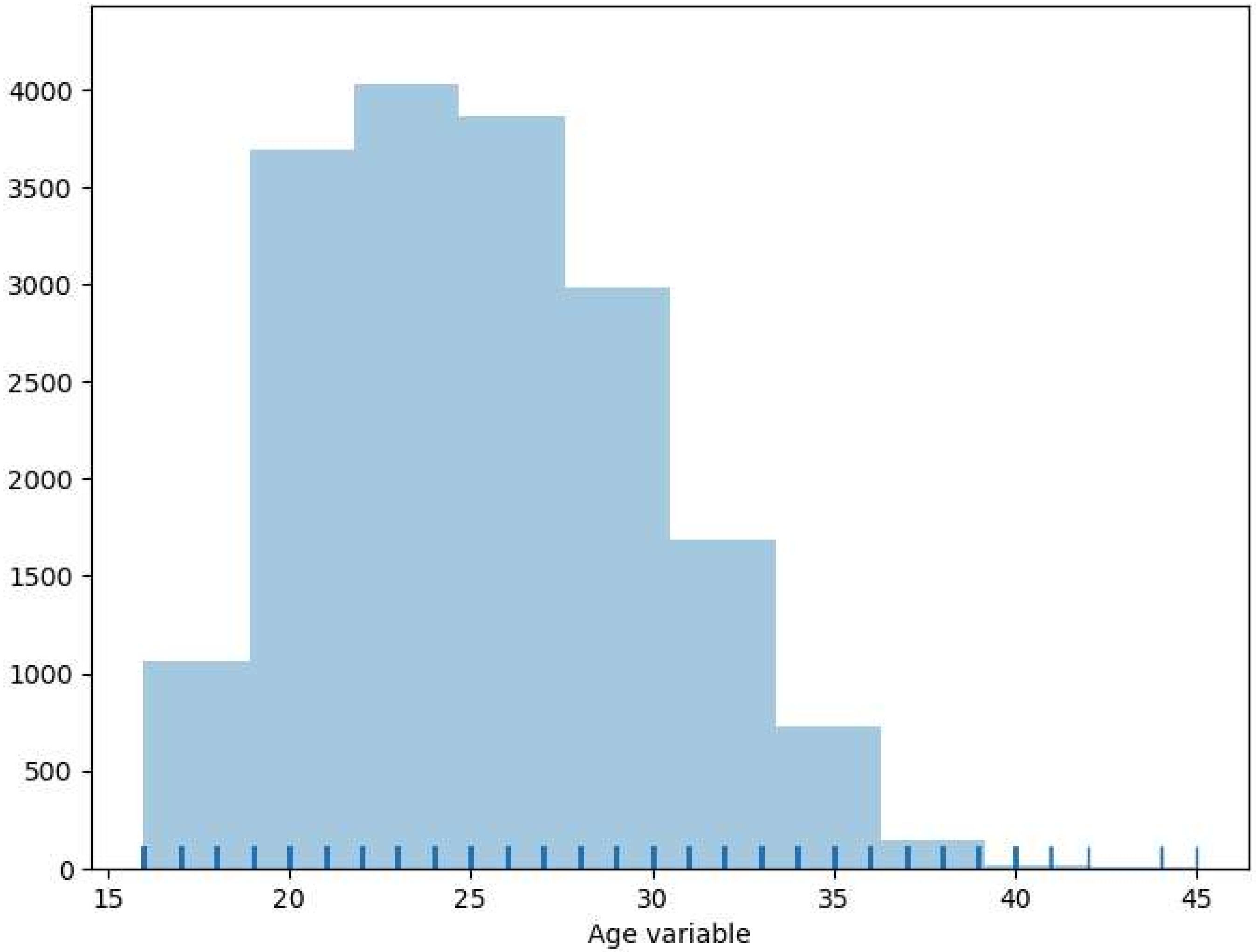


In [13]:

```
f, ax = plt.subplots(figsize = (8,6))
x = fifa['Age']
x = pd.Series(x, name = "Age variable")
ax = sns.kdeplot(x, shade = True, color = 'r')
plt.show()
```

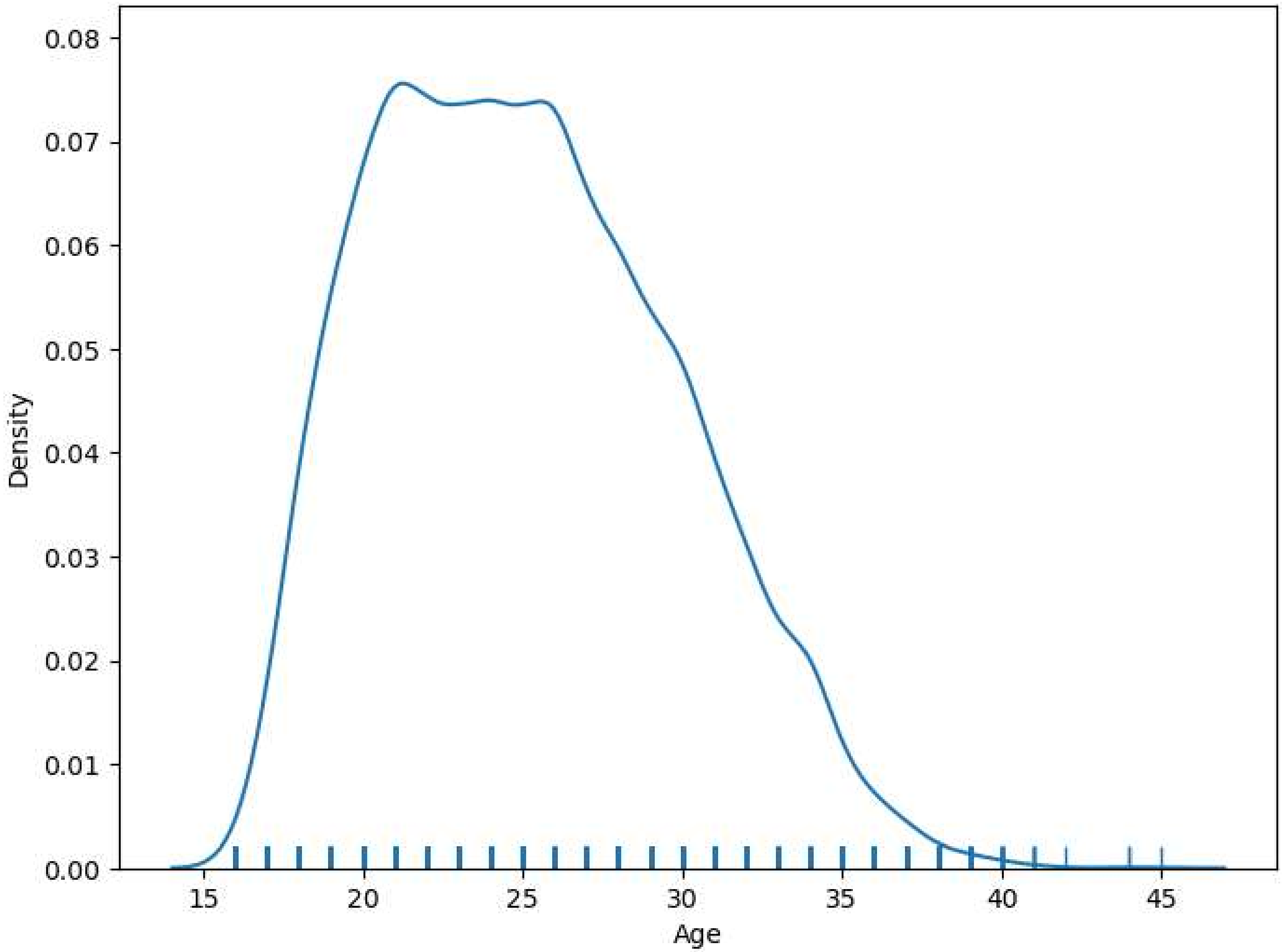


In [14]:
```
f, ax = plt.subplots(figsize = (8,6))
x = fifa['Age']
x = pd.Series(x, name = "Age variable")
ax = sns.distplot(x, kde = False, rug = True, bins = 10)
plt.show()
```



In [17]:

```
f, ax = plt.subplots(figsize = (8,6))
x = fifa['Age']
ax = sns.distplot(x, hist = False, rug = True, bins = 10)
plt.show()
```



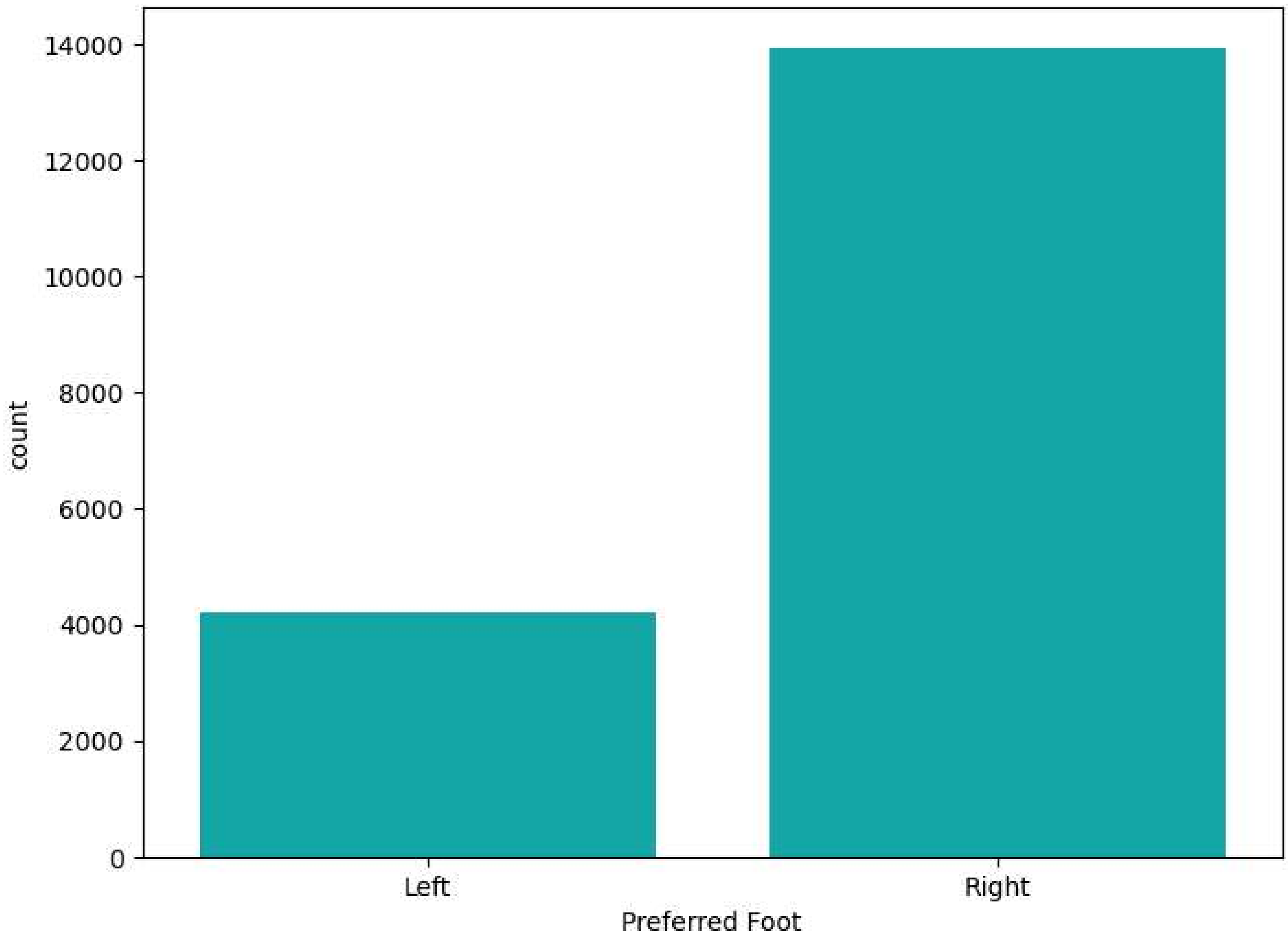
In [18]:
```
fifa['Preferred Foot'].nunique()
```

Out[18]: 2

In [19]:
```
fifa['Preferred Foot'].value_counts()
```

Out[19]:
```
Preferred Foot
Right    13948
Left      4211
Name: count, dtype: int64
```

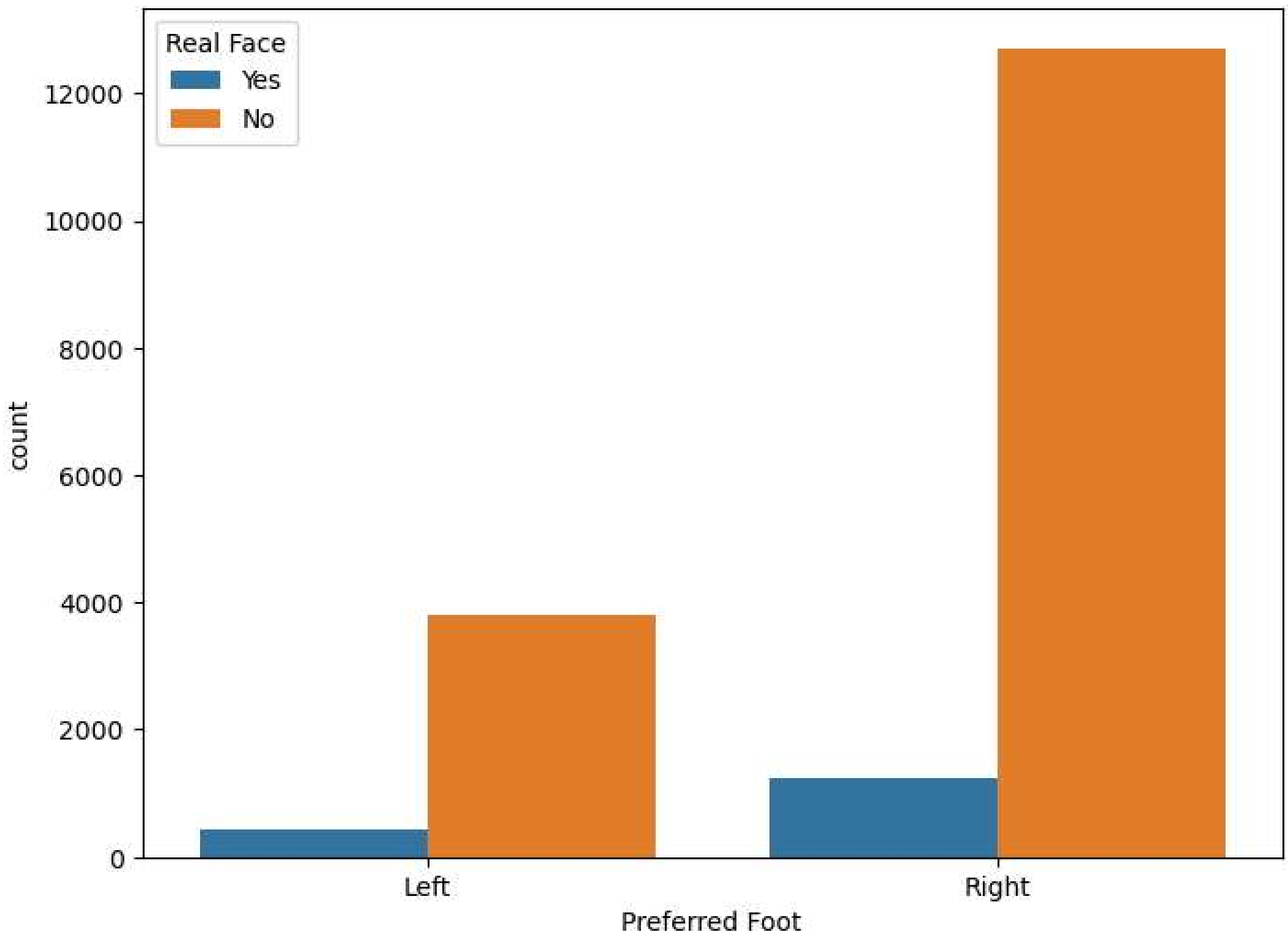
In [20]:
```
f, ax = plt.subplots(figsize = (8,6))
ax = sns.countplot(x = "Preferred Foot", data = fifa, color = 'c')
plt.show()
```



In [22]:
```
fifa.groupby('Preferred Foot')['Real Face'].value_counts()
```

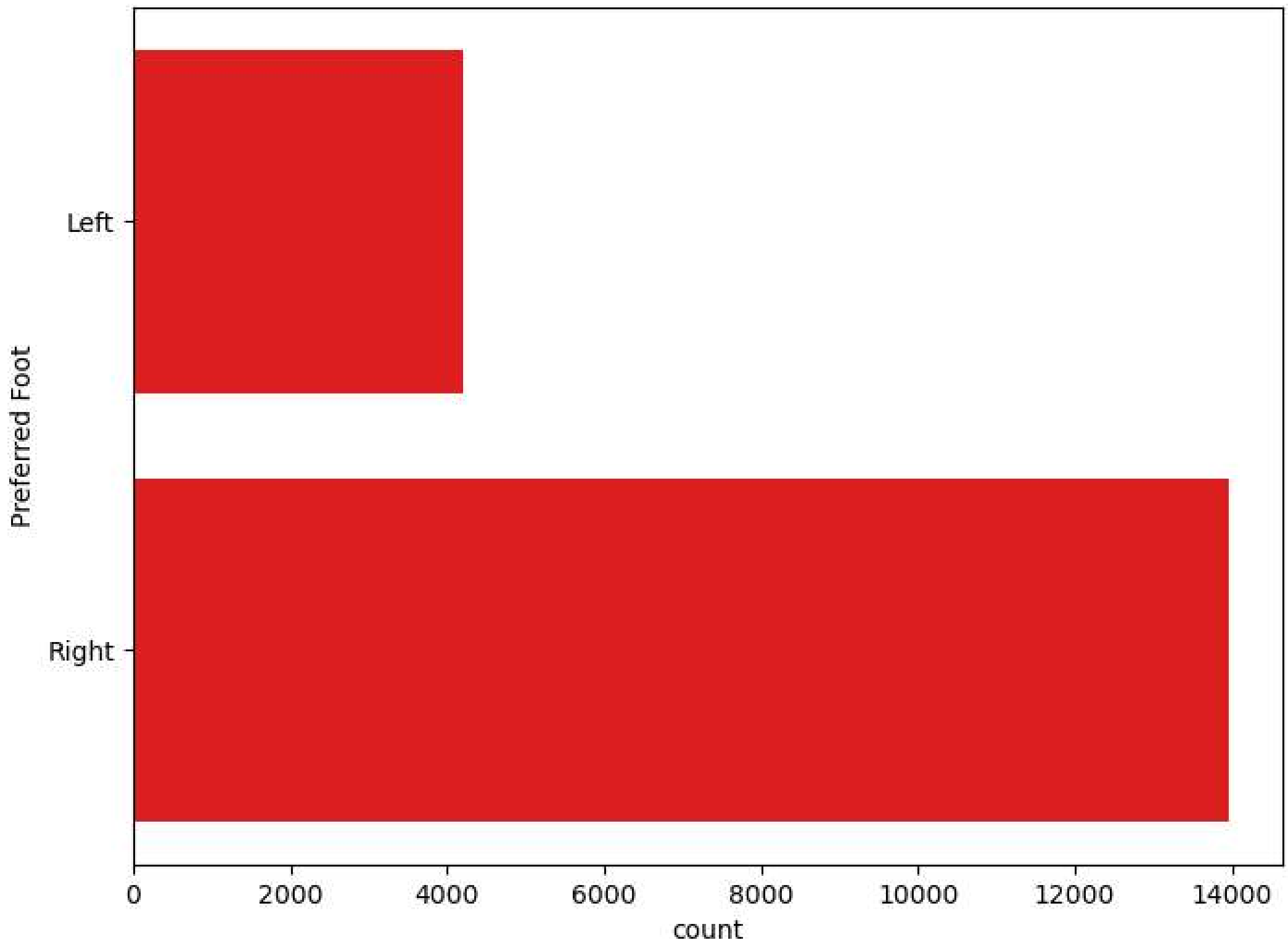
Out[22]:
```
Preferred Foot  Real Face
Left            No           3796
                Yes           415
Right           No          12709
                Yes          1239
Name: count, dtype: int64
```

In [23]:
```python
f, ax = plt.subplots(figsize = (8, 6))
sns.countplot(x = "Preferred Foot", hue = "Real Face", data = fifa)
plt.show()
```



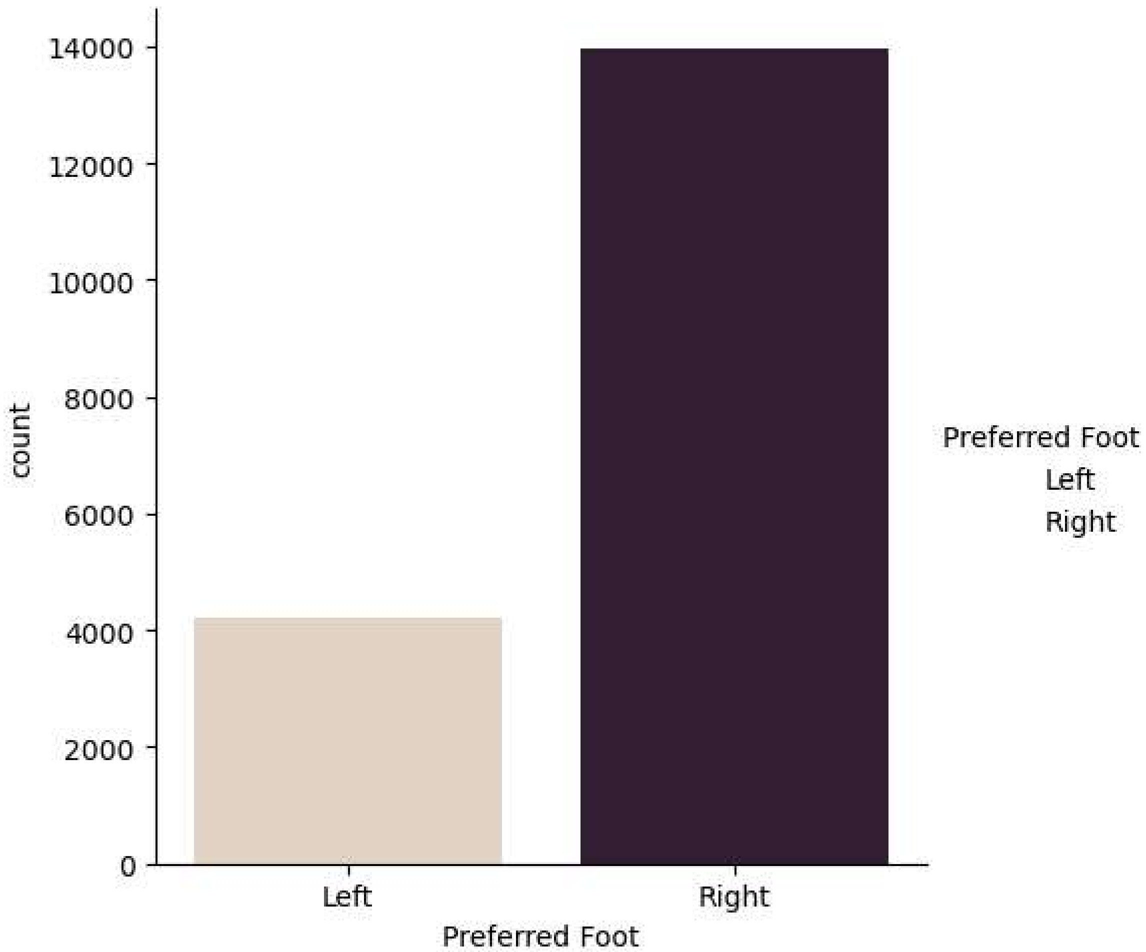
In [24]:

```
f, ax = plt.subplots(figsize = (8, 6))
sns.countplot(y = "Preferred Foot", data = fifa, color = 'r')
plt.show()
```



In [25]:

```
g = sns.catplot(x = "Preferred Foot", kind = 'count', data = fifa, palette = 'ch:0.25')
plt.show()
```



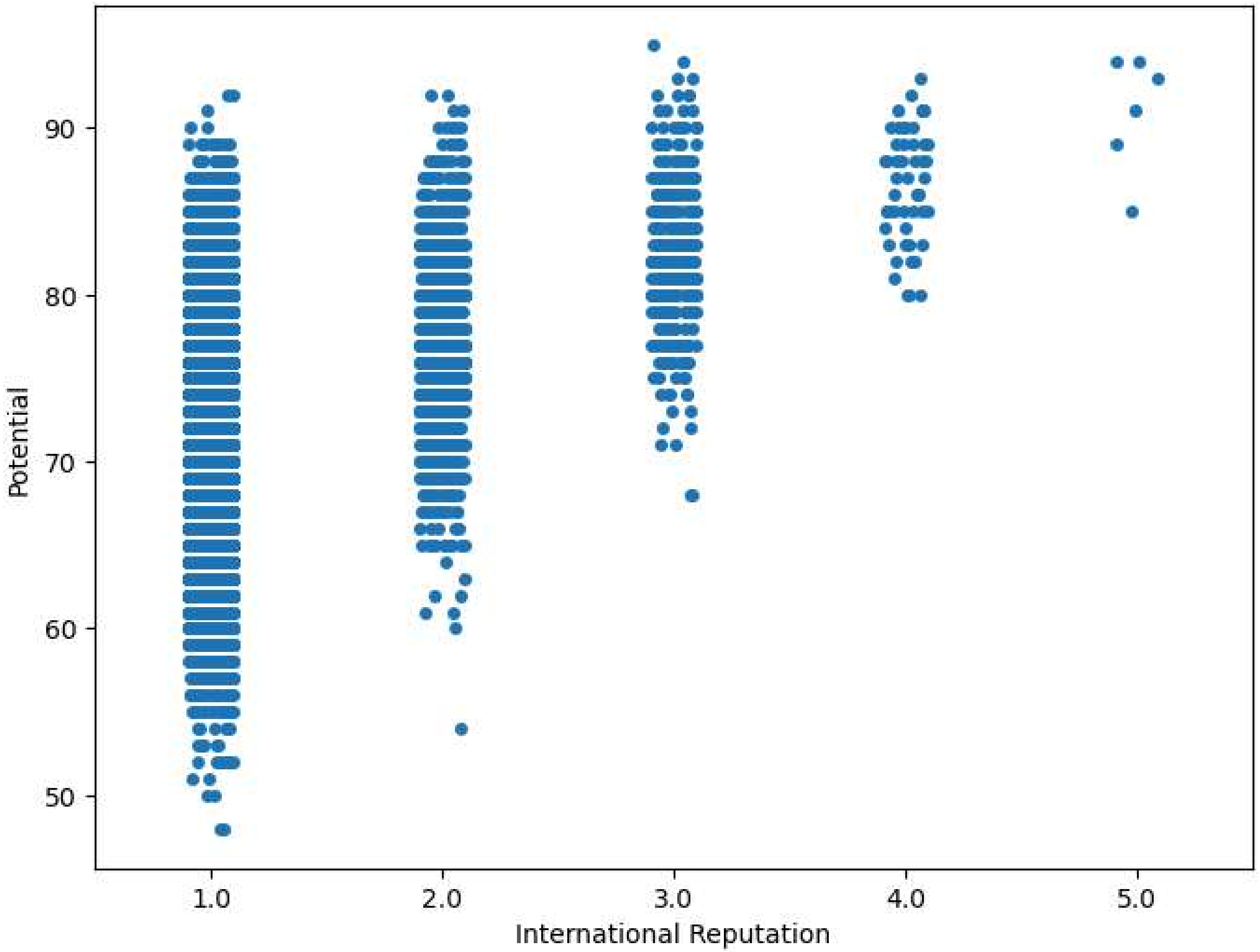
In [26]:
```
fifa['International Reputation'].nunique()
```

Out[26]: 5

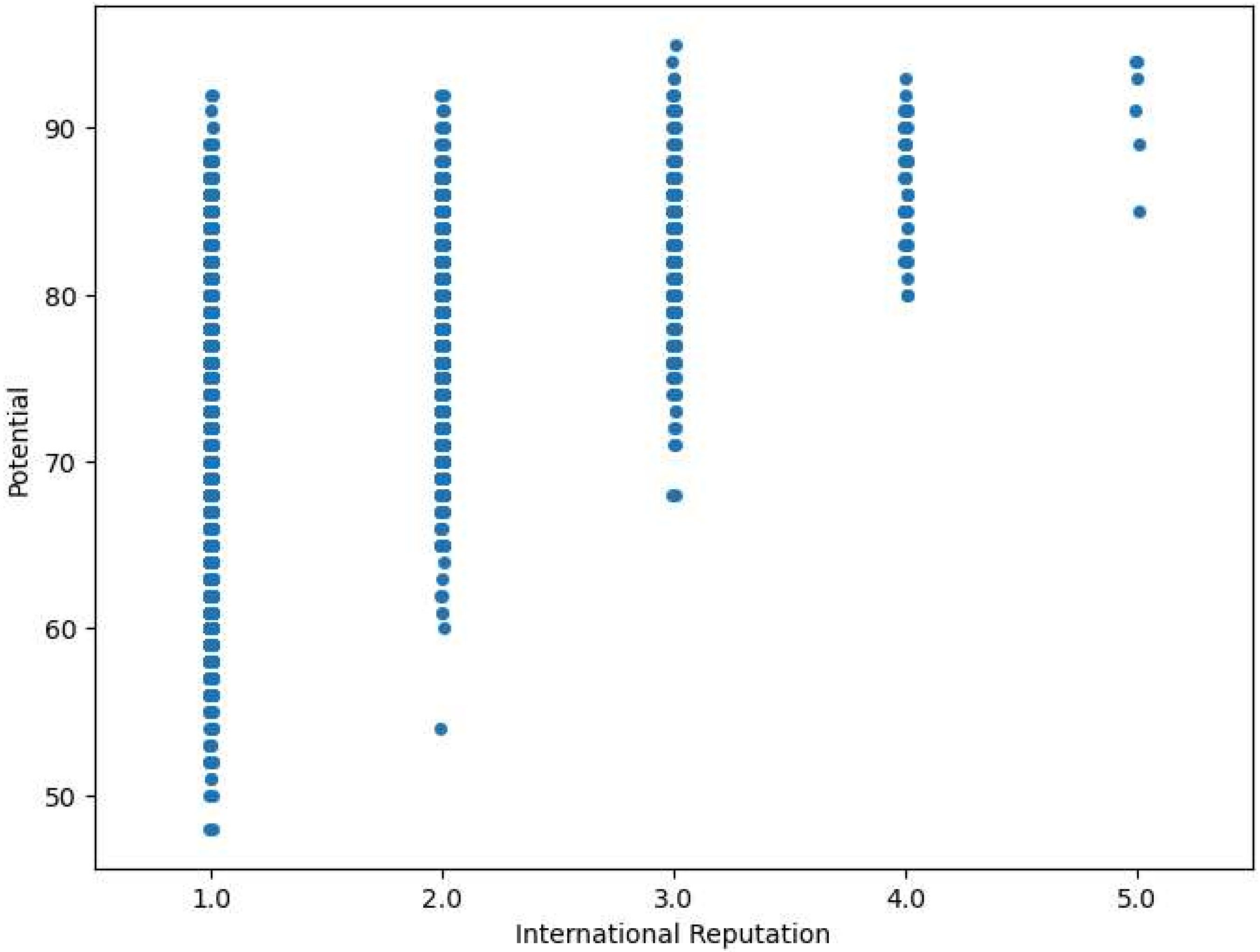
In [27]:
```
fifa['International Reputation'].value_counts()
```

Out[27]:
```
International Reputation
1.0    16532
2.0     1261
3.0      309
4.0       51
5.0        6
Name: count, dtype: int64
```

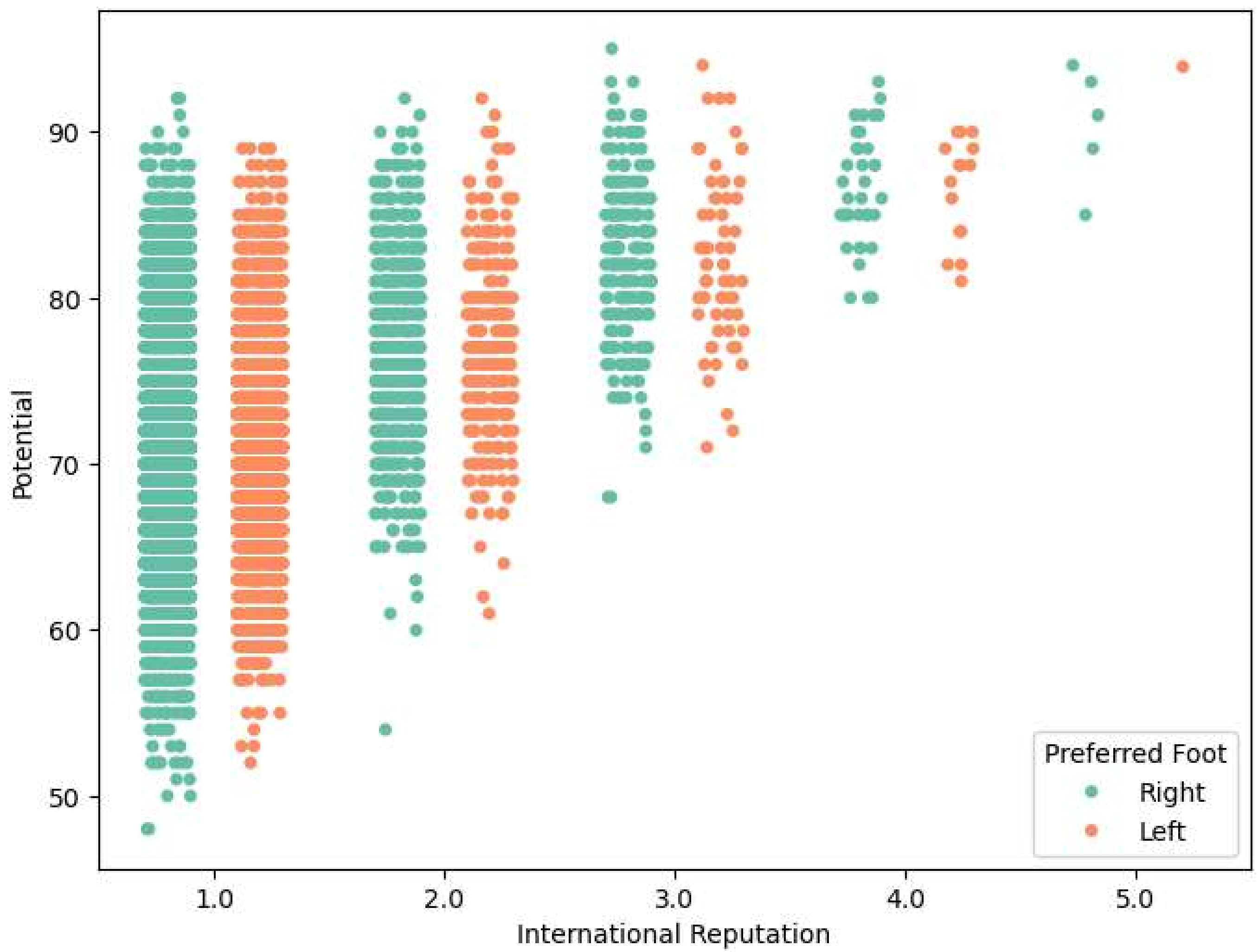
In [29]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.stripplot(x = "International Reputation", y = "Potential", data = fifa)
plt.show()
```



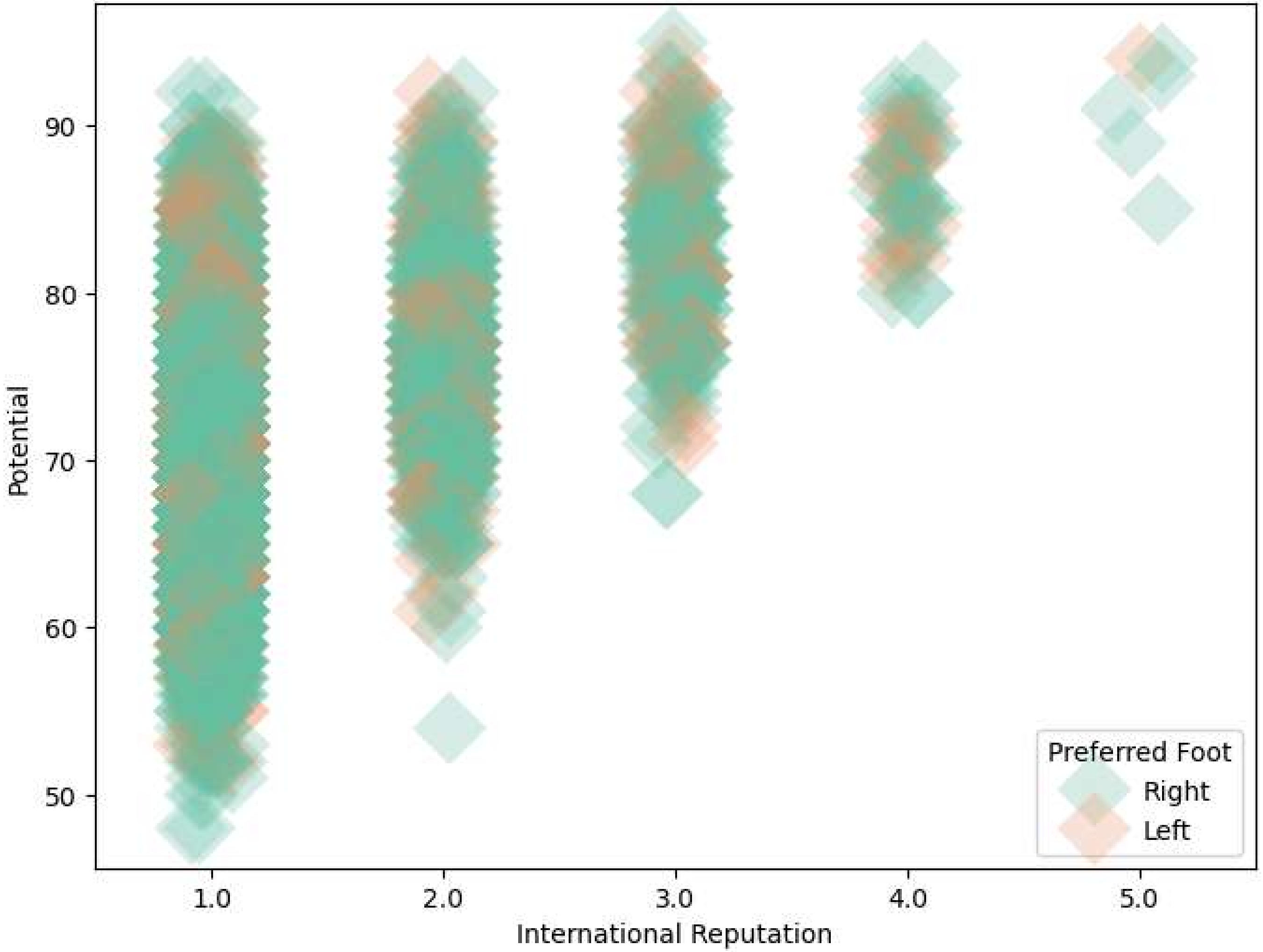
In [30]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.stripplot(x = "International Reputation", y = "Potential", data = fifa, jitter = 0.01)
plt.show()
```



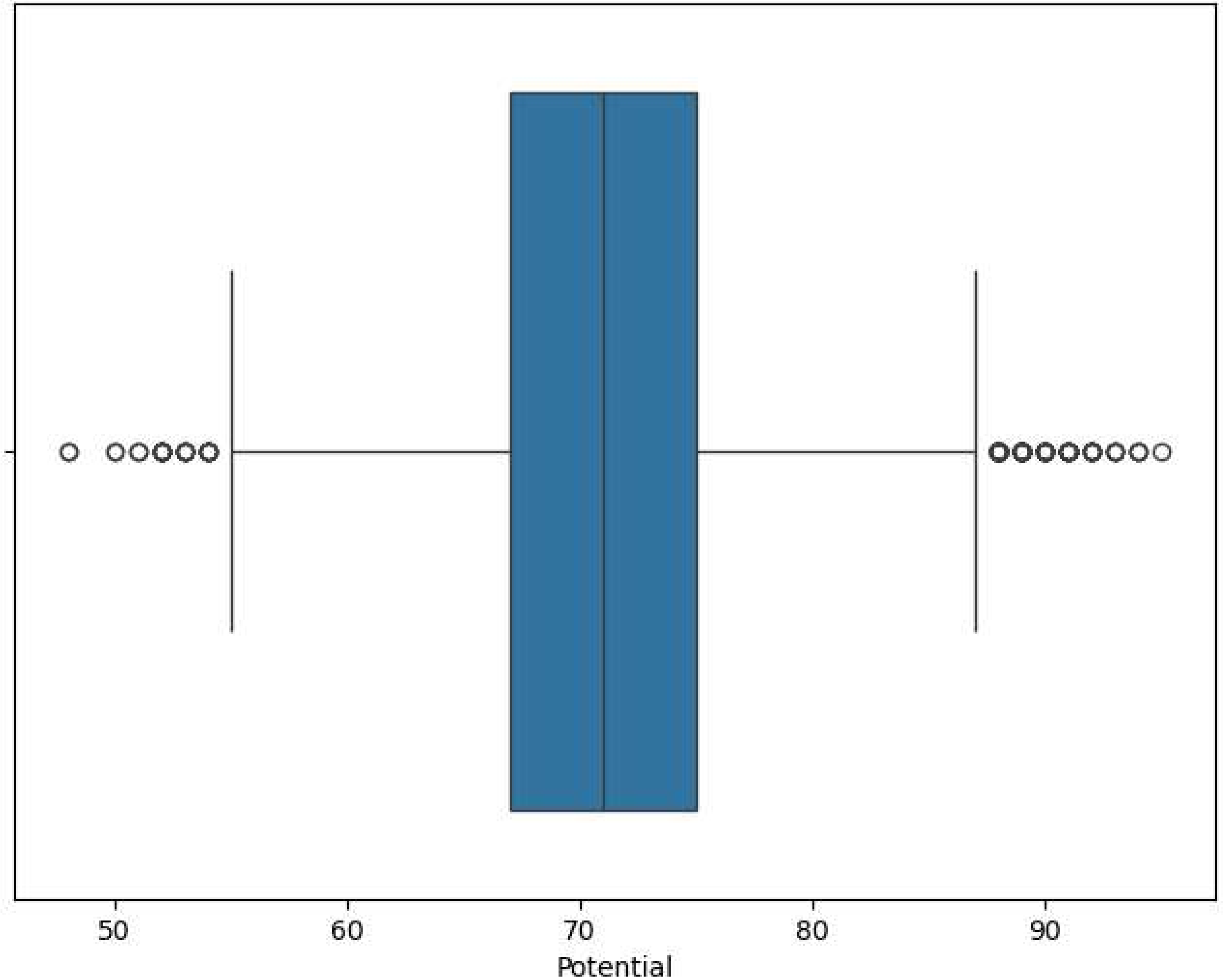
In [31]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.stripplot(x = "International Reputation", y = "Potential", hue = "Preferred Foot", data = fifa, jitter = 0.2, pa
plt.show()
```



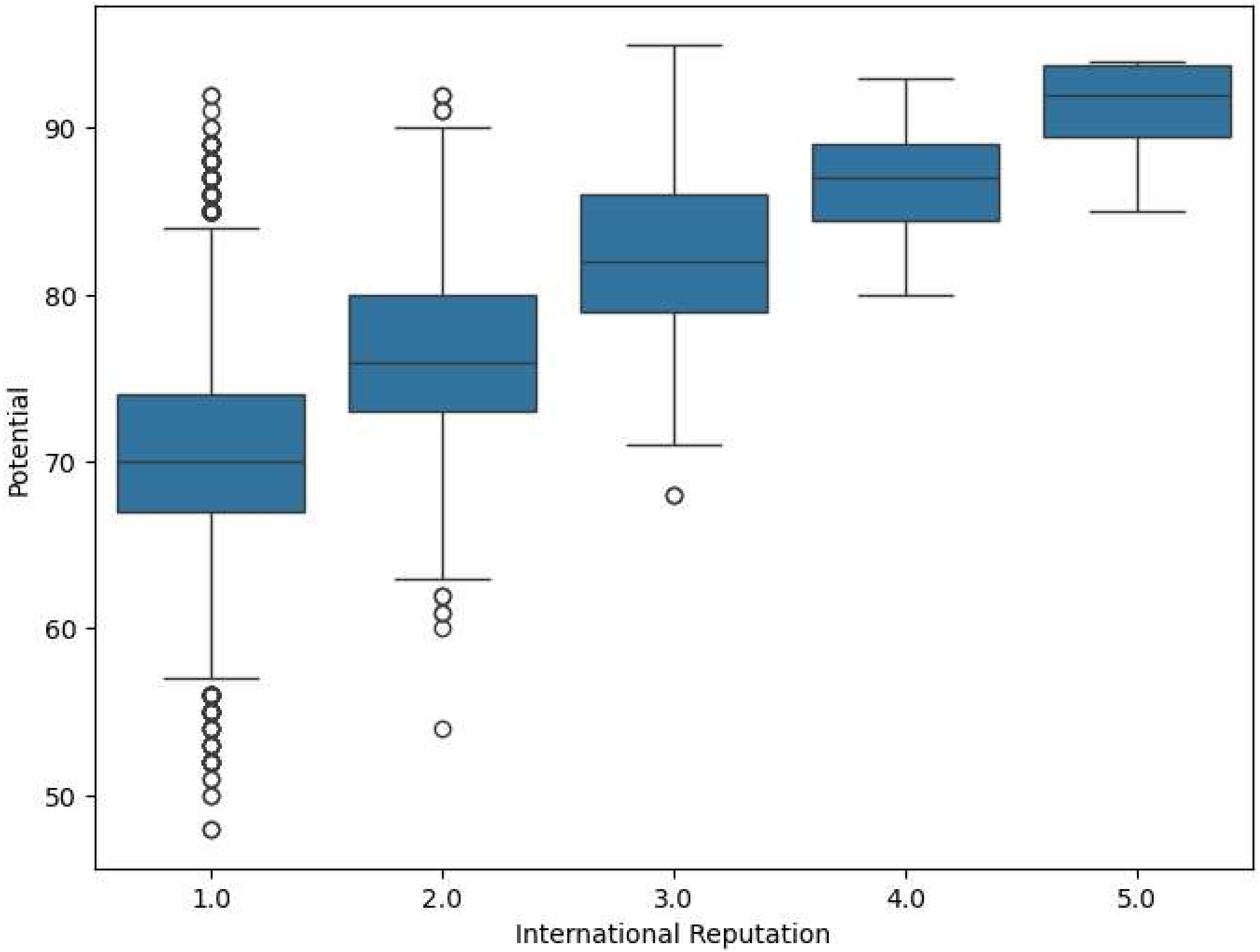
In [35]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.stripplot(x = "International Reputation", y = "Potential", hue = "Preferred Foot", data = fifa, size = 20, palett
plt.show()
```



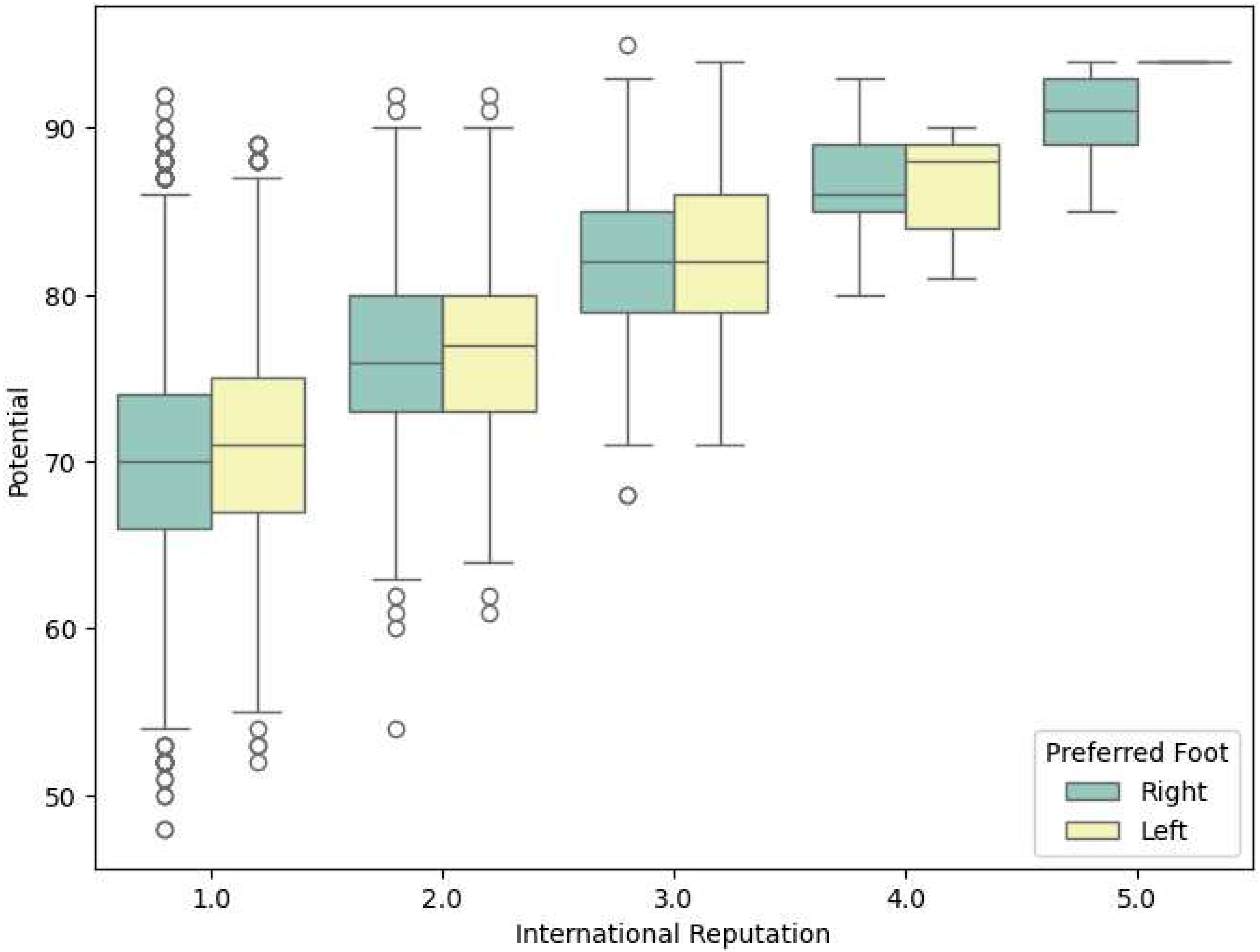
In [36]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.boxplot(x = fifa['Potential'])
plt.show()
```



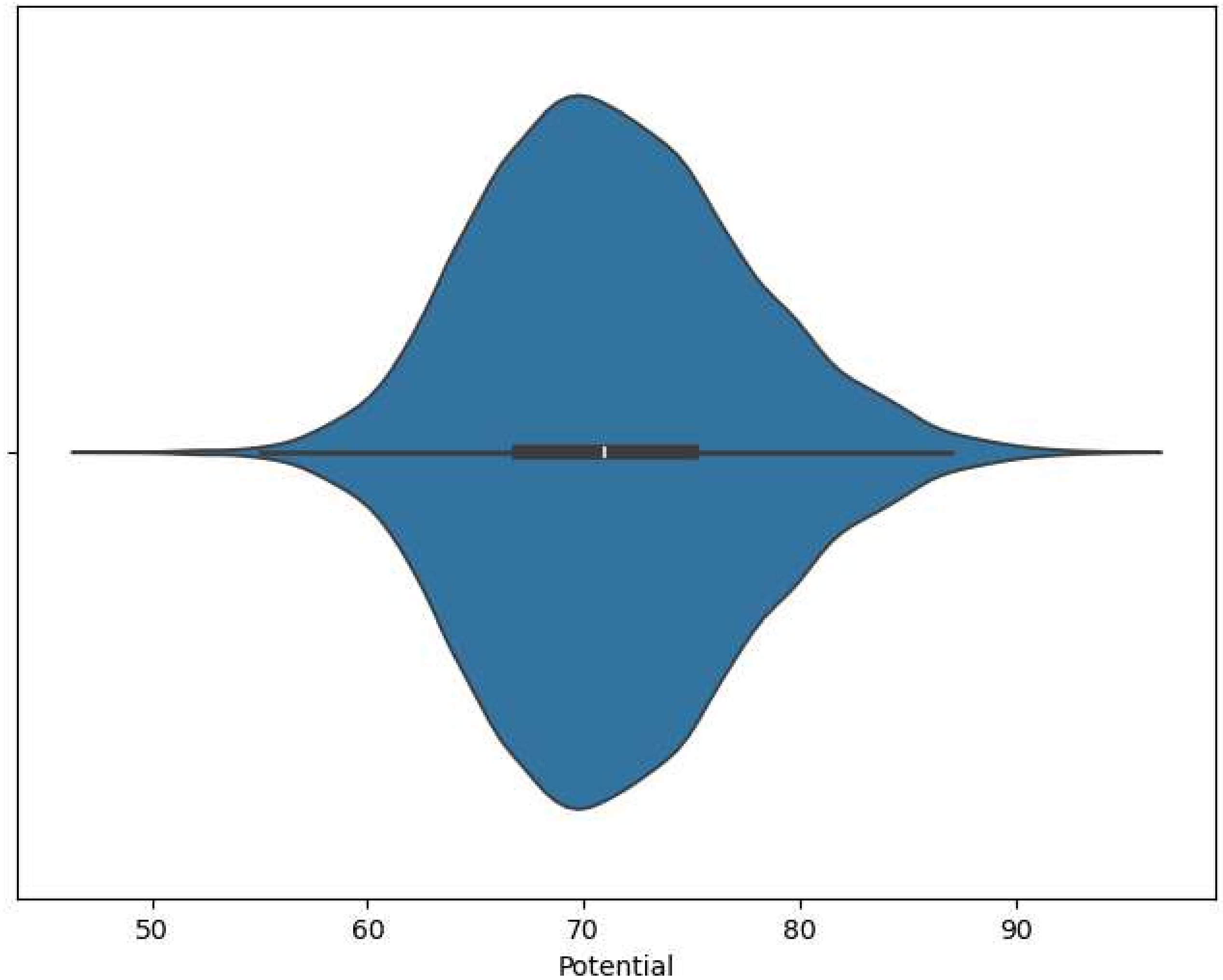
In [37]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.boxplot(x = "International Reputation", y = "Potential", data = fifa)
plt.show()
```



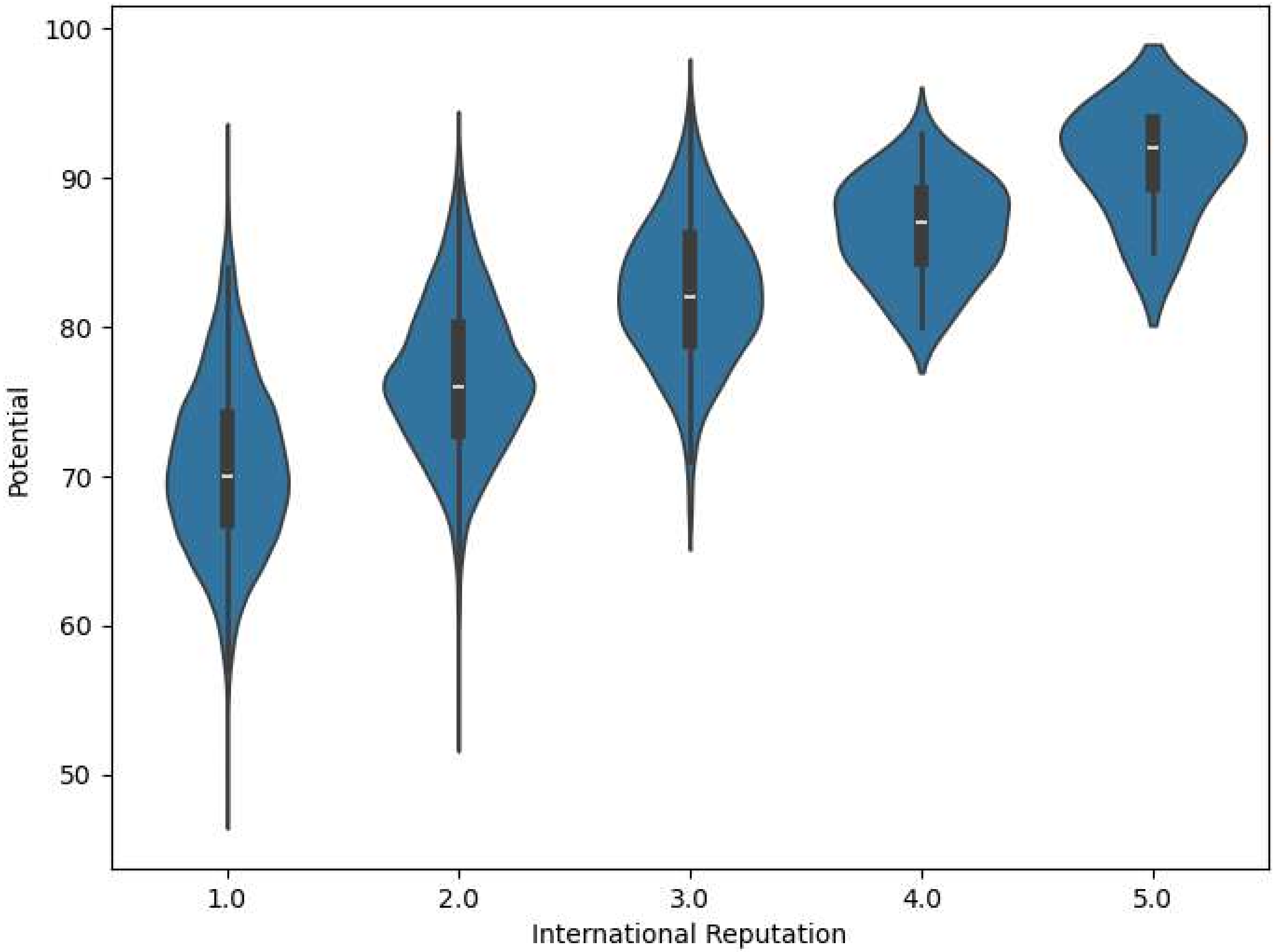
In [38]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.boxplot(x = "International Reputation", y = "Potential", hue = "Preferred Foot", palette = "Set3", data = fifa)
plt.show()
```



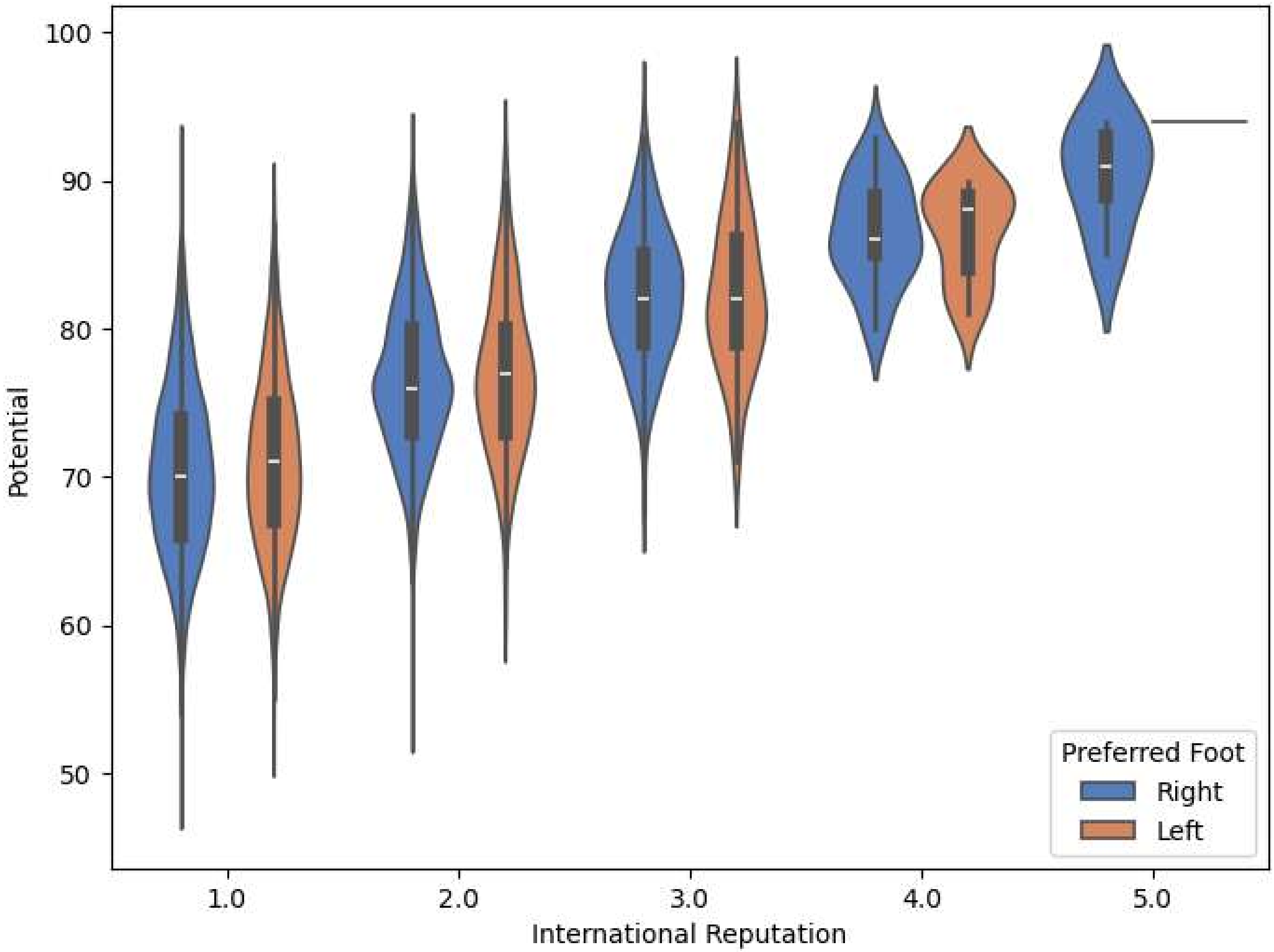
In [39]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.violinplot(x = fifa['Potential'])
plt.show()
```



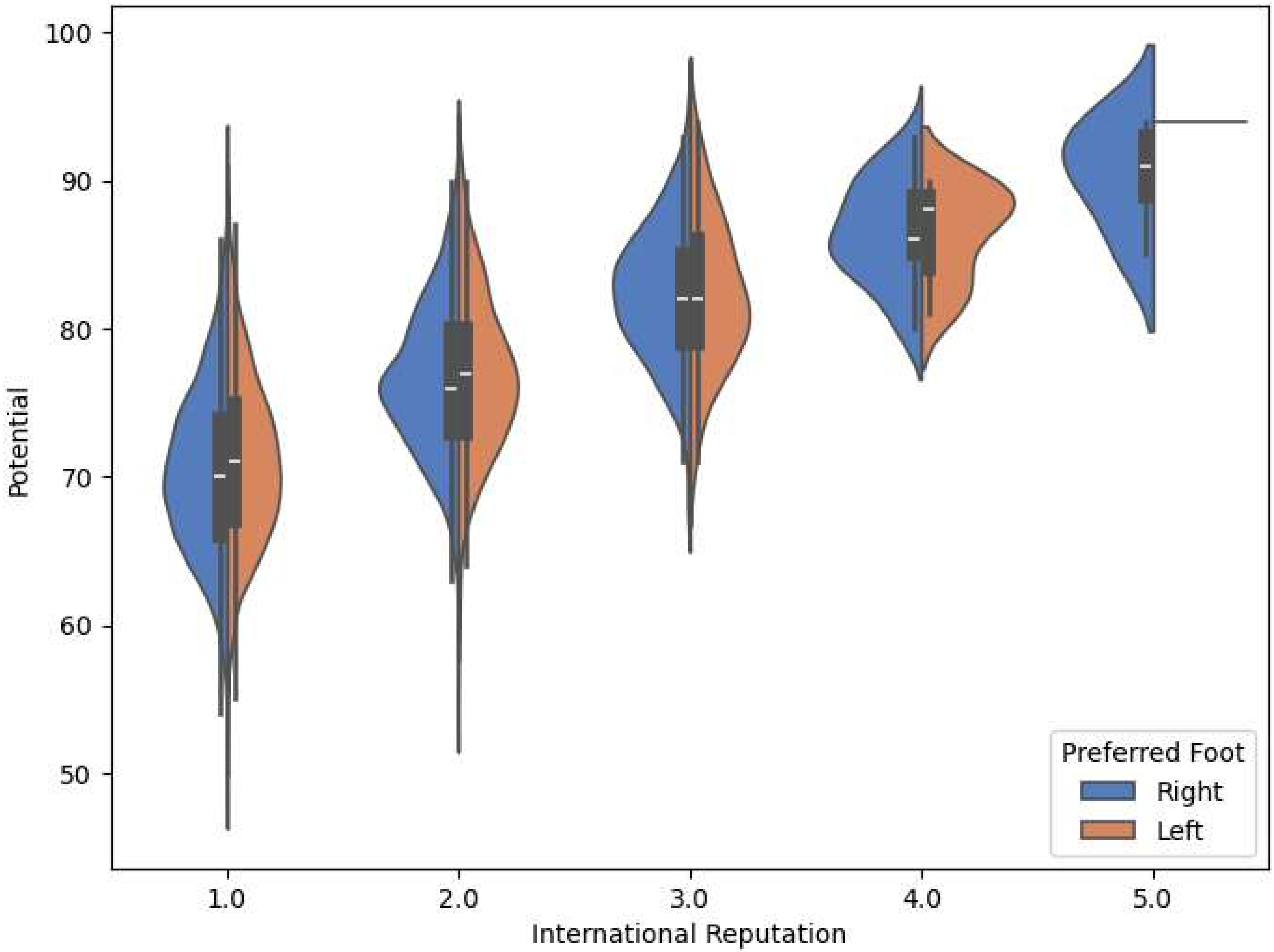
In [40]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.violinplot(x = "International Reputation", y = "Potential", data = fifa)
plt.show()
```



In [41]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.violinplot(x = "International Reputation", y = "Potential", hue = "Preferred Foot", data = fifa, palette = "mutec
plt.show()
```

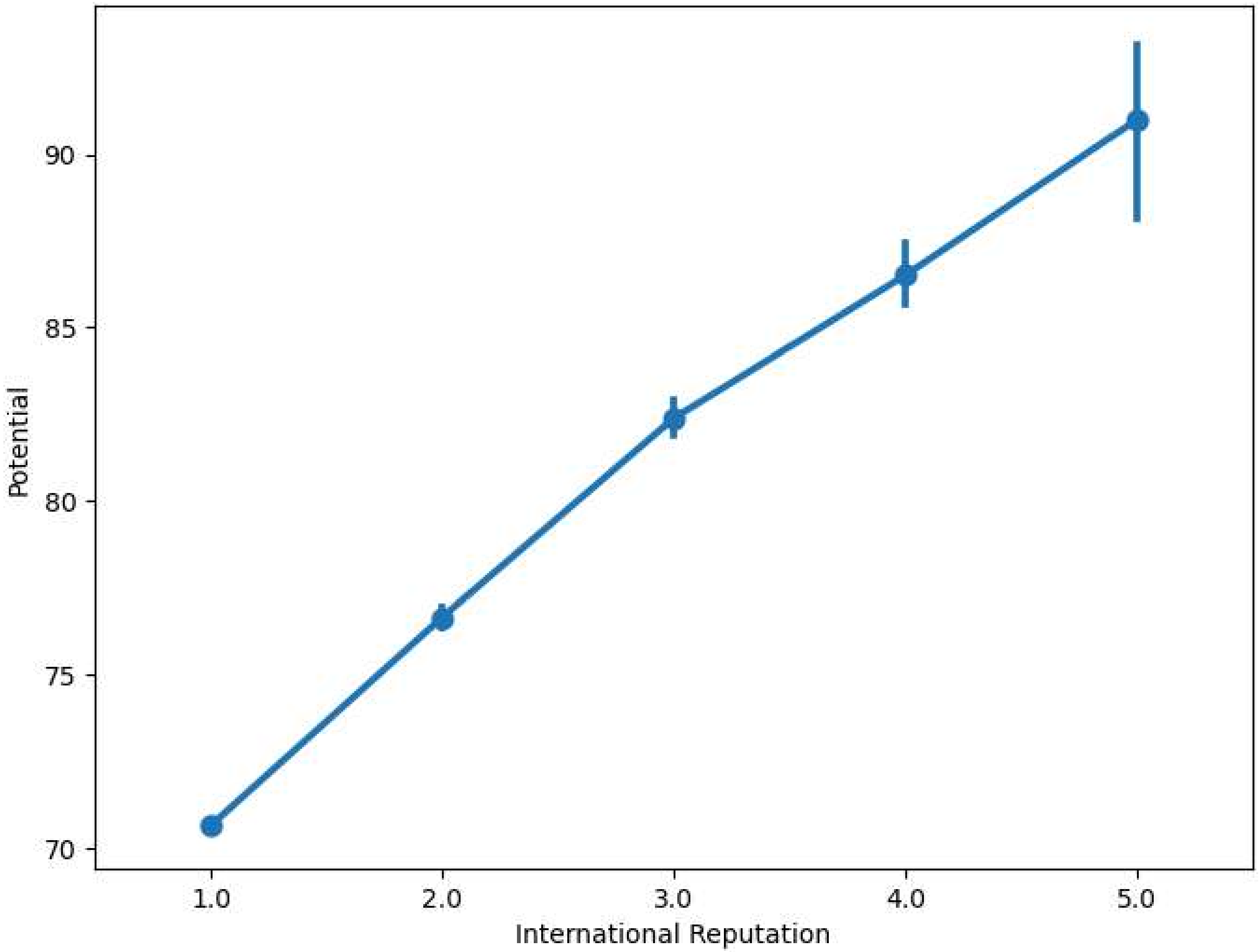


In [42]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.violinplot(x = "International Reputation", y = "Potential", hue = "Preferred Foot", data = fifa, palette = "mutec
plt.show()
```

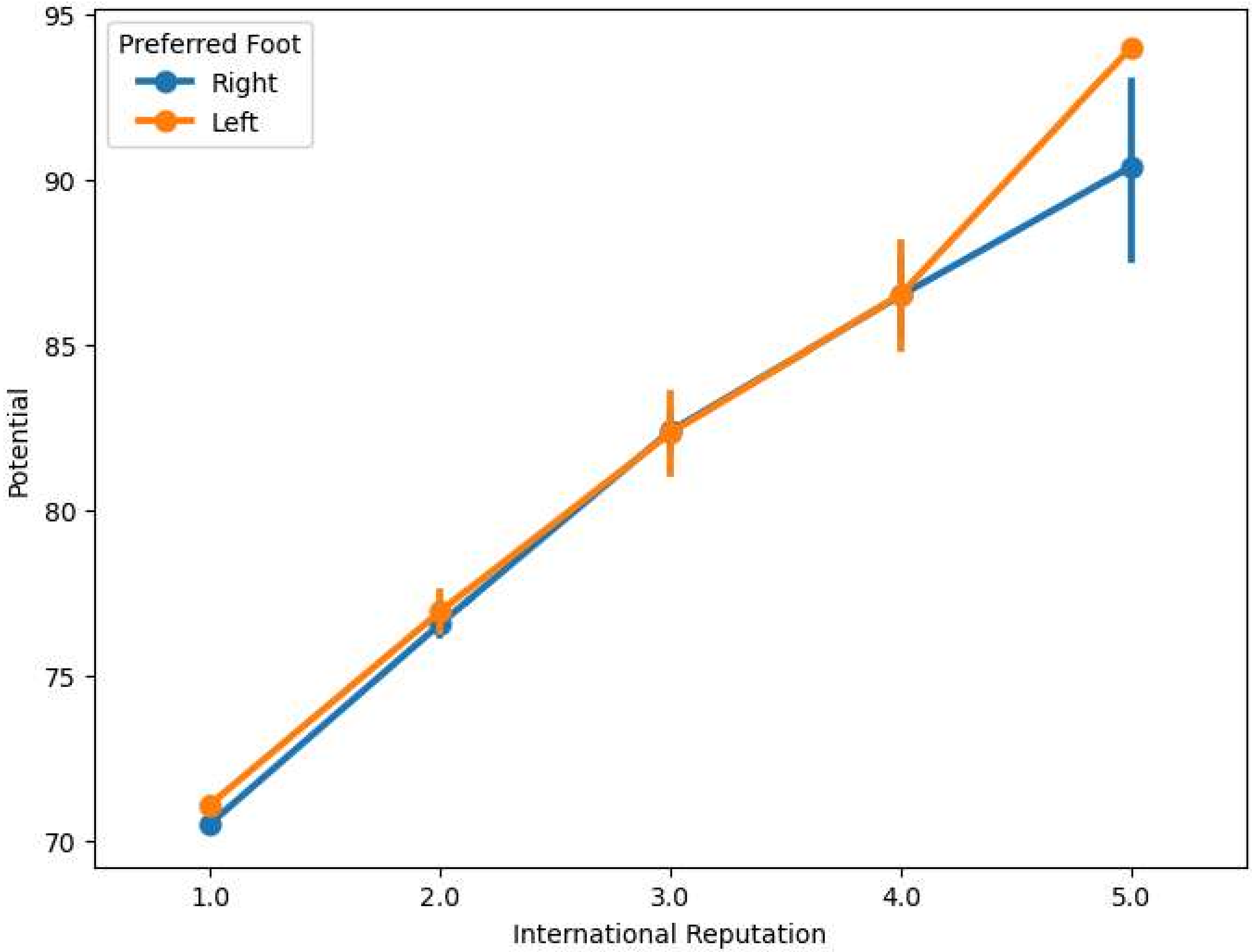


In [43]:

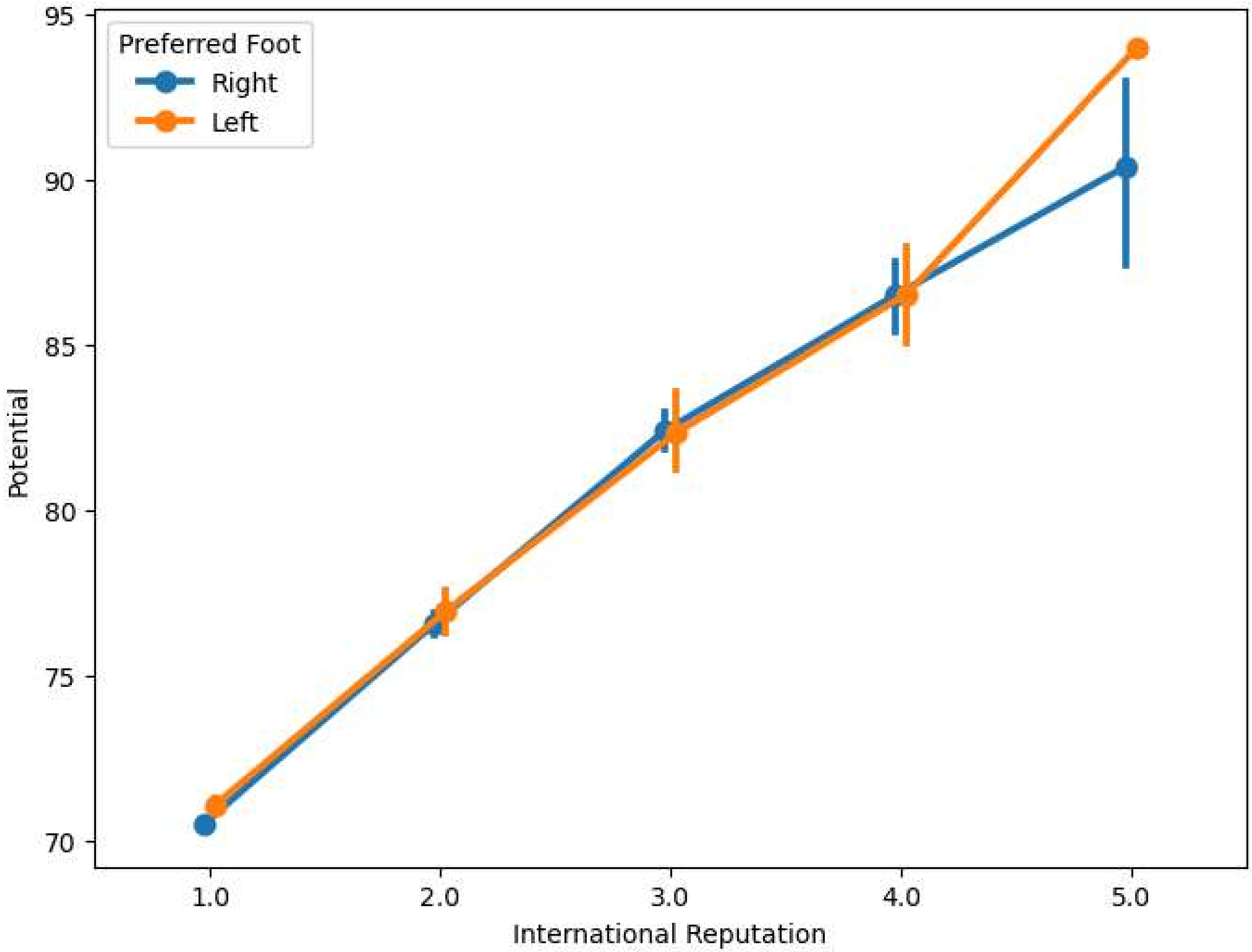
```
f, ax = plt.subplots(figsize = (8, 6))
sns.pointplot(x = "International Reputation", y = "Potential", data = fifa)
plt.show()
```



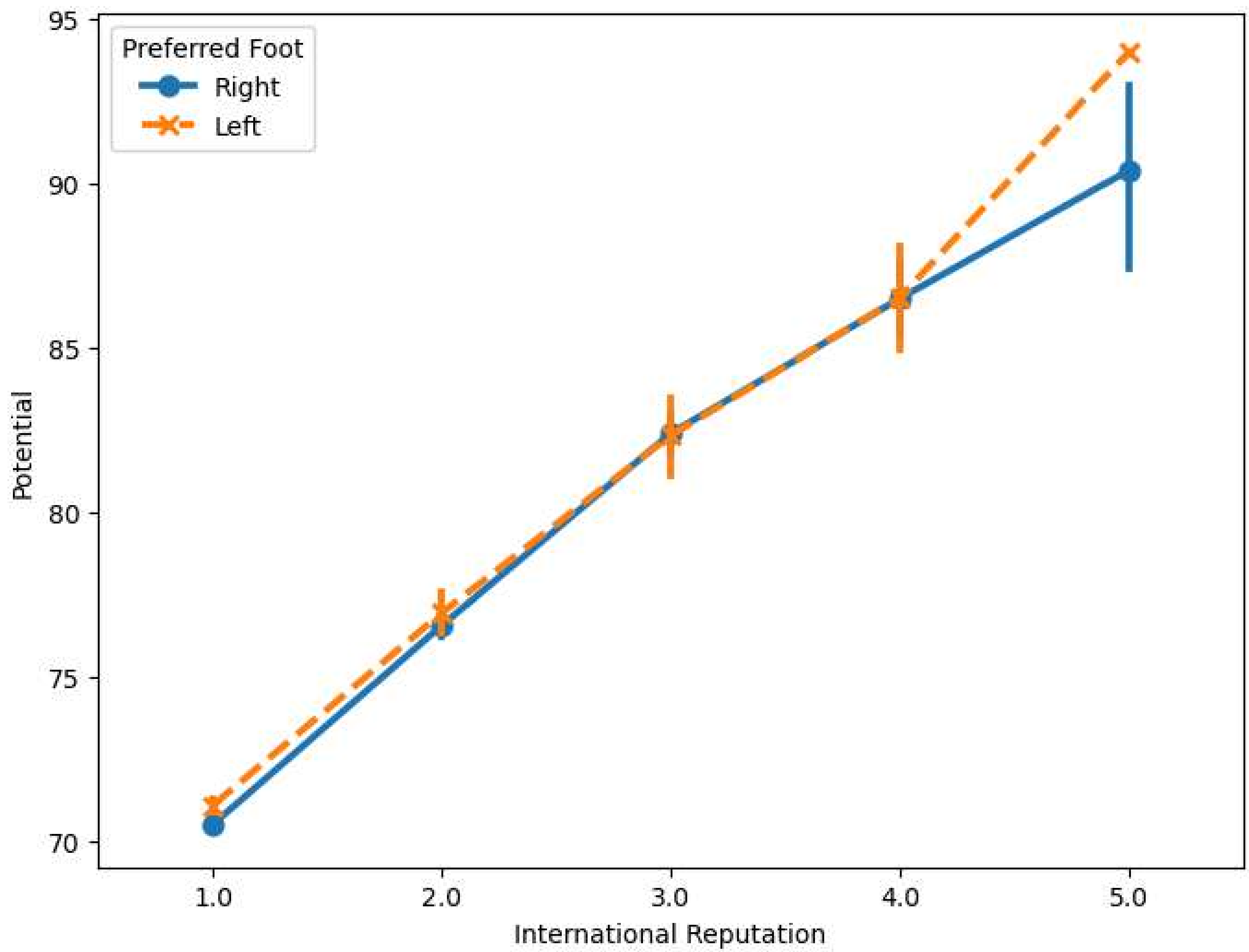
In [44]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.pointplot(x = "International Reputation", y = "Potential", hue = "Preferred Foot", data = fifa)
plt.show()
```



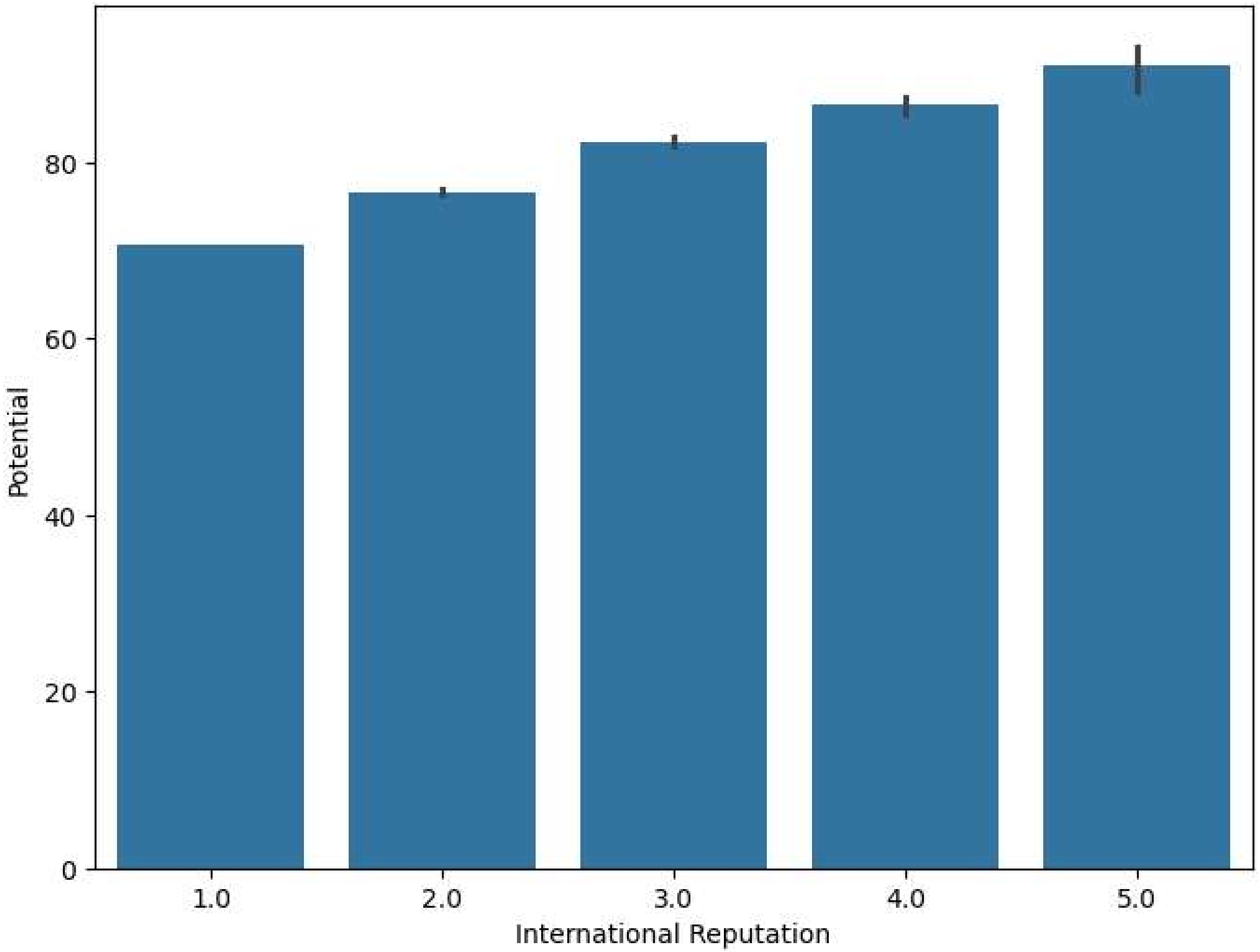
In [45]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.pointplot(x = "International Reputation", y = "Potential", hue = "Preferred Foot", data = fifa, dodge = True)
plt.show()
```



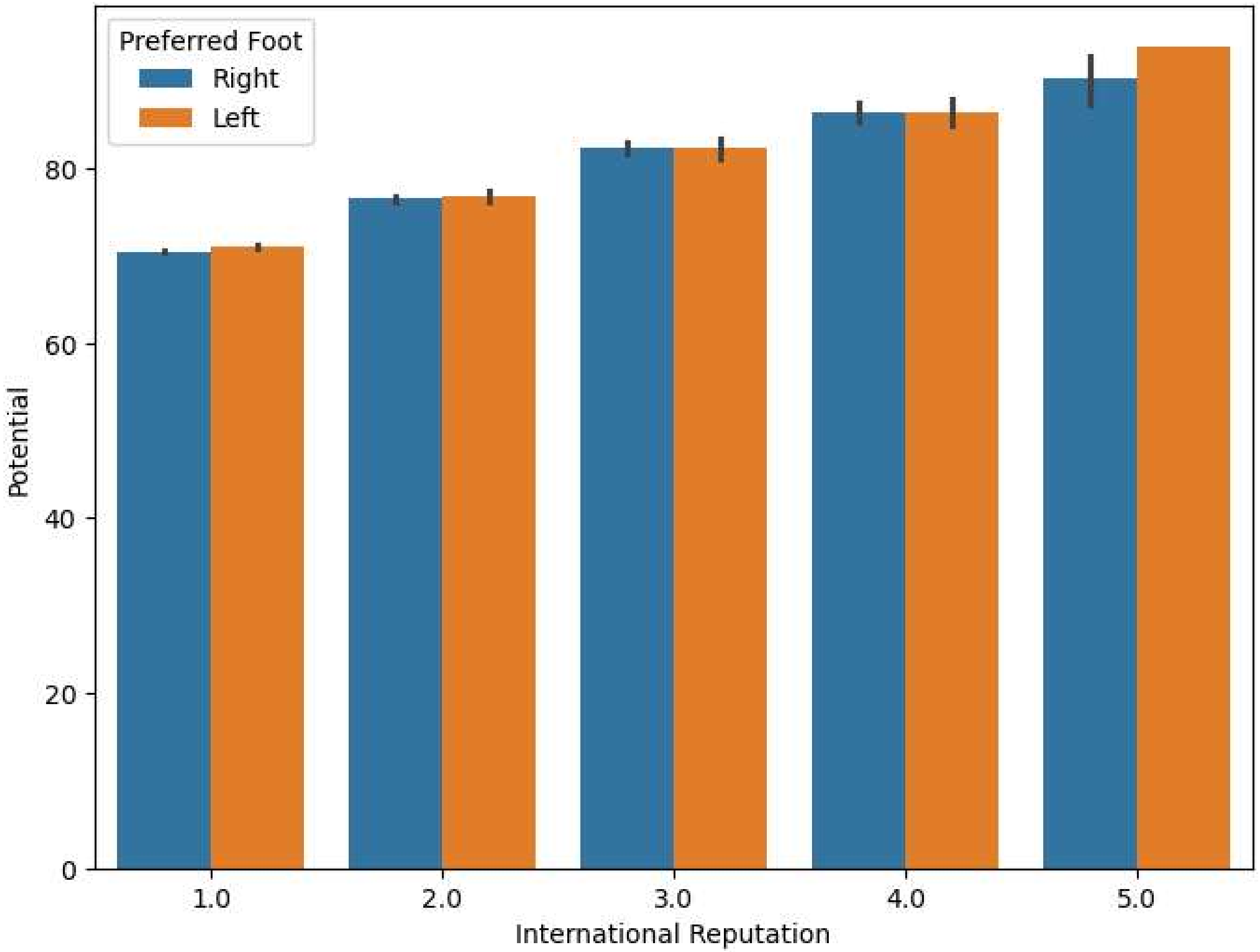
In [46]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.pointplot(x = "International Reputation", y = "Potential", hue = "Preferred Foot", data = fifa, markers = ["o","x
plt.show()
```



In [47]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.barplot(x = "International Reputation", y = "Potential", data = fifa)
plt.show()
```

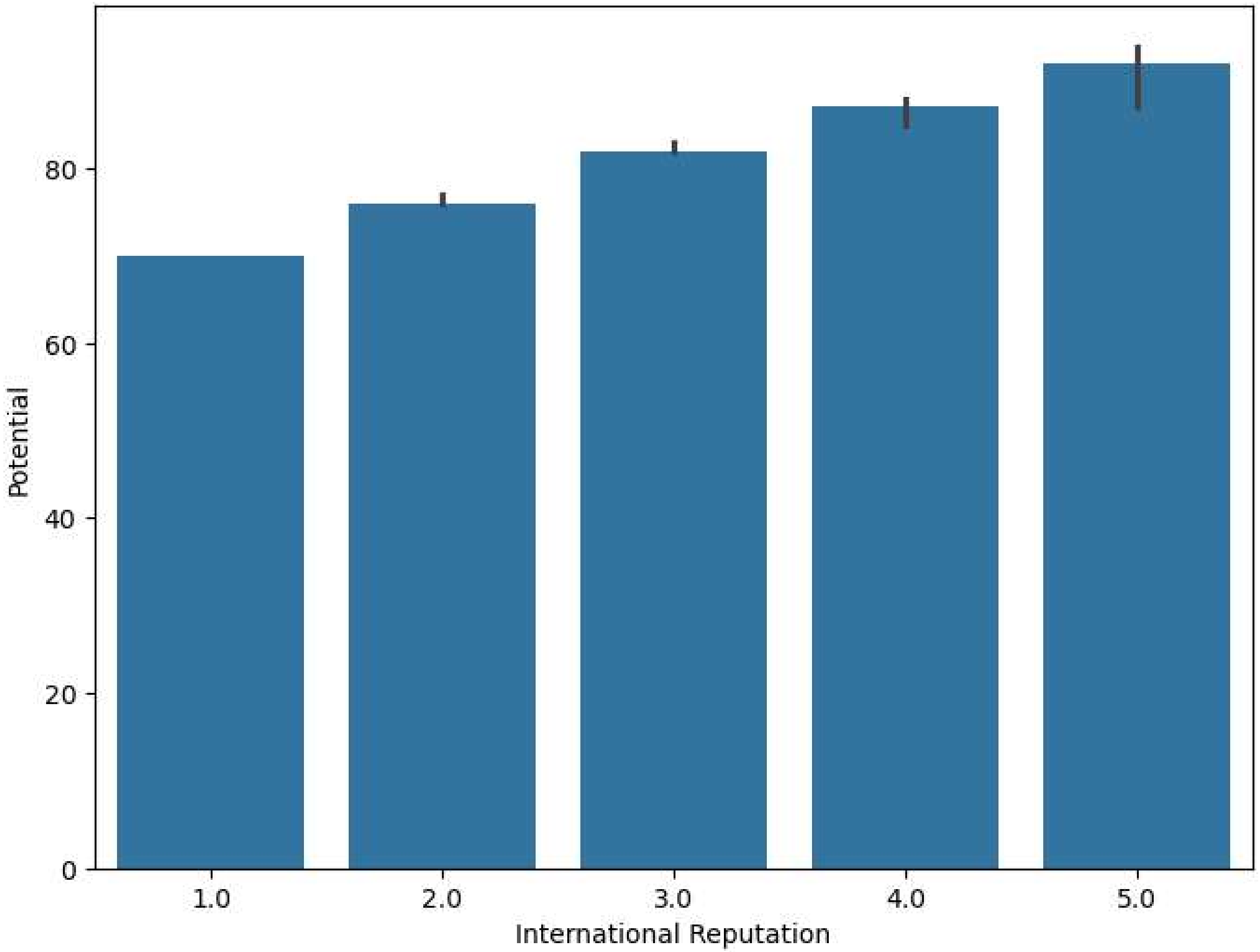


In [48]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.barplot(x = "International Reputation", y = "Potential", hue = "Preferred Foot", data = fifa)
plt.show()
```

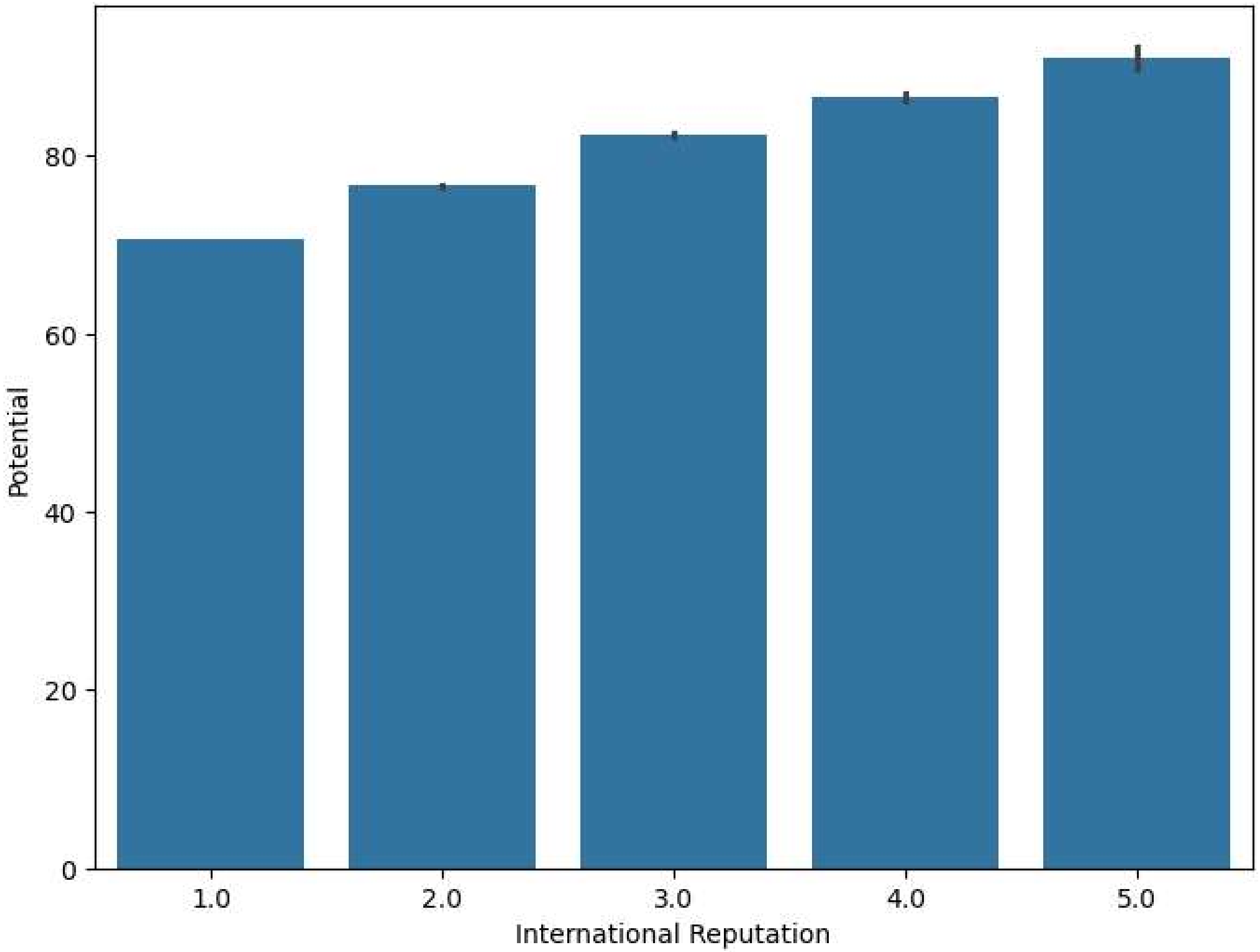


In [49]:

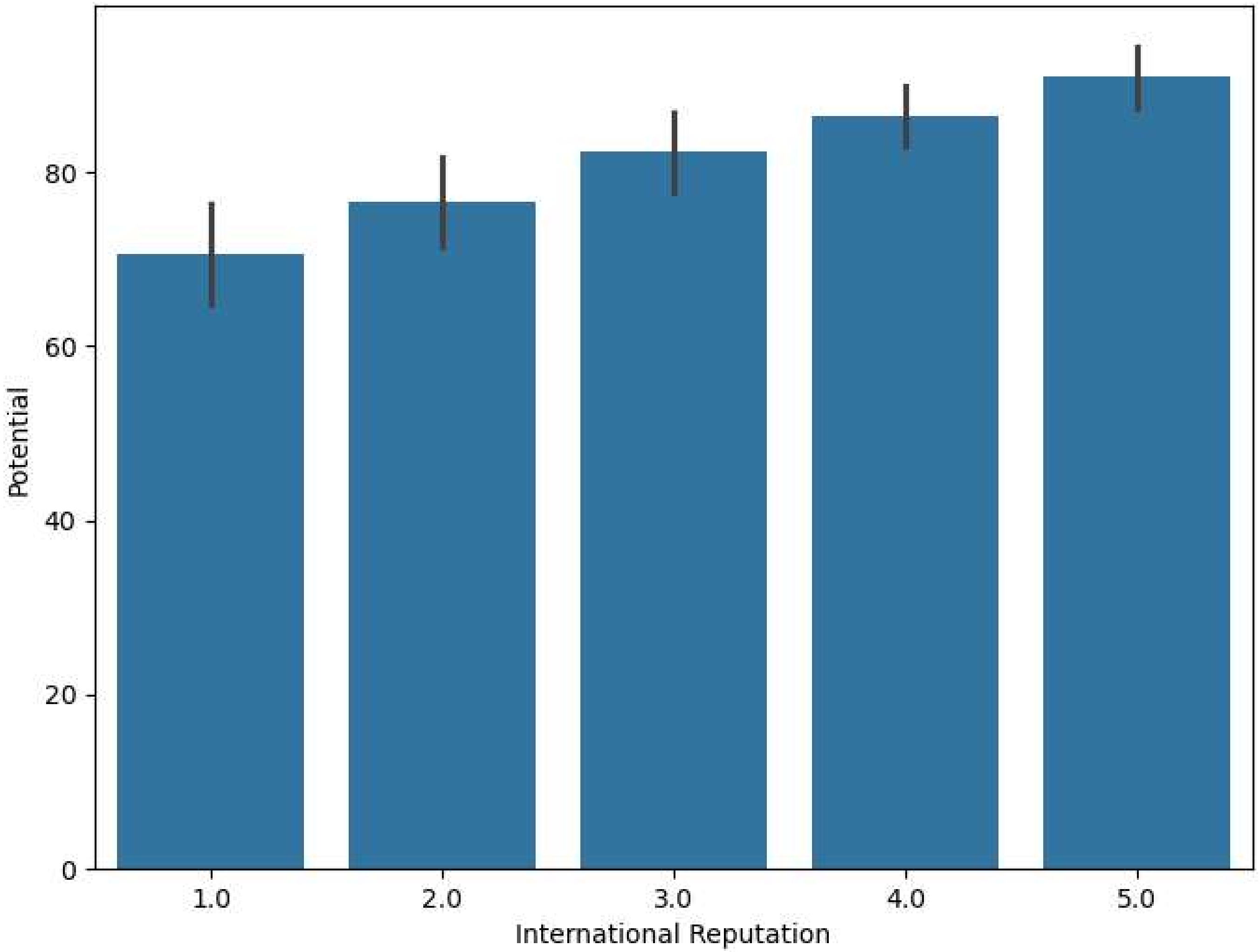
```
from numpy import median
f, ax = plt.subplots(figsize = (8, 6))
sns.barplot(x = "International Reputation", y = "Potential", data = fifa, estimator = median)
plt.show()
```



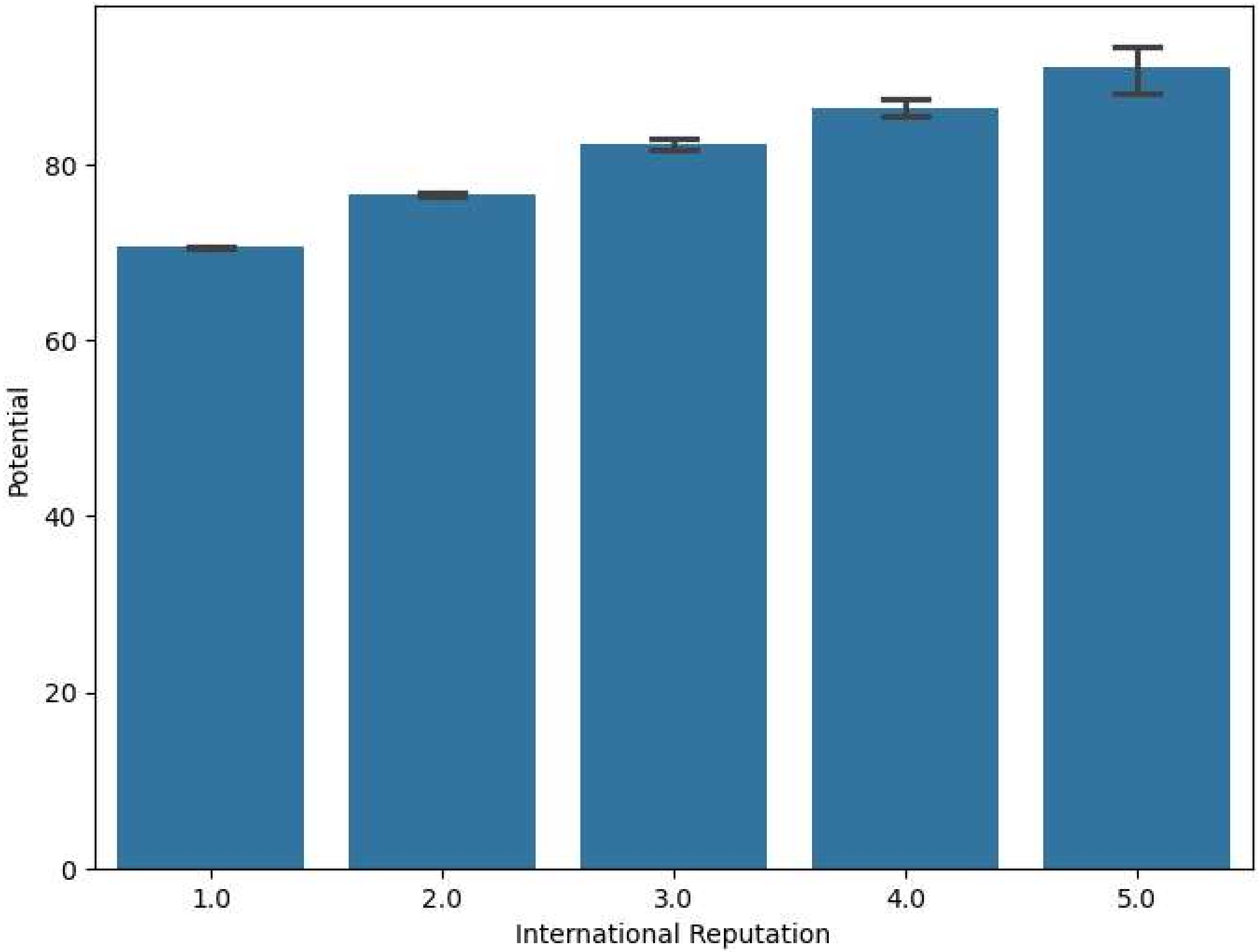
In [50]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.barplot(x = "International Reputation", y = "Potential", data = fifa, ci = 68)
plt.show()
```



In [52]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.barplot(x = "International Reputation", y = "Potential", data = fifa, ci = 'sd')
plt.show()
```

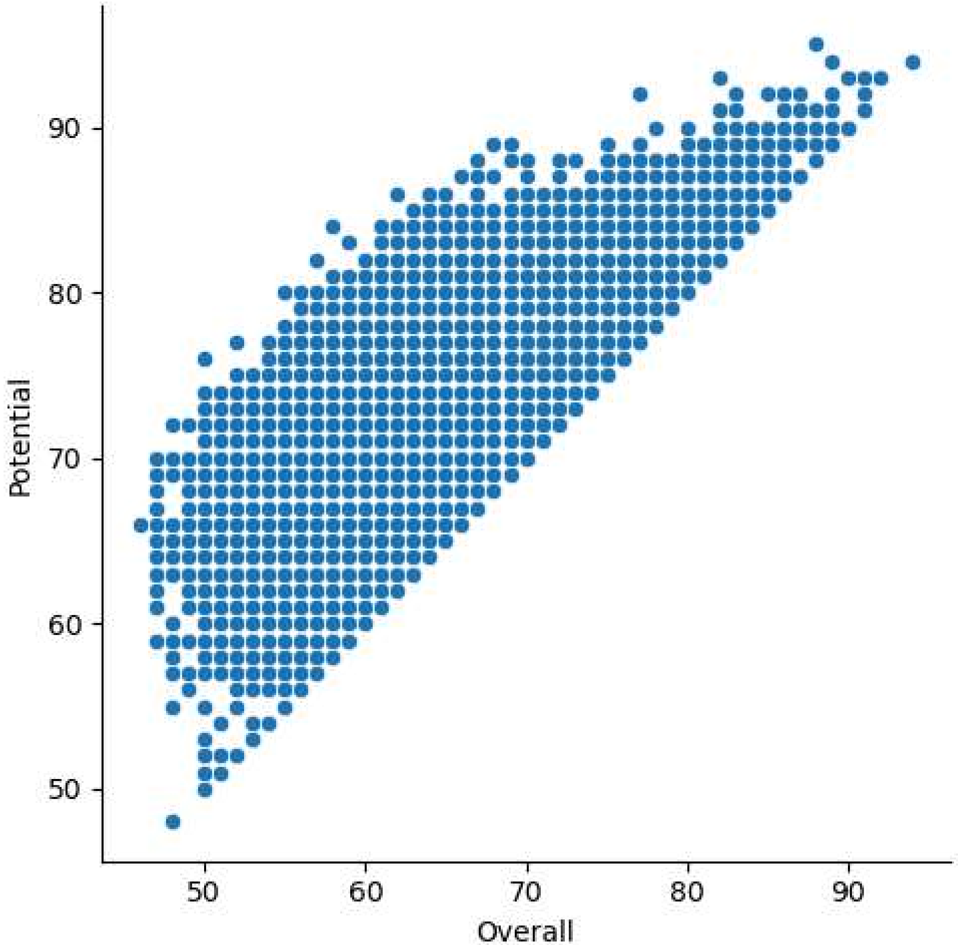


In [53]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.barplot(x = "International Reputation", y = "Potential", data = fifa, capsize = 0.2)
plt.show()
```

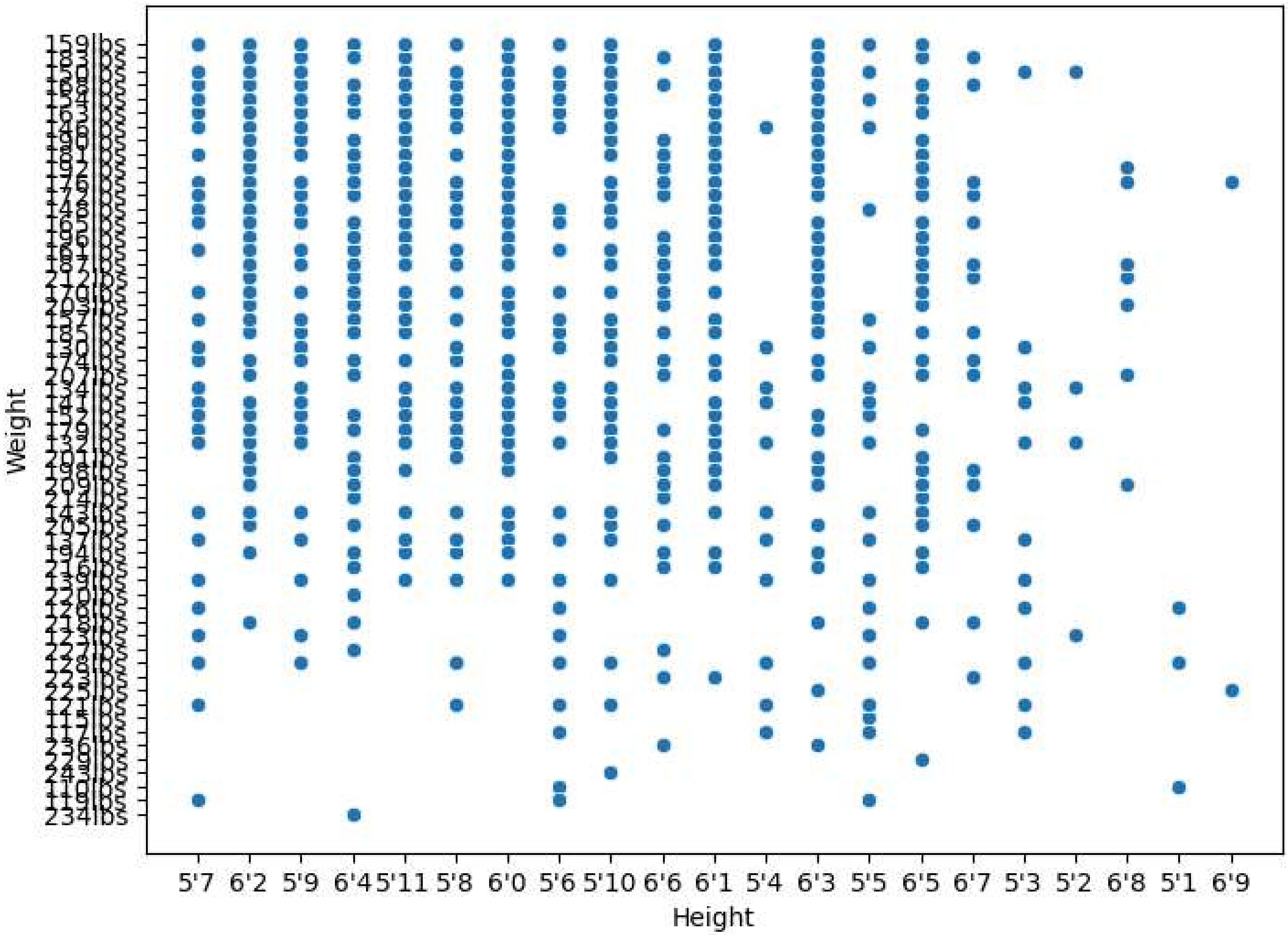


In [54]:
```
g = sns.relplot(x = "Overall", y = "Potential", data = fifa)
plt.plot()
```

Out[54]: []

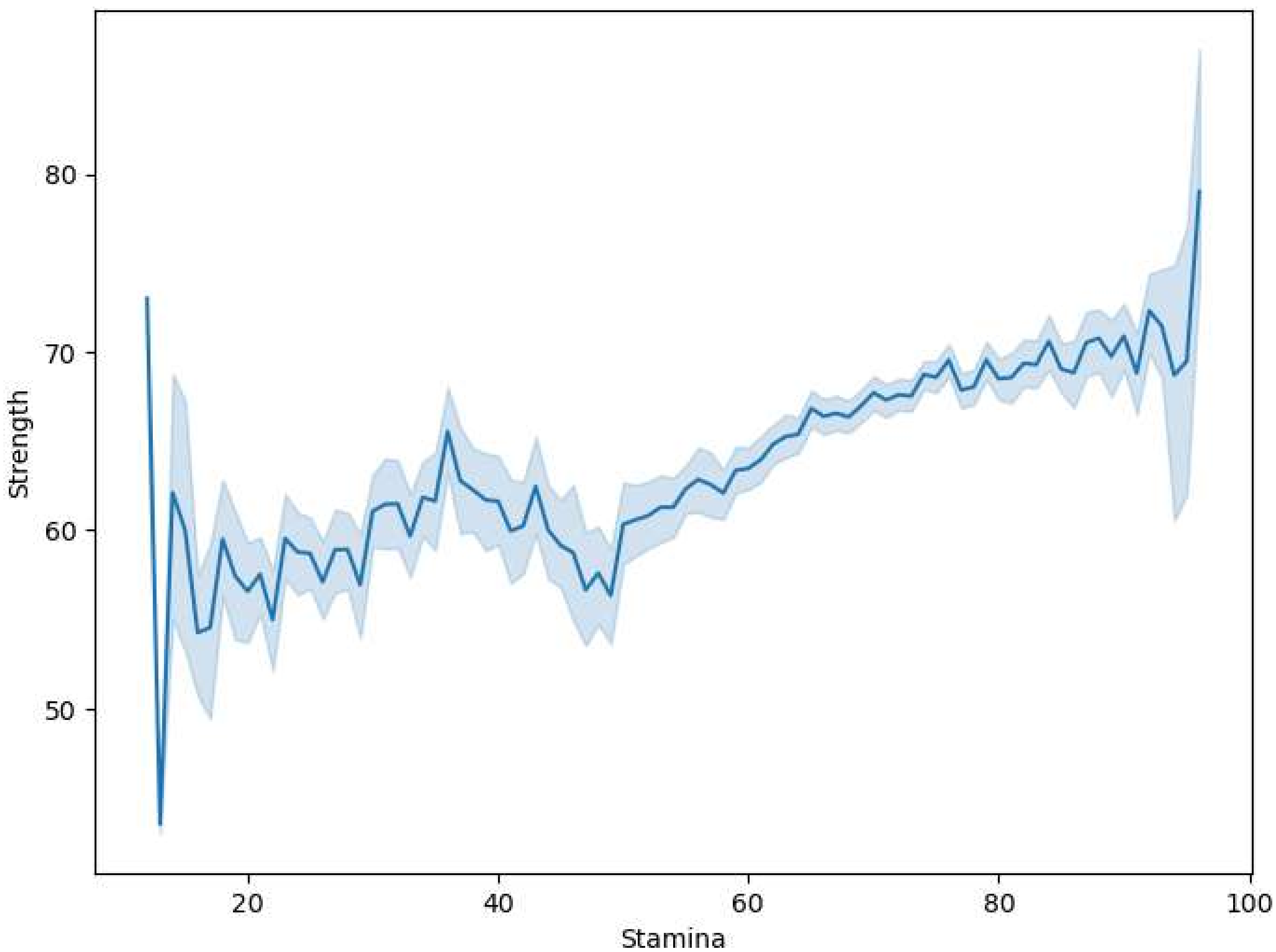


In [55]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.scatterplot(x = "Height", y = "Weight", data = fifa)
plt.show()
```

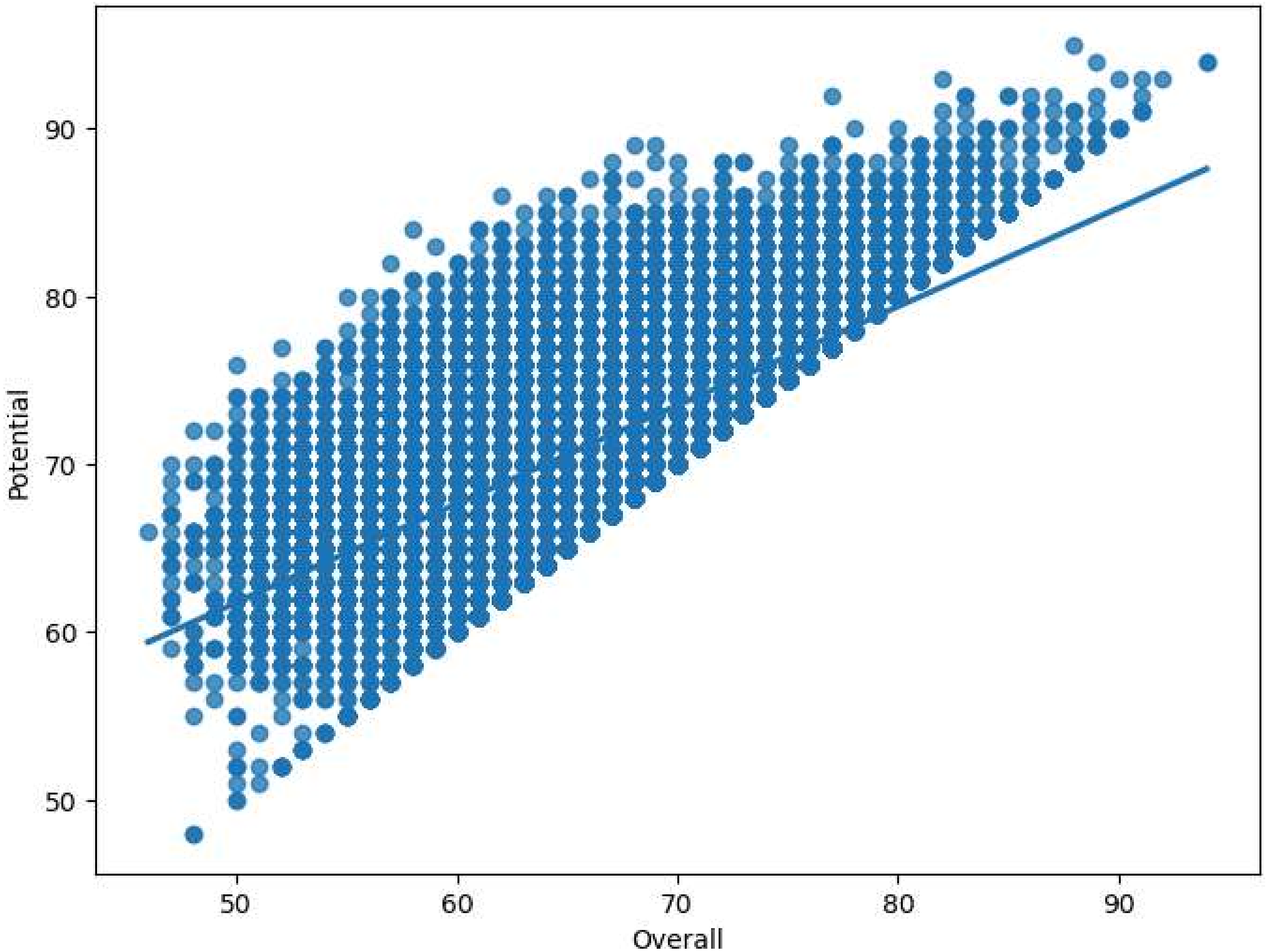


In [56]:

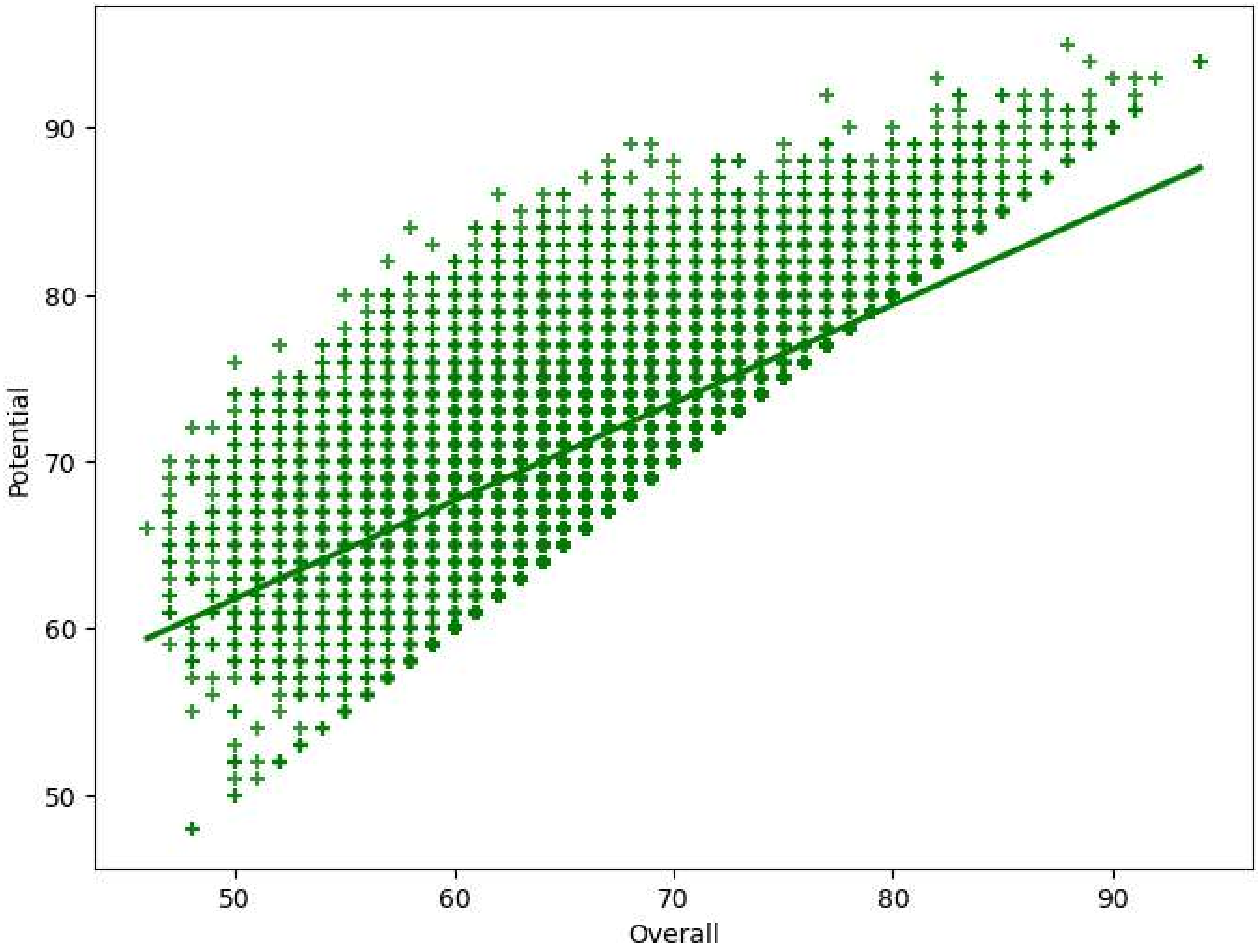
```
f, ax = plt.subplots(figsize = (8, 6))
sns.lineplot(x = "Stamina", y = "Strength", data = fifa)
plt.show()
```



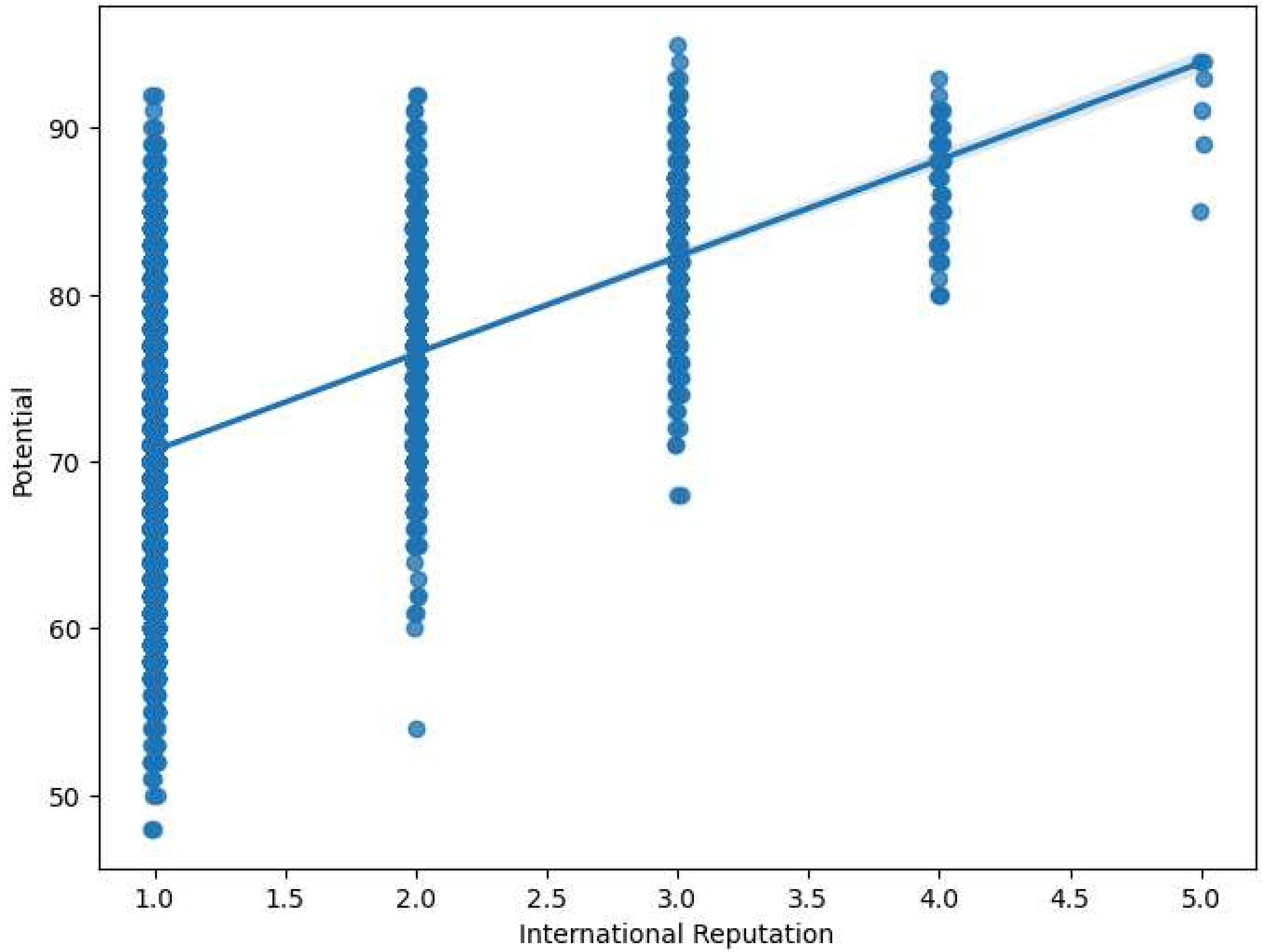
In [57]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.regplot(x = "Overall", y = "Potential", data = fifa)
plt.show()
```



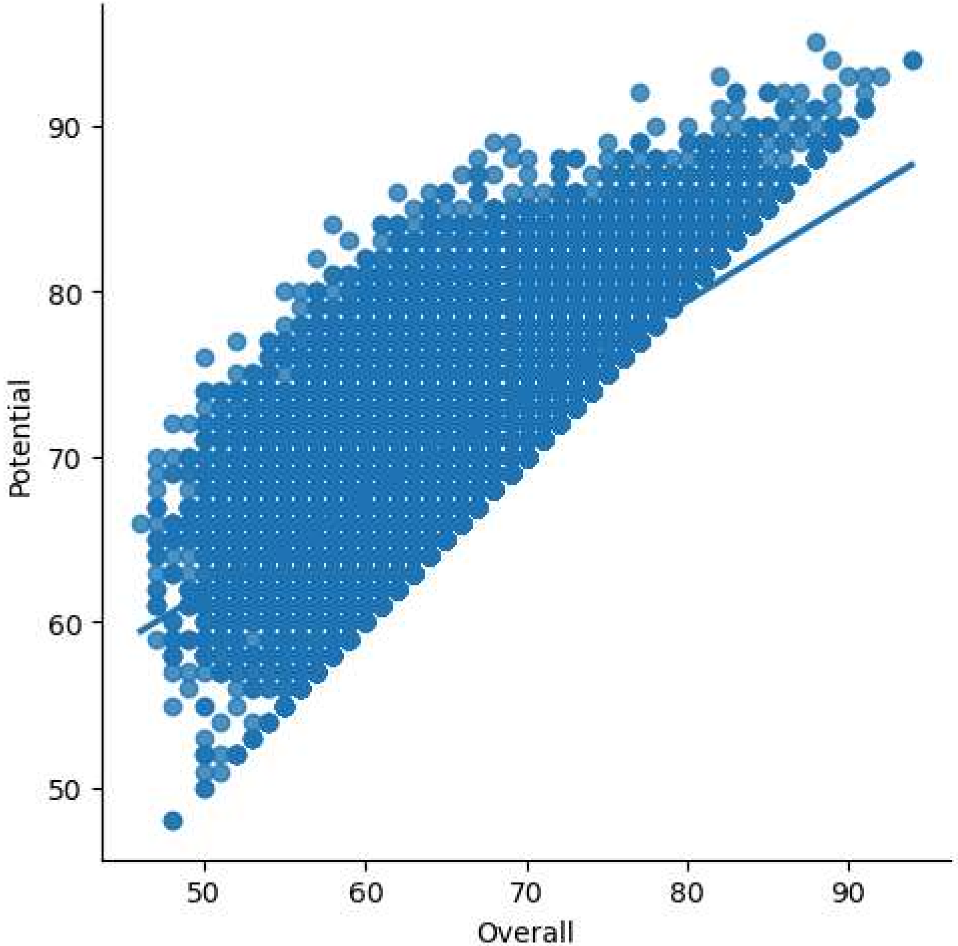
In [58]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.regplot(x = "Overall", y = "Potential", data = fifa, color = 'g', marker = '+')
plt.show()
```



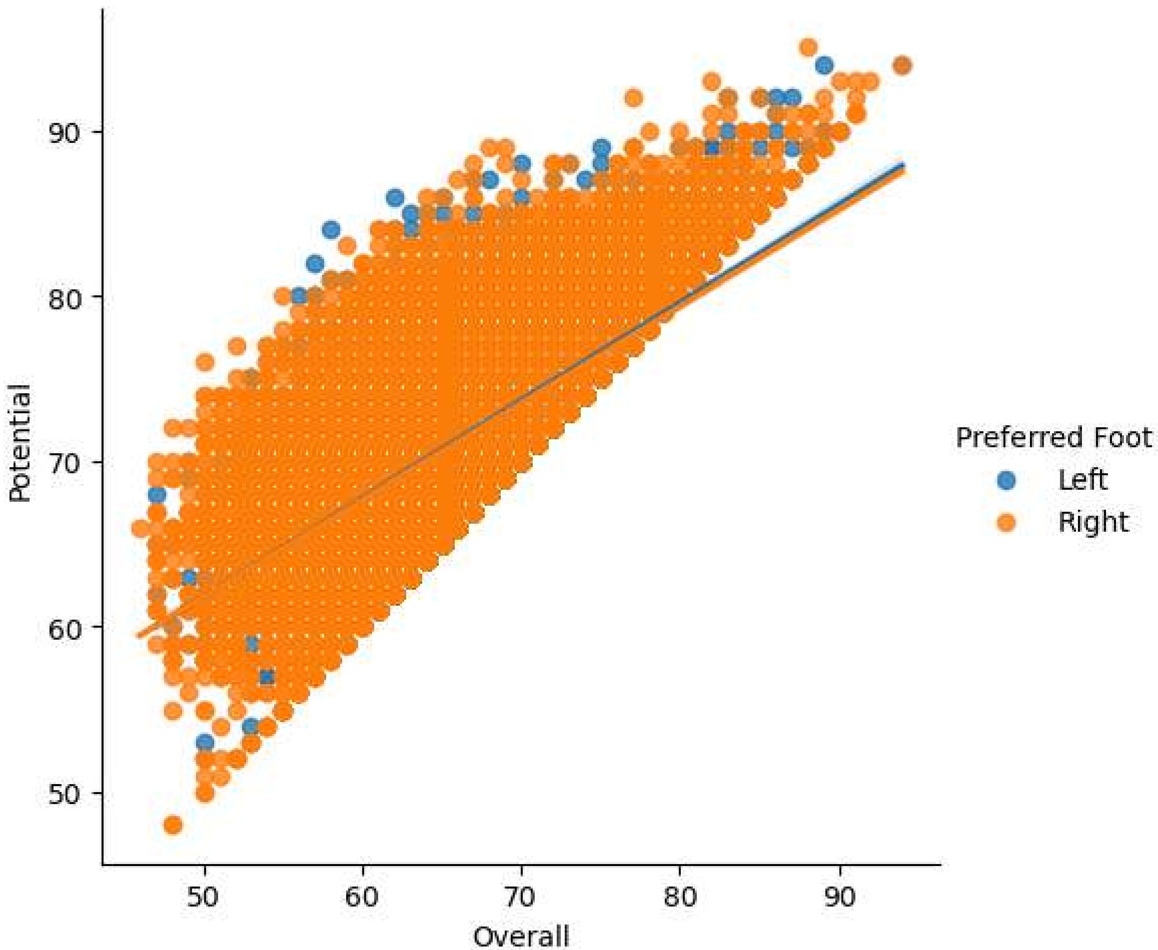
In [59]:
```
f, ax = plt.subplots(figsize = (8, 6))
sns.regplot(x = "International Reputation", y = "Potential", data = fifa, x_jitter = 0.01)
plt.show()
```



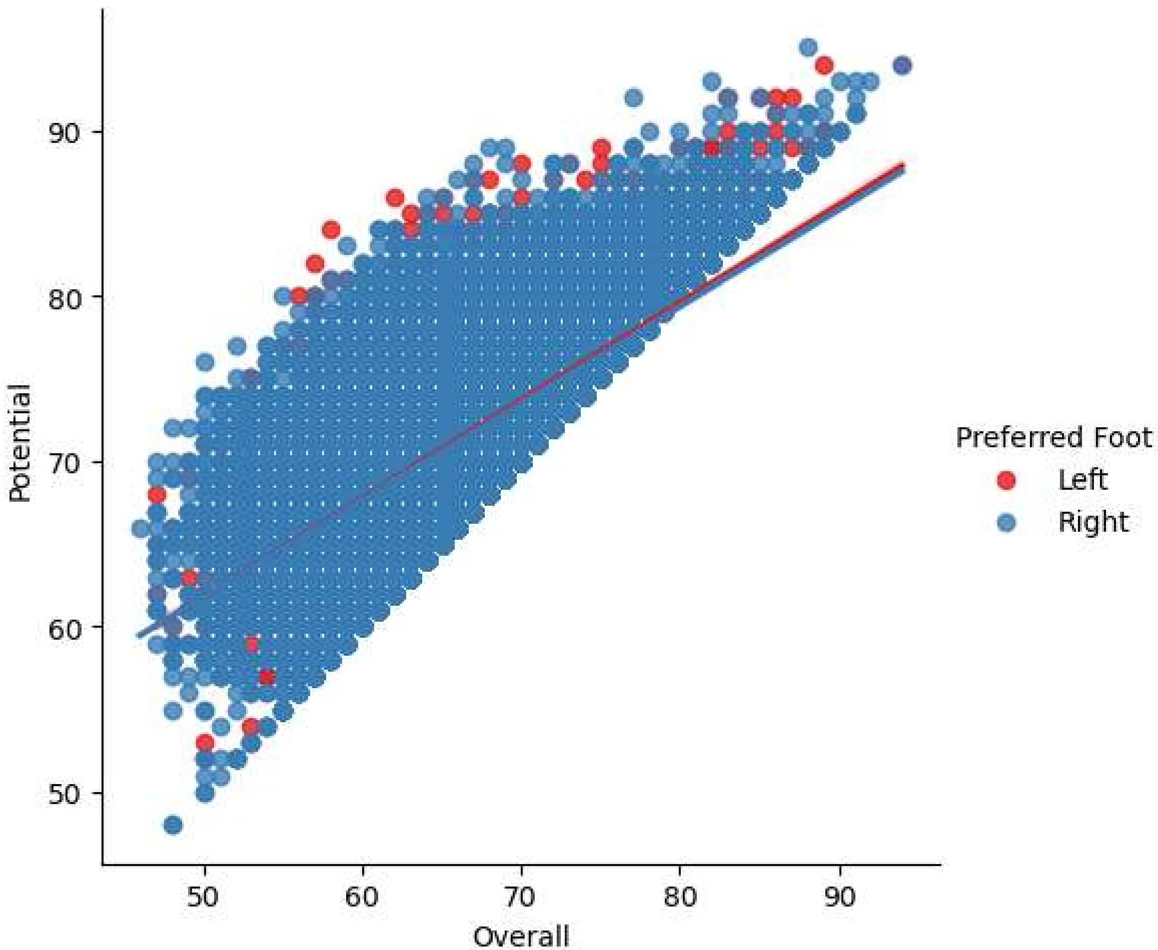
In [60]:
```
g = sns.lmplot(x = "Overall", y = "Potential", data = fifa)
plt.show()
```



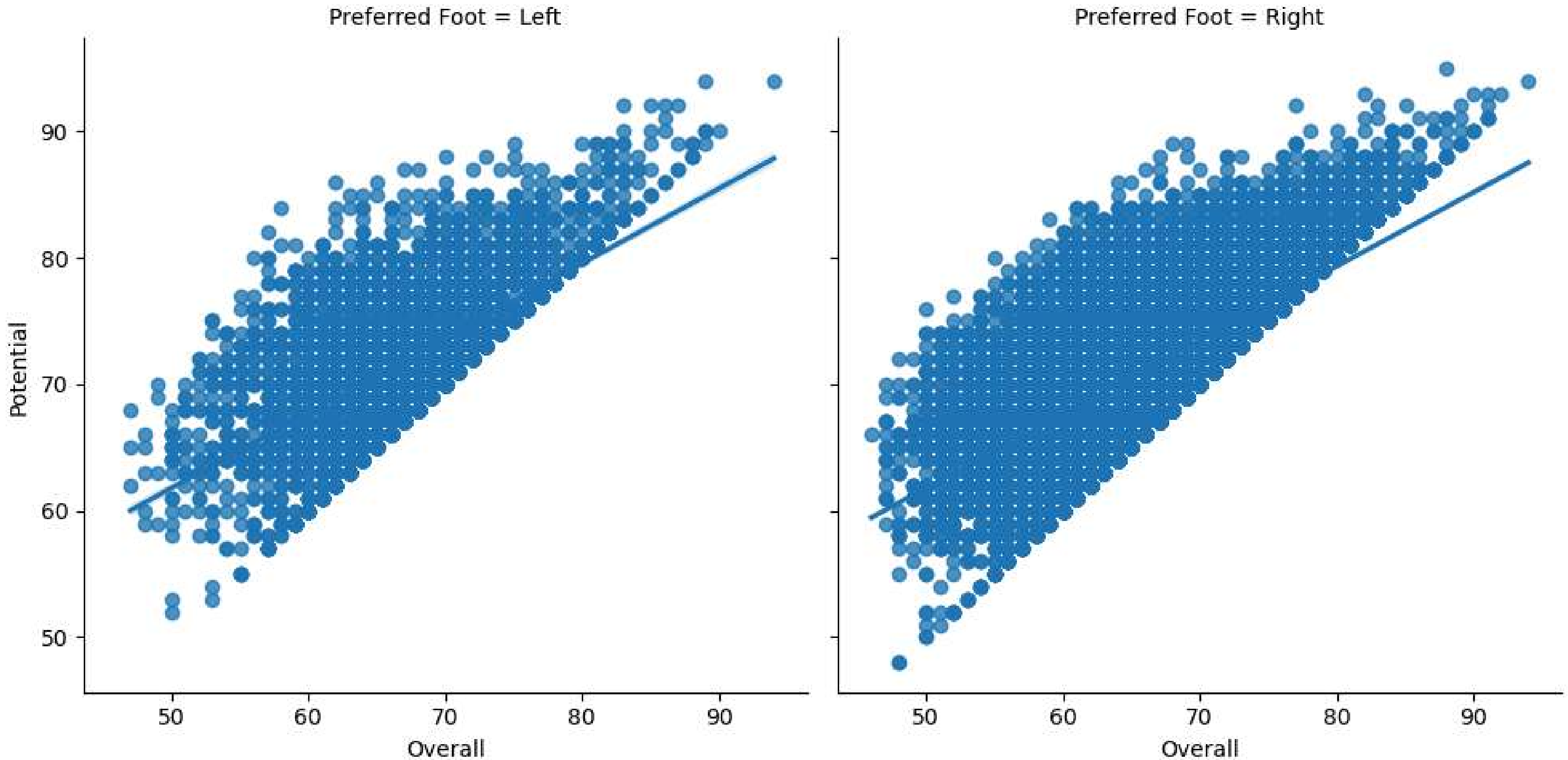
In [61]:
```
g = sns.lmplot(x = "Overall", y = "Potential", hue = "Preferred Foot", data = fifa)
plt.show()
```



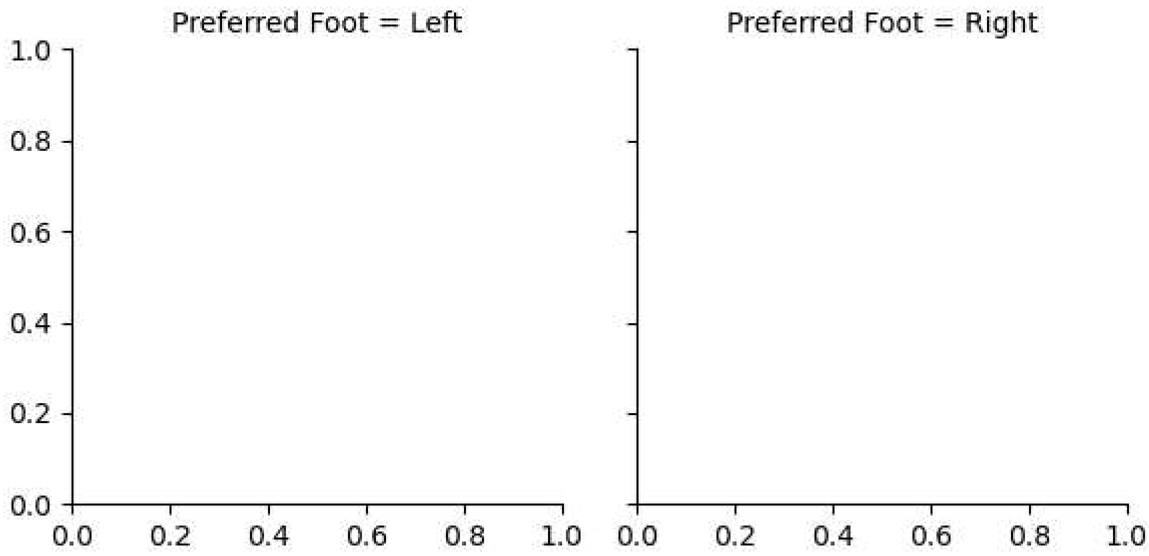
In [62]:
```
g = sns.lmplot(x = "Overall", y = "Potential", hue = "Preferred Foot", palette = "Set1", data = fifa)
plt.show()
```



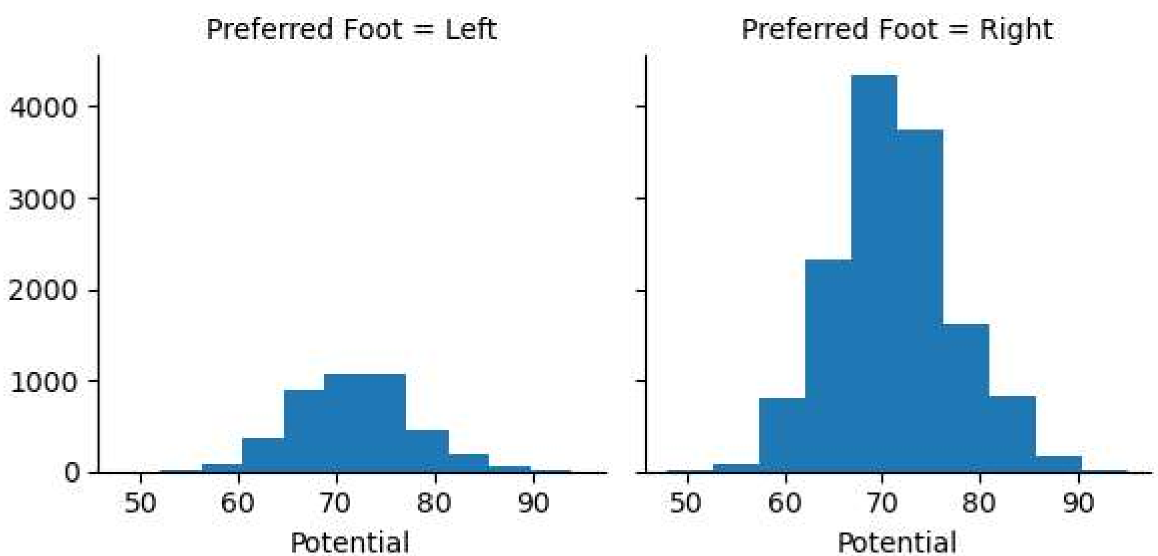
In [63]:
```
g = sns.lmplot(x = "Overall", y = "Potential", col = "Preferred Foot", data = fifa)
plt.show()
```



In [64]:
```
g = sns.FacetGrid(fifa, col = "Preferred Foot")
plt.show()
```

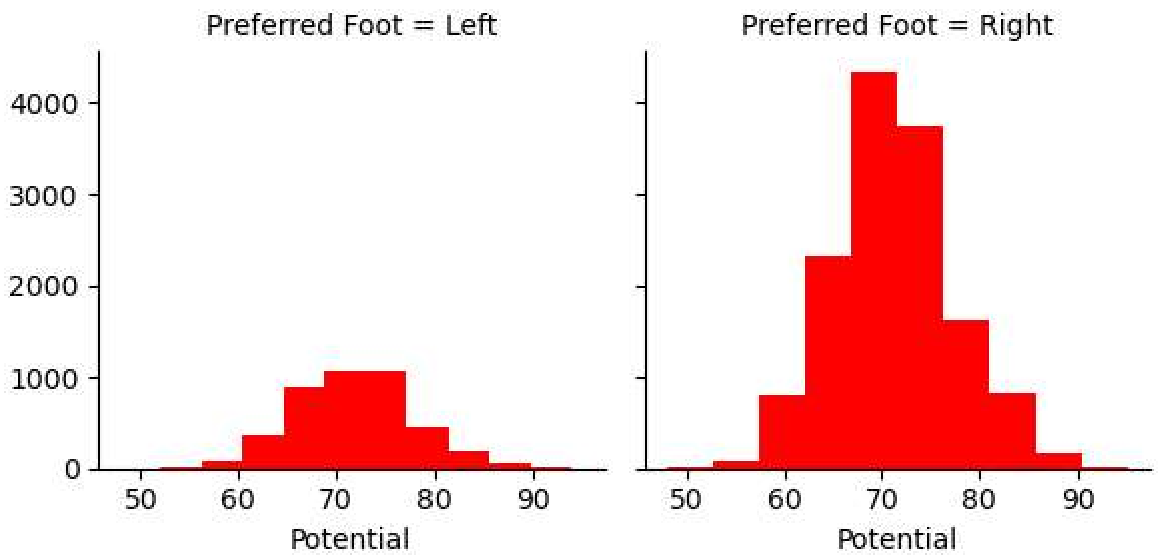


In [65]:
```
g = sns.FacetGrid(fifa, col = "Preferred Foot")
g = g.map(plt.hist, "Potential")
plt.show()
```

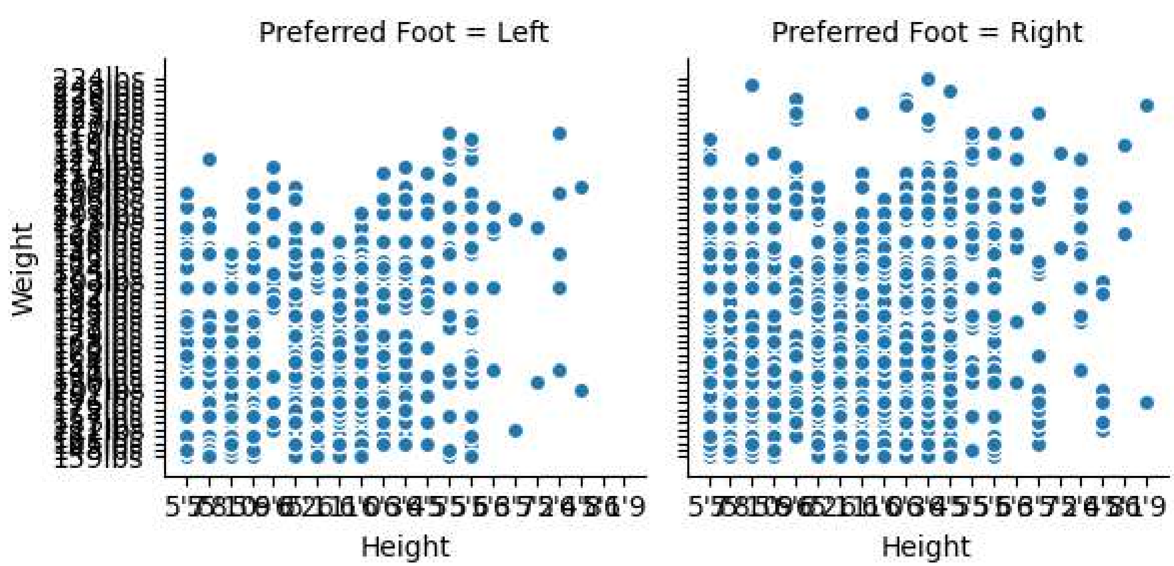


In [66]:
```
g = sns.FacetGrid(fifa, col = "Preferred Foot")
g = g.map(plt.hist, "Potential", bins = 10, color = 'r')
```

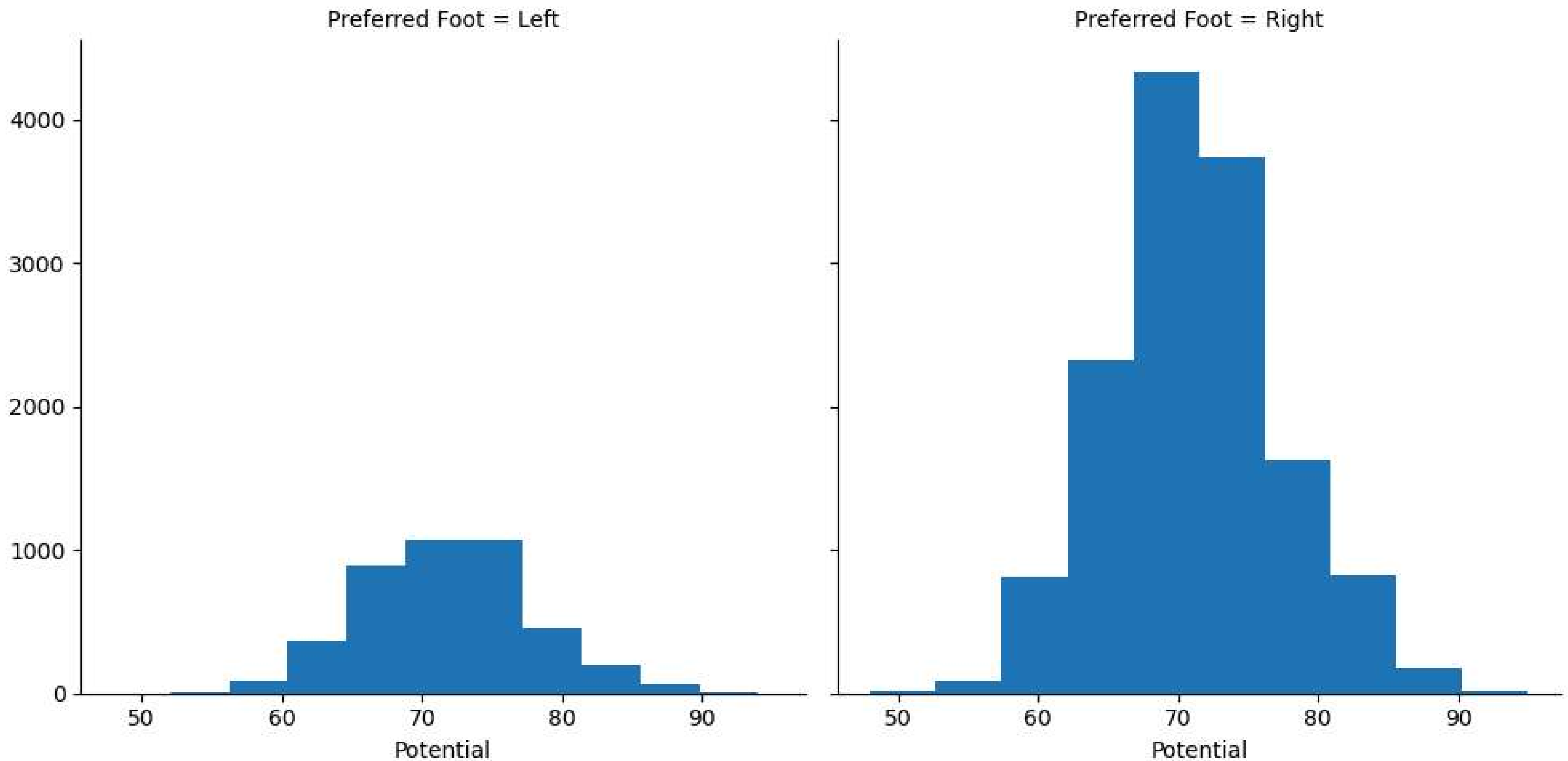
```
plt.show()
```



In [68]:
```
g = sns.FacetGrid(fifa, col = "Preferred Foot")
g = g.map(plt.scatter, "Height", "Weight", edgecolor = 'w').add_legend()
plt.show()
```

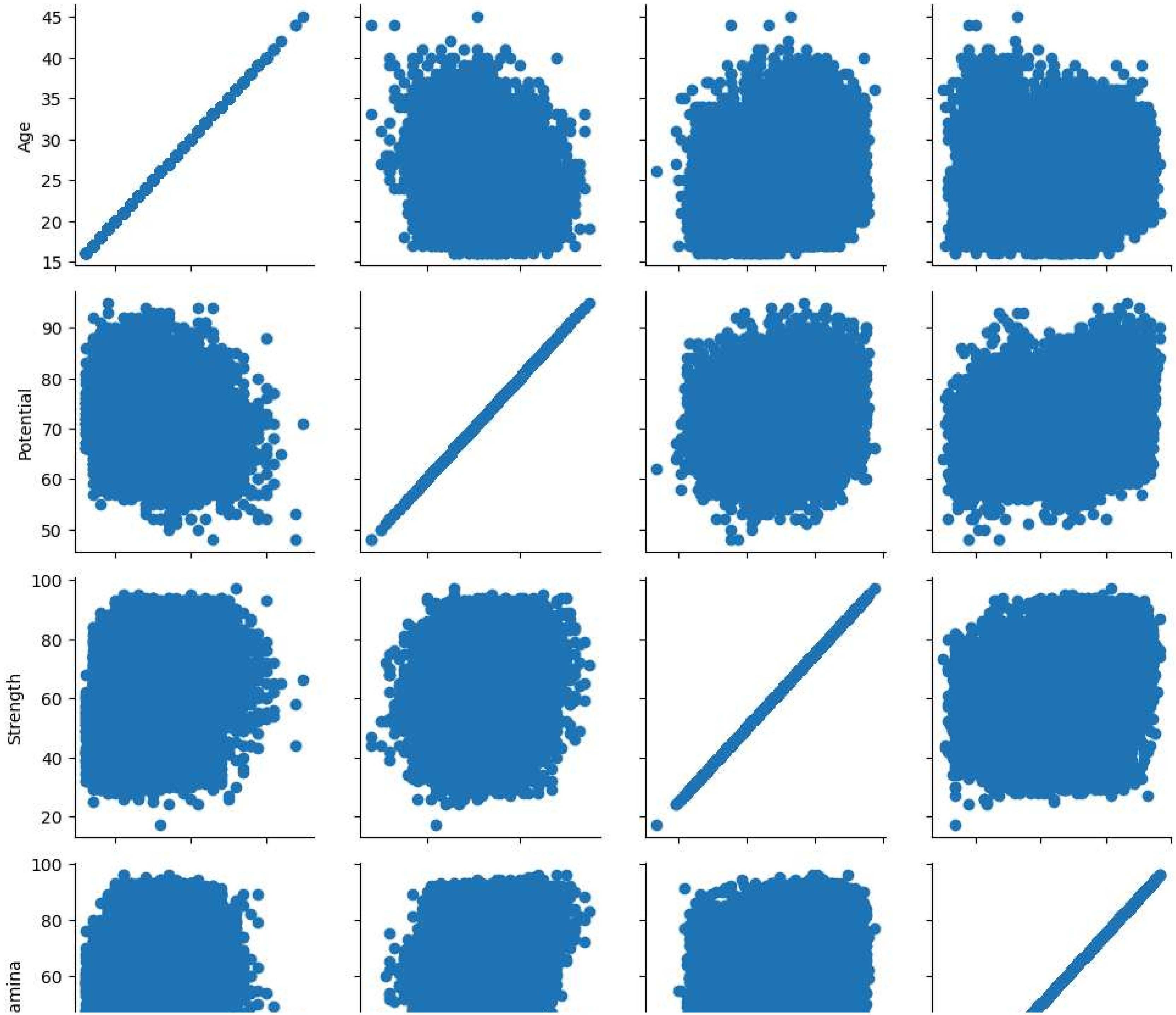


In [69]:
```
g = sns.FacetGrid(fifa, col = "Preferred Foot", height = 5, aspect = 1)
g = g.map(plt.hist, "Potential")
plt.show()
```



In [70]:
```
fifa_new = fifa[['Age', 'Potential', 'Strength', 'Stamina', 'Preferred Foot']]
```

In [71]:
```
g = sns.PairGrid(fifa_new)
g = g.map(plt.scatter)
plt.show()
```


Age
Potential
Strength
amina
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40
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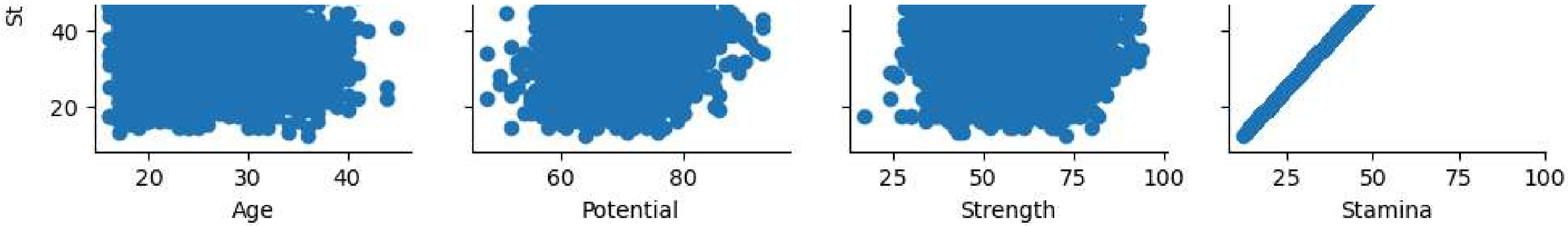
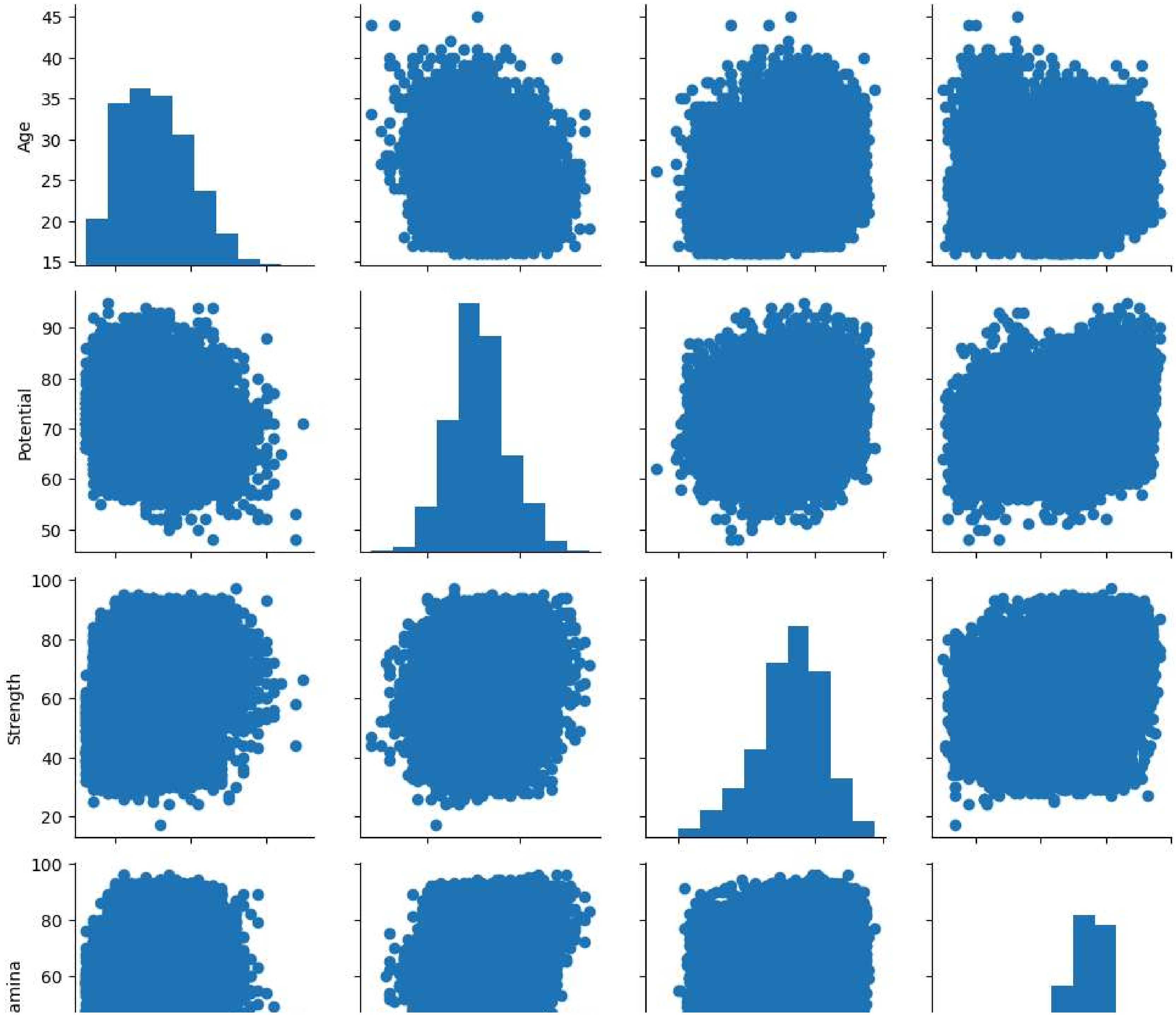
70
60
50
100
80
60
40
20
100
80
60



In [72]:

```
g = sns.PairGrid(fifa_new)
g = g.map_diag(plt.hist)
g = g.map_offdiag(plt.scatter)
plt.show()
```



Age
Potential
Strength
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100
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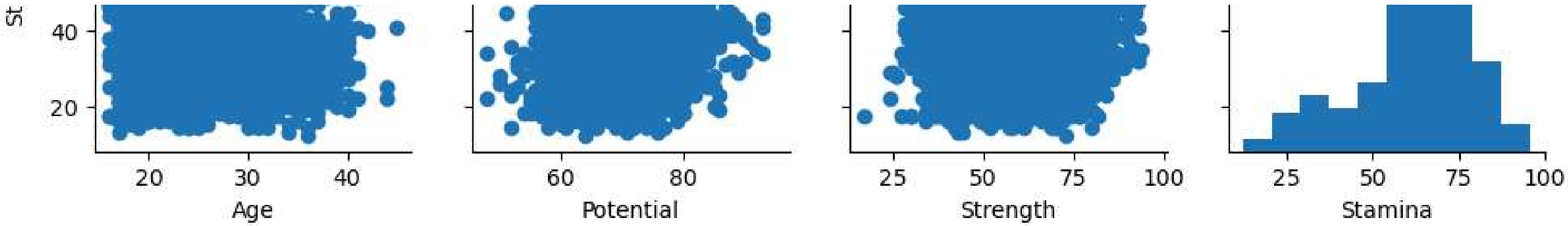
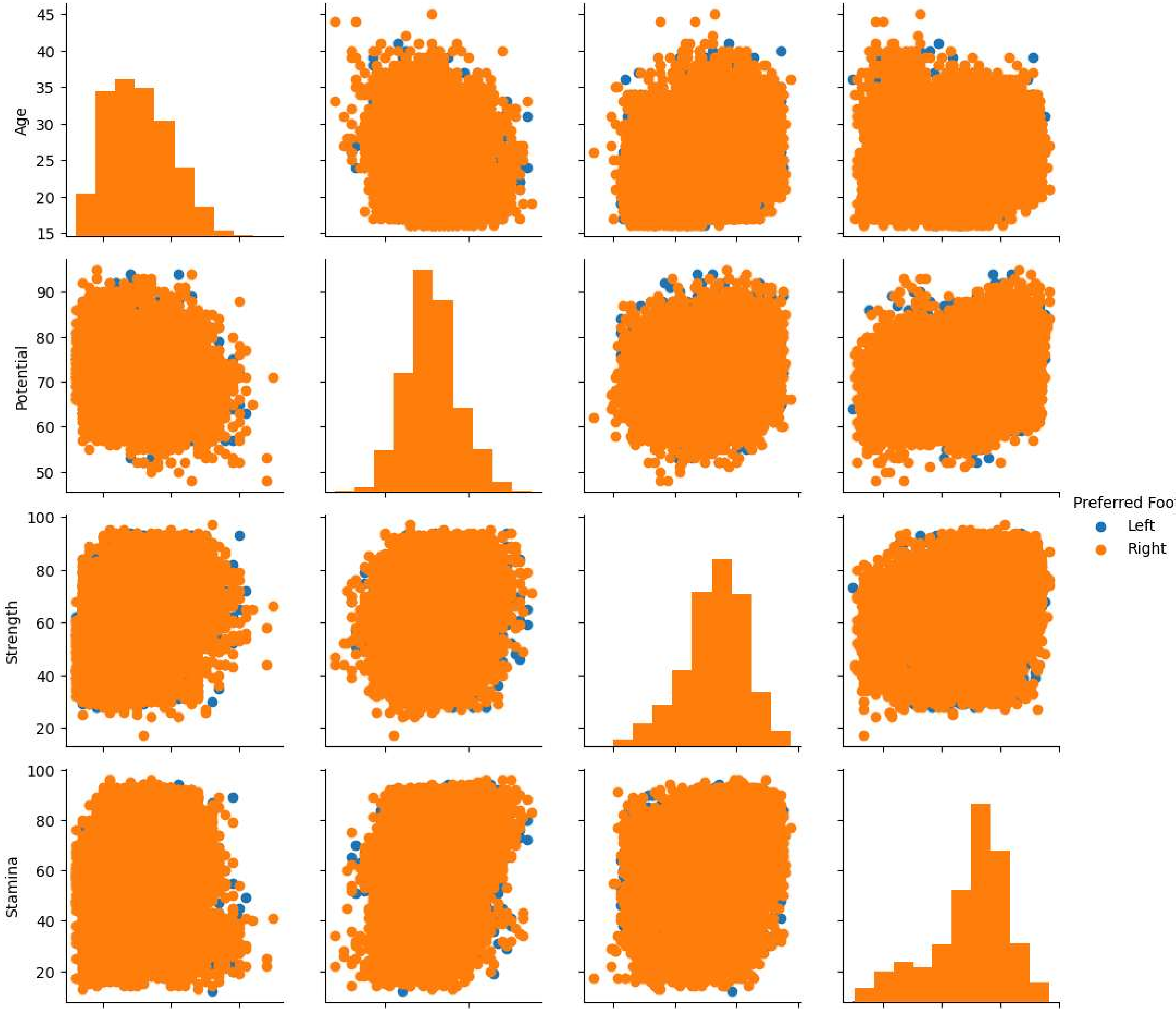
60



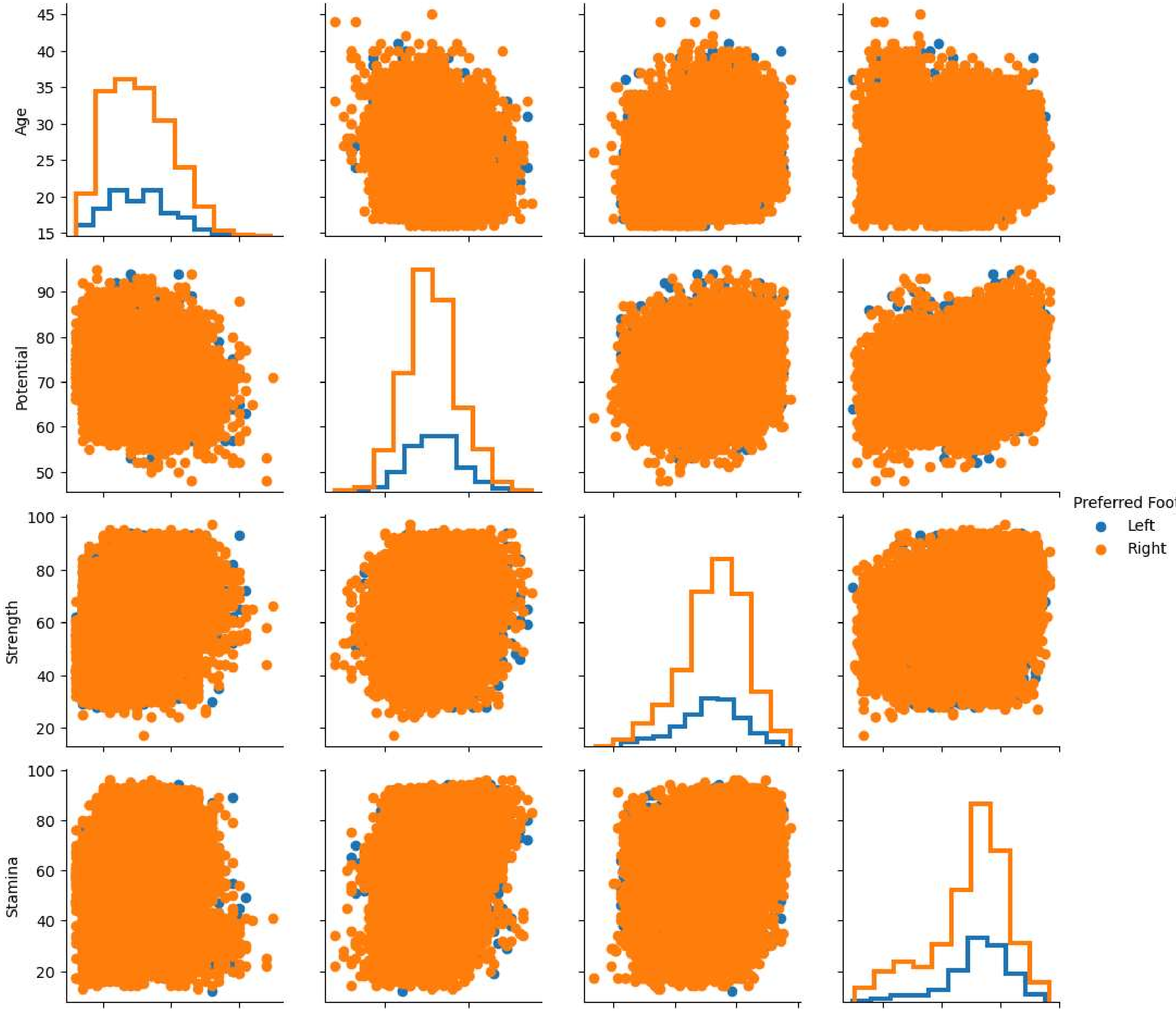
In [73]:
```
g = sns.PairGrid(fifa_new, hue = "Preferred Foot")
g = g.map_diag(plt.hist)
g = g.map_offdiag(plt.scatter)
g = g.add_legend()
plt.show()
```



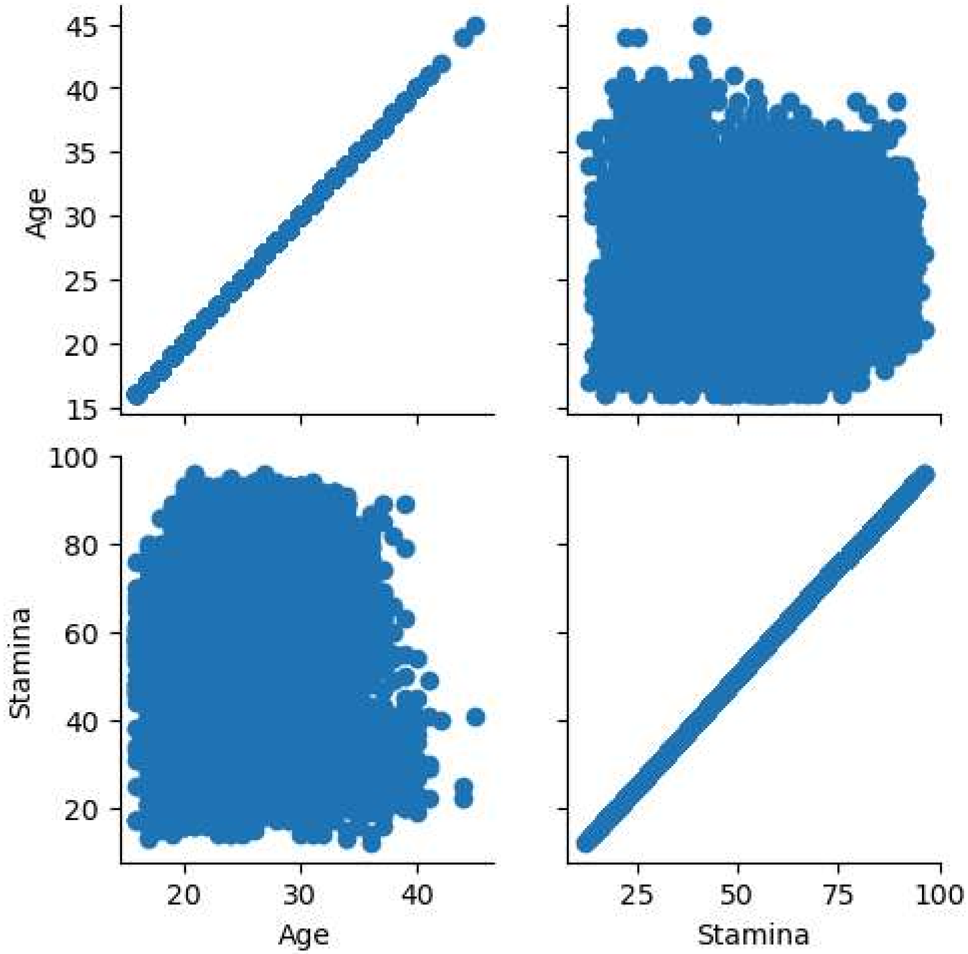
Age
Potential
Strength
Stamina
Preferred Foot
Left
Right

In [74]:

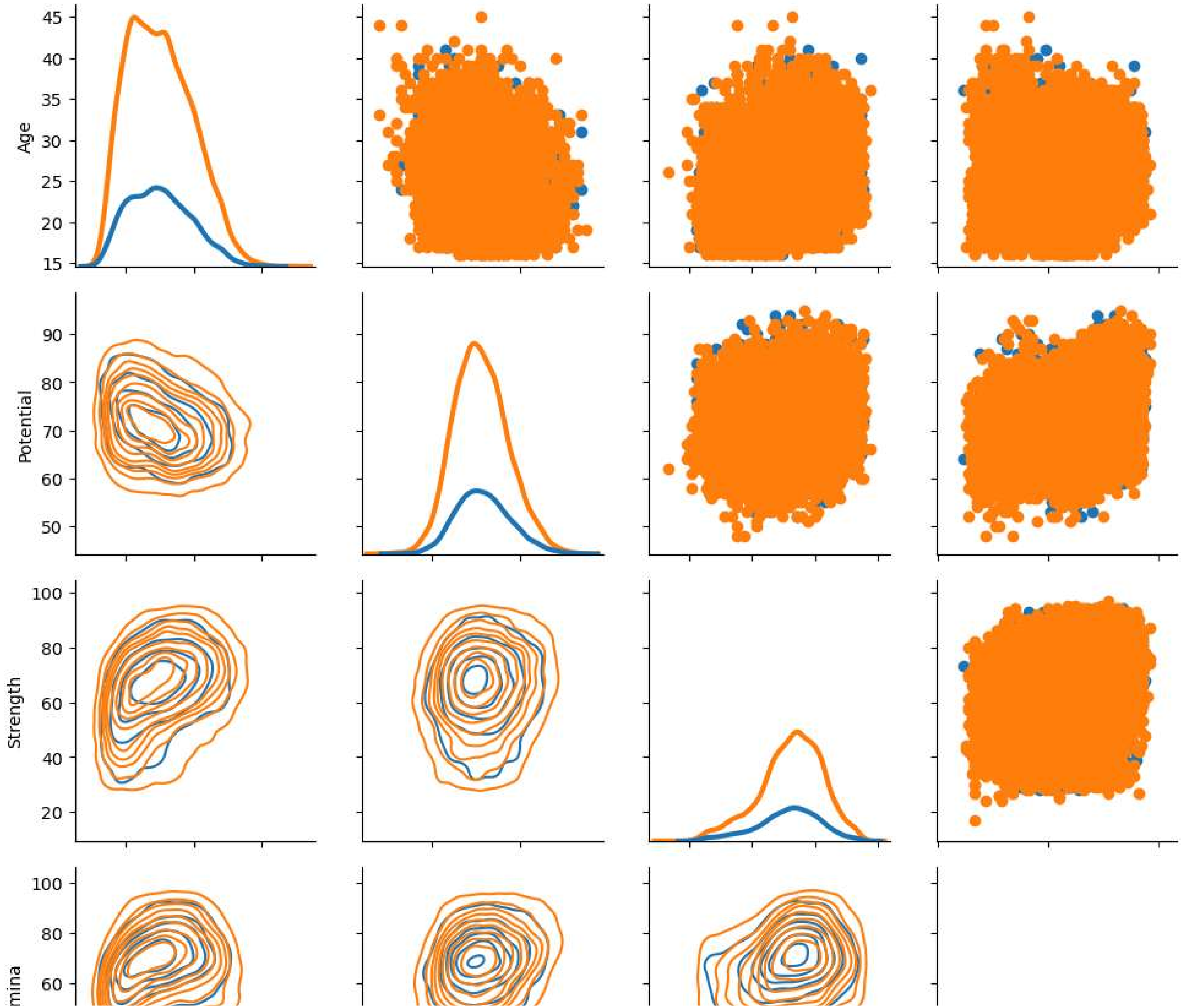
```
g = sns.PairGrid(fifa_new, hue = "Preferred Foot")
g = g.map_diag(plt.hist, histtype = 'step', linewidth = 3)
g = g.map_offdiag(plt.scatter)
g = g.add_legend()
plt.show()
```


Age
Potential
Strength
Stamina
Preferred Foot
Left
Right

In [75]:
```
g = sns.PairGrid(fifa_new, vars = ['Age', 'Stamina'])
g = g.map(plt.scatter)
plt.show()
```



In [76]:
```
g = sns.PairGrid(fifa_new, hue = "Preferred Foot")
g = g.map_upper(plt.scatter)
g = g.map_lower(sns.kdeplot, cmap = "Blues_d")
g = g.map_diag(sns.kdeplot, lw = 3, legend = False)
plt.show()
```



Age
Potential
Strength
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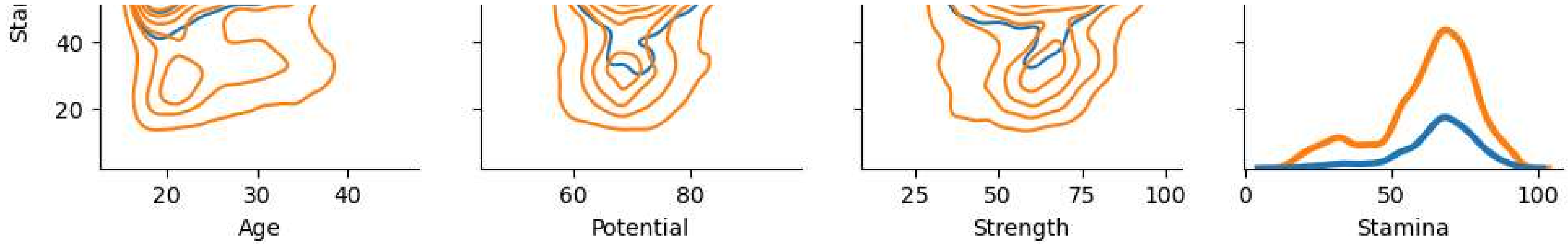
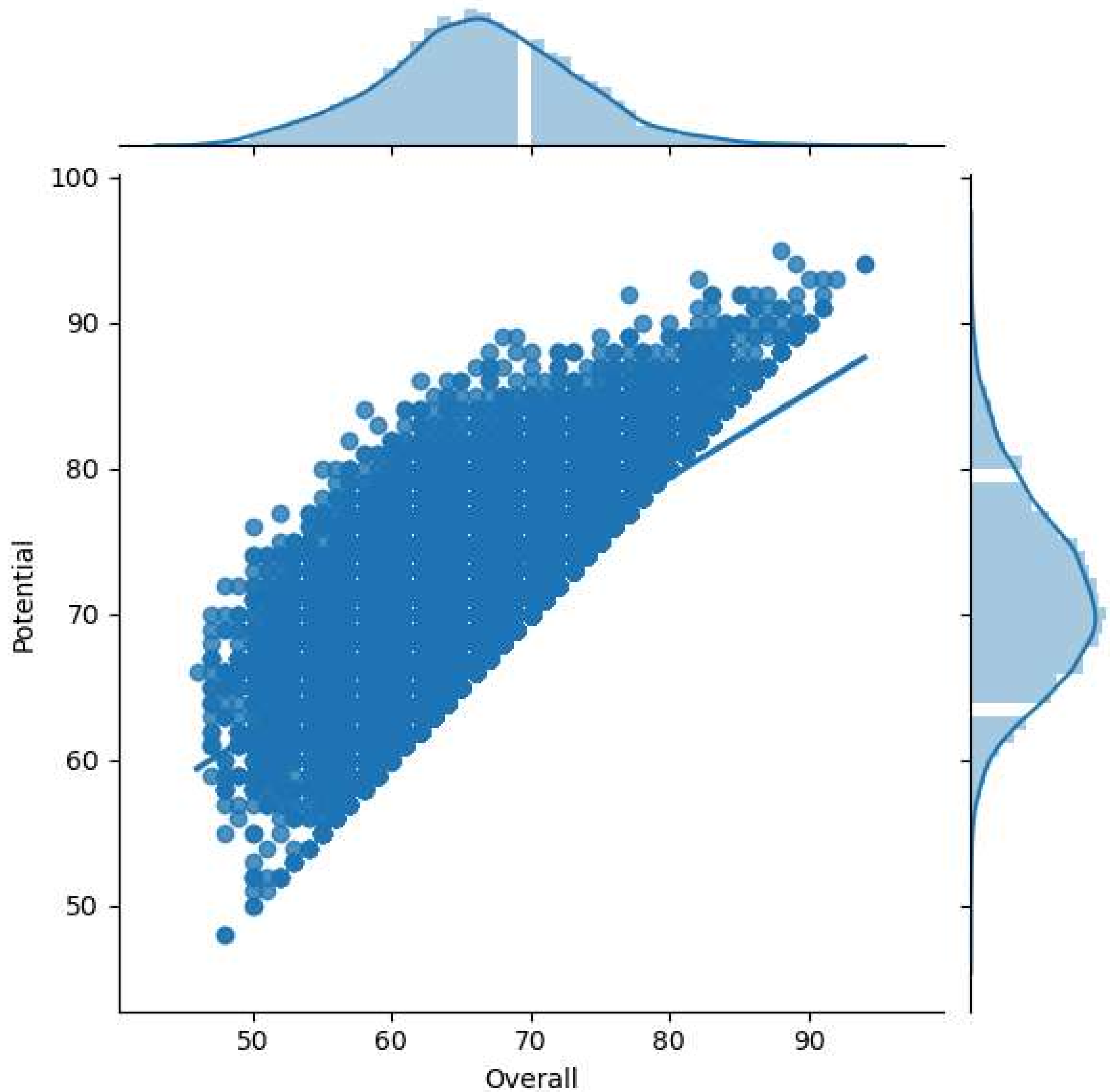
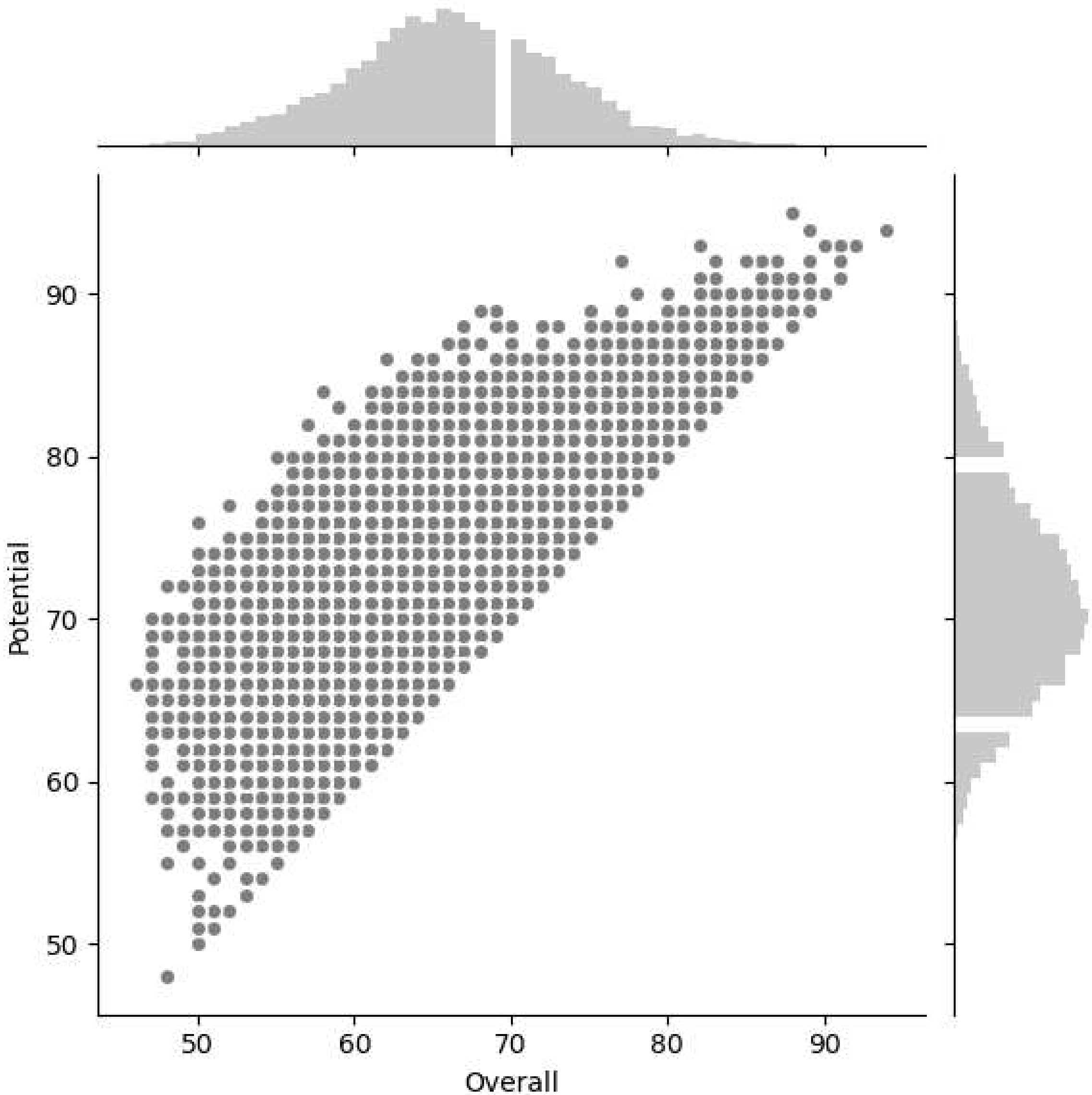
50
100
80
60
40
20
100
80
60



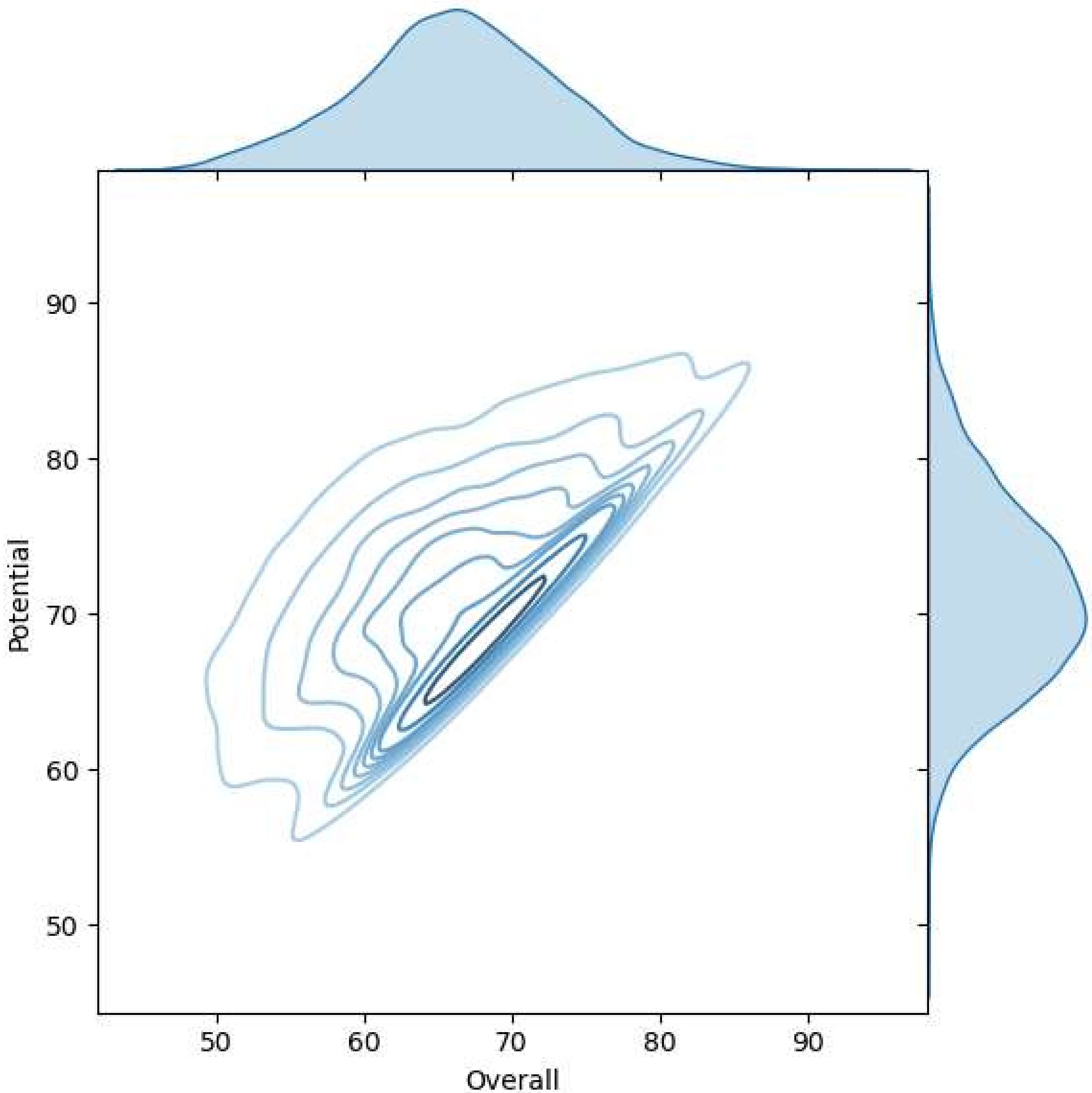
In [77]:
```
g = sns.JointGrid(x = "Overall", y = "Potential", data = fifa)
g = g.plot(sns.regplot, sns.distplot)
```



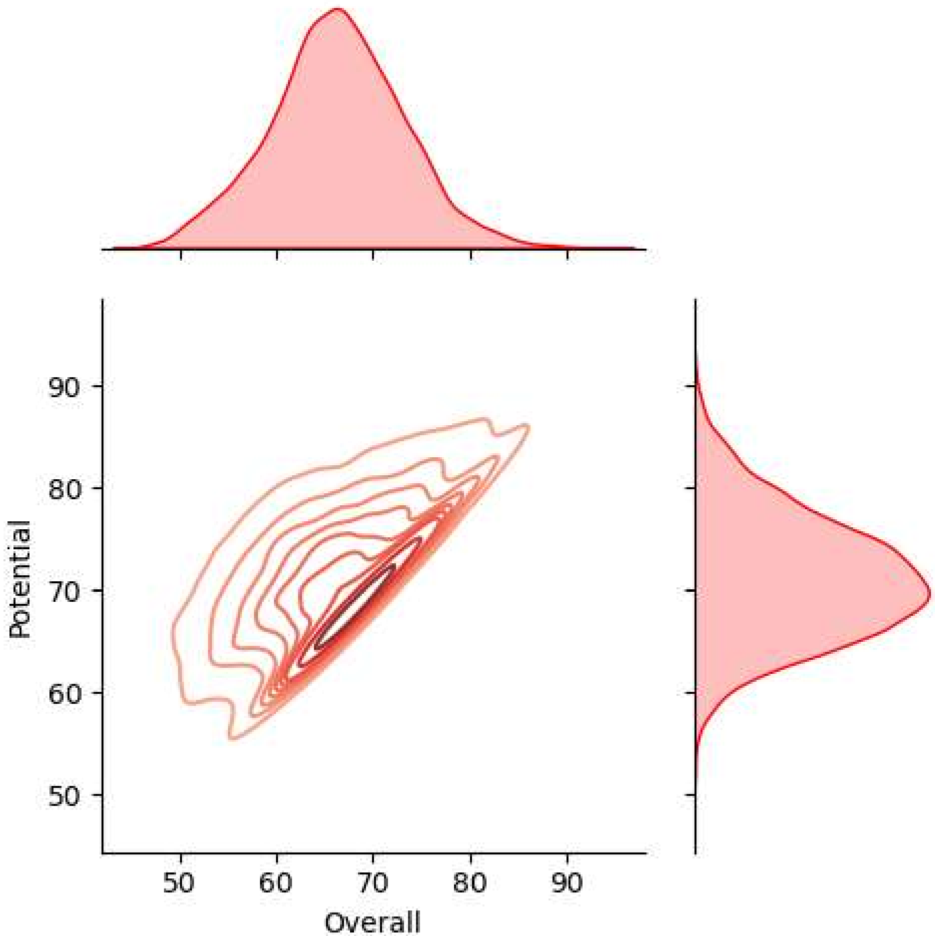
In [79]:
```
g = sns.JointGrid(x = "Overall", y = "Potential", data = fifa)
g = g.plot_joint(plt.scatter, color = "0.5", edgecolor = "white")
g = g.plot_marginals(sns.distplot, kde = False, color = "0.5")
plt.show()
```



In [81]:
```
g = sns.JointGrid(x = "Overall", y = "Potential", data = fifa, space = 0)
g = g.plot_joint(sns.kdeplot, cmap = "Blues_d")
g = g.plot_marginals(sns.kdeplot, shade = True)
plt.show()
```

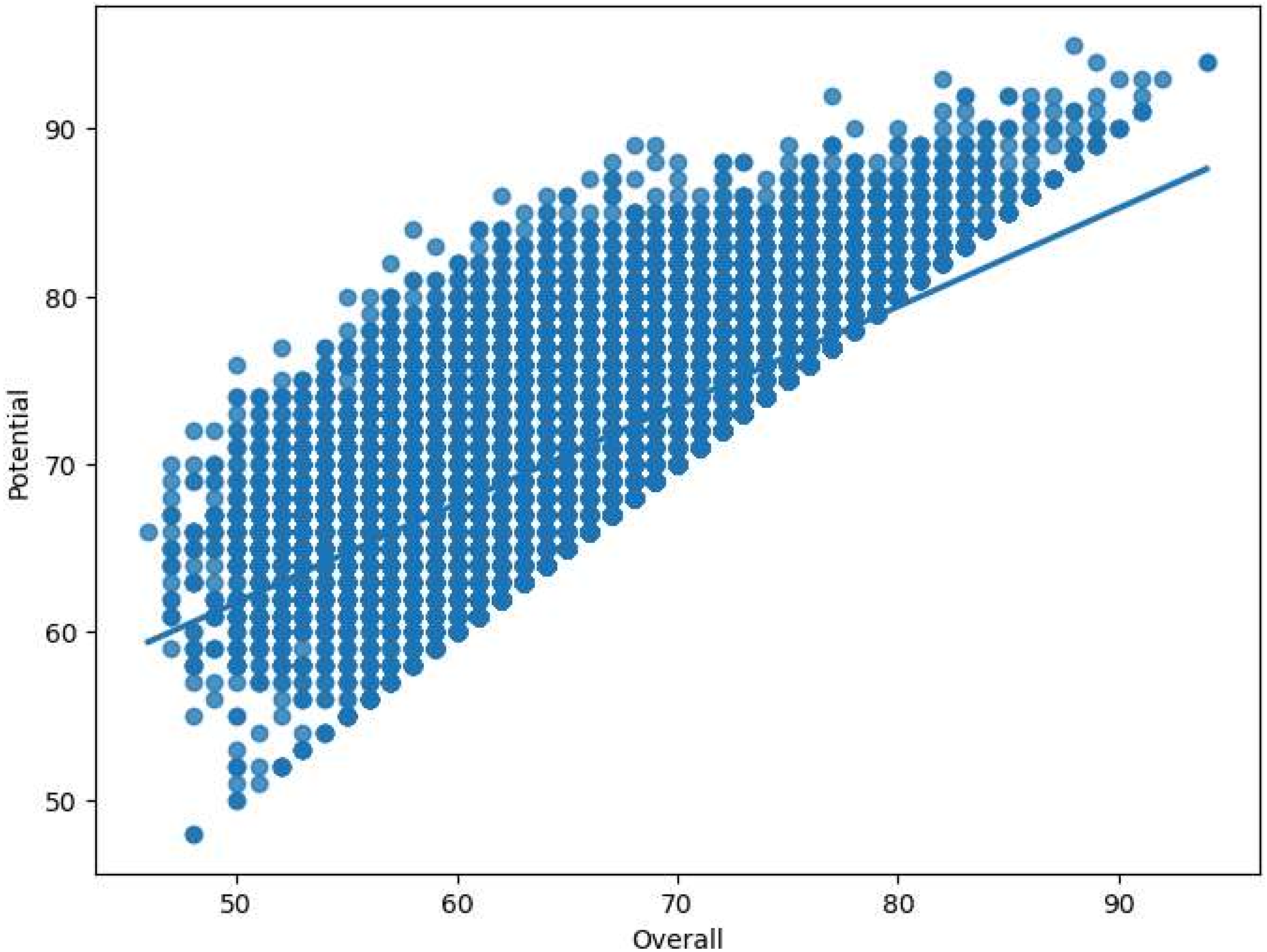



```
In [82]: g = sns.JointGrid(x = "Overall", y = "Potential", data = fifa, height = 5, ratio = 2)
         g = g.plot_joint(sns.kdeplot, cmap = "Reds_d")
         g = g.plot_marginals(sns.kdeplot, color = 'r', shade = True)
         plt.show()
```



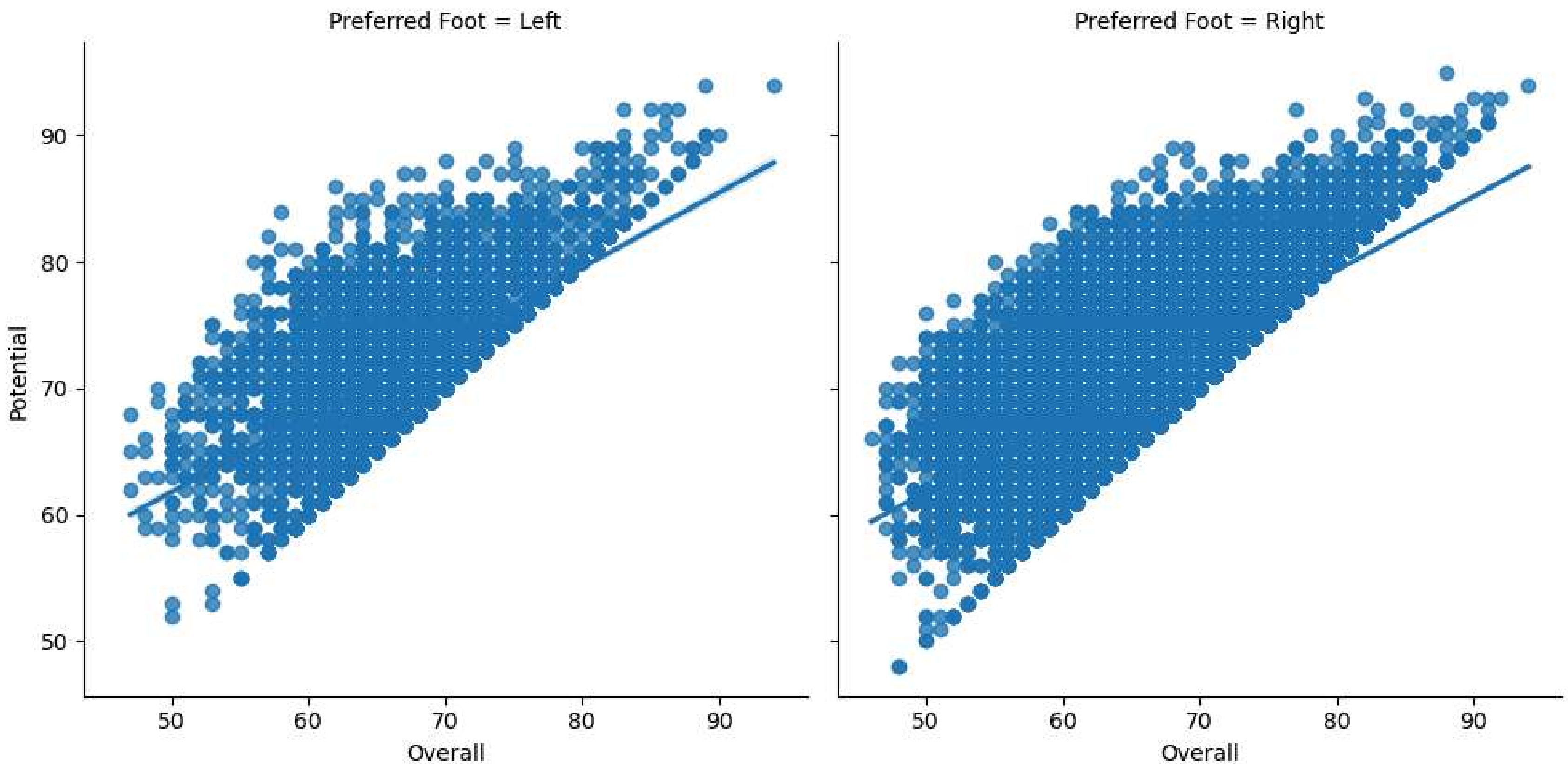
In [83]:

```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.regplot(x = "Overall", y = "Potential", data = fifa)
plt.show()
```

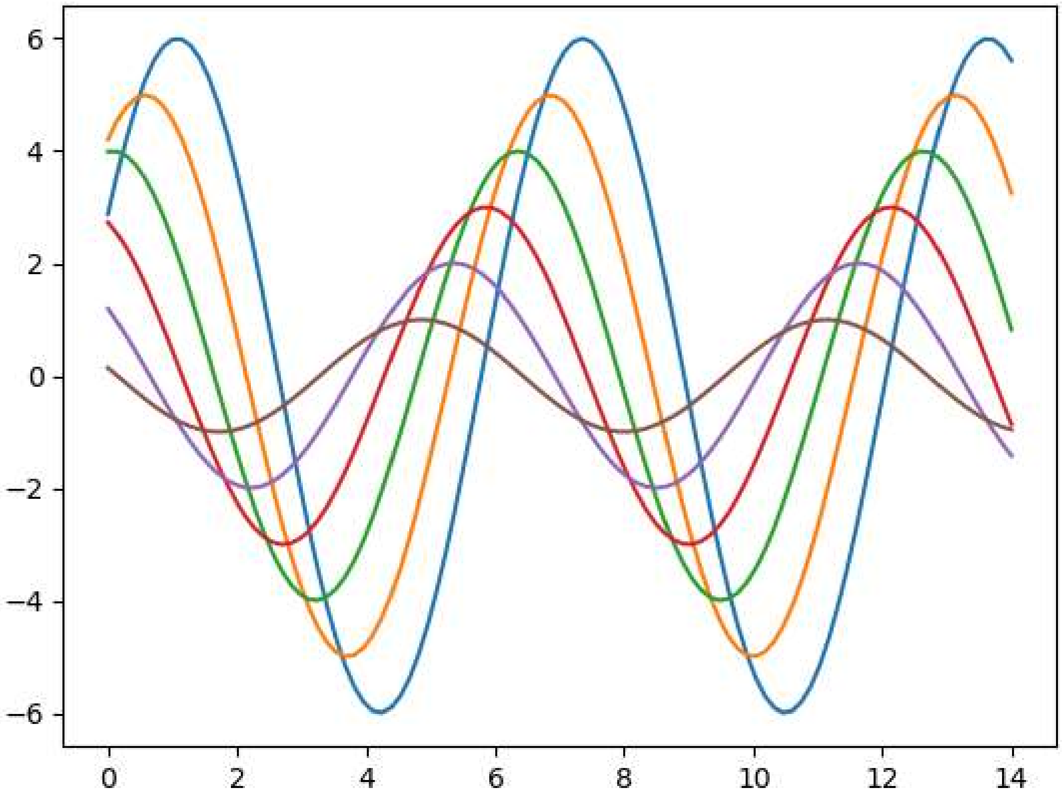


In [84]:

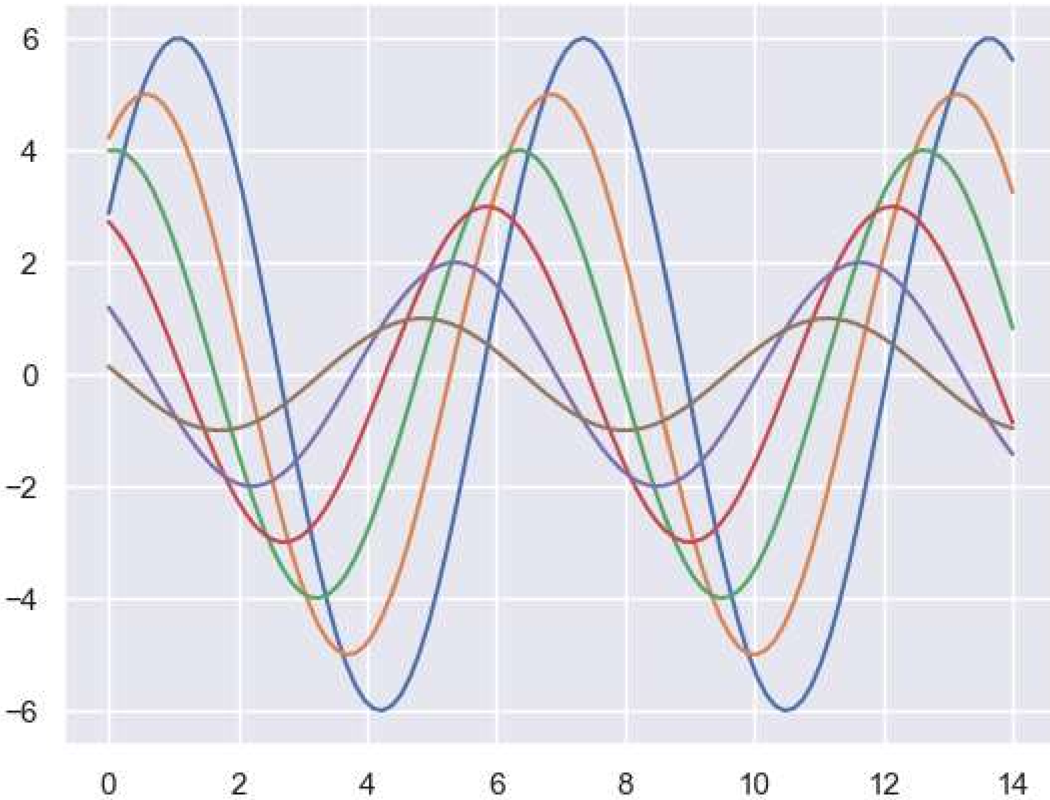
```
sns.lmplot(x = "Overall", y = "Potential", col = "Preferred Foot", data = fifa, col_wrap = 2, height = 5, aspect = 1)
plt.show()
```



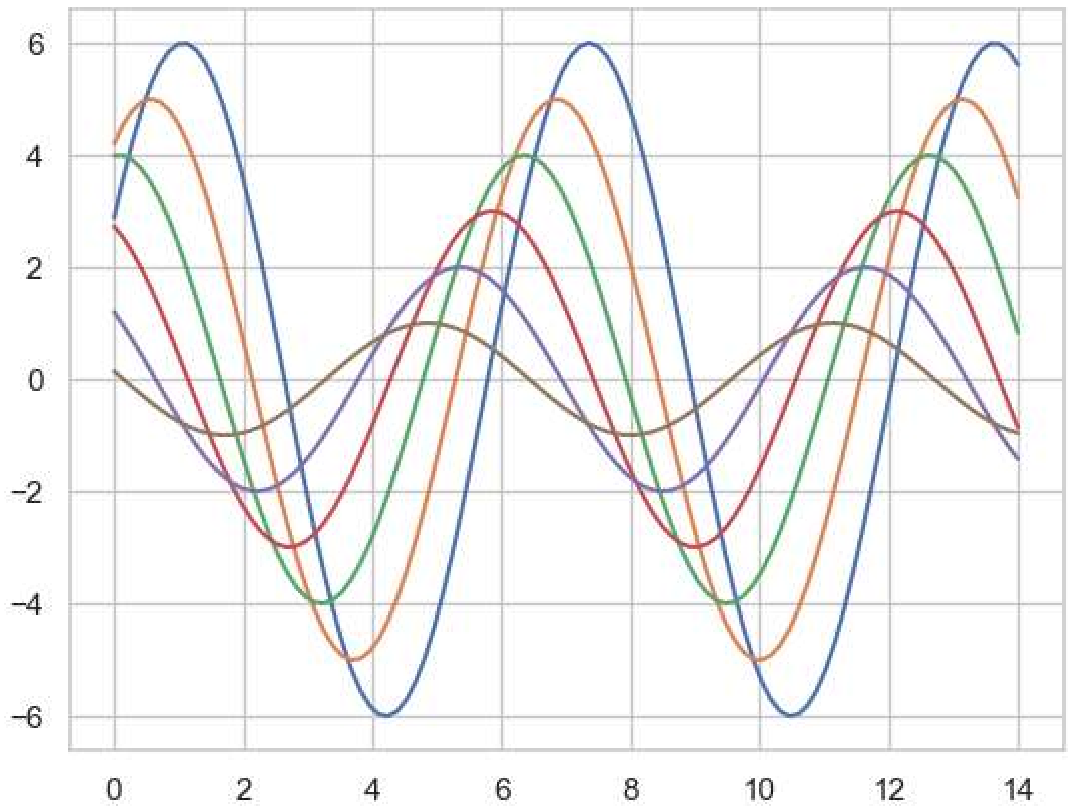
In [85]:
```
def sinplot(flip=1):
    x = np.linspace(0, 14, 100)
    for i in range(1, 7):
        plt.plot(x, np.sin(x + i * .5) * (7 - i) * flip)

sinplot()
```



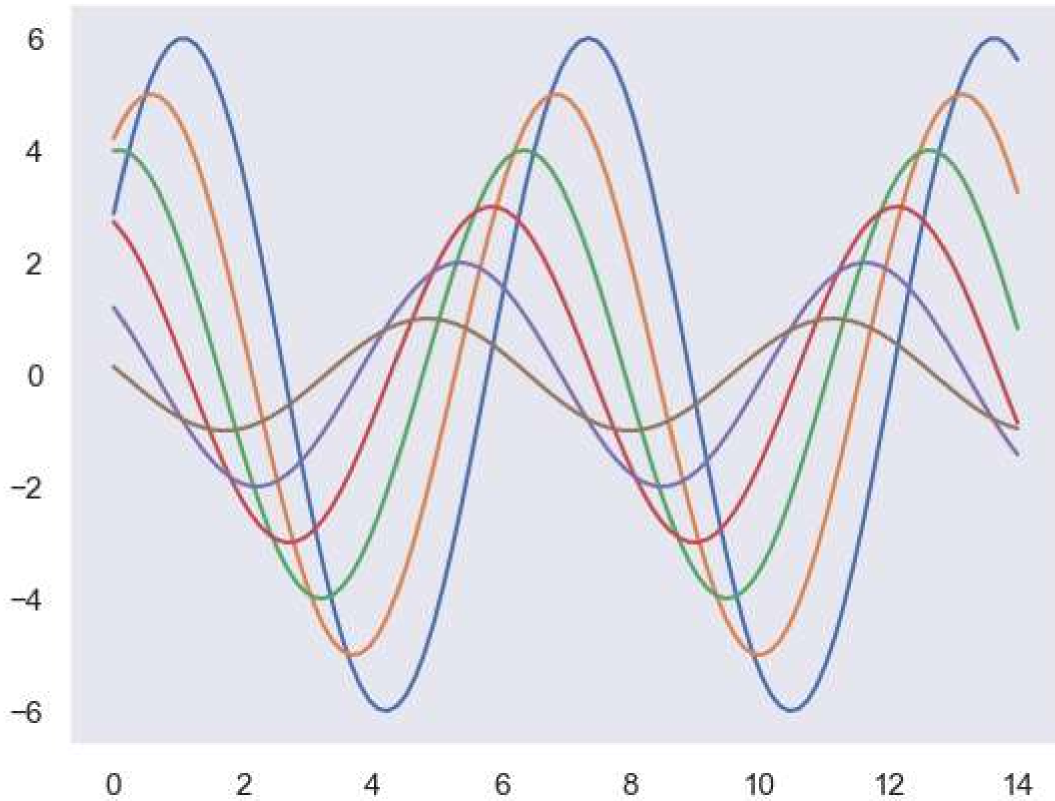
In [86]:
```
sns.set()
sinplot()
```



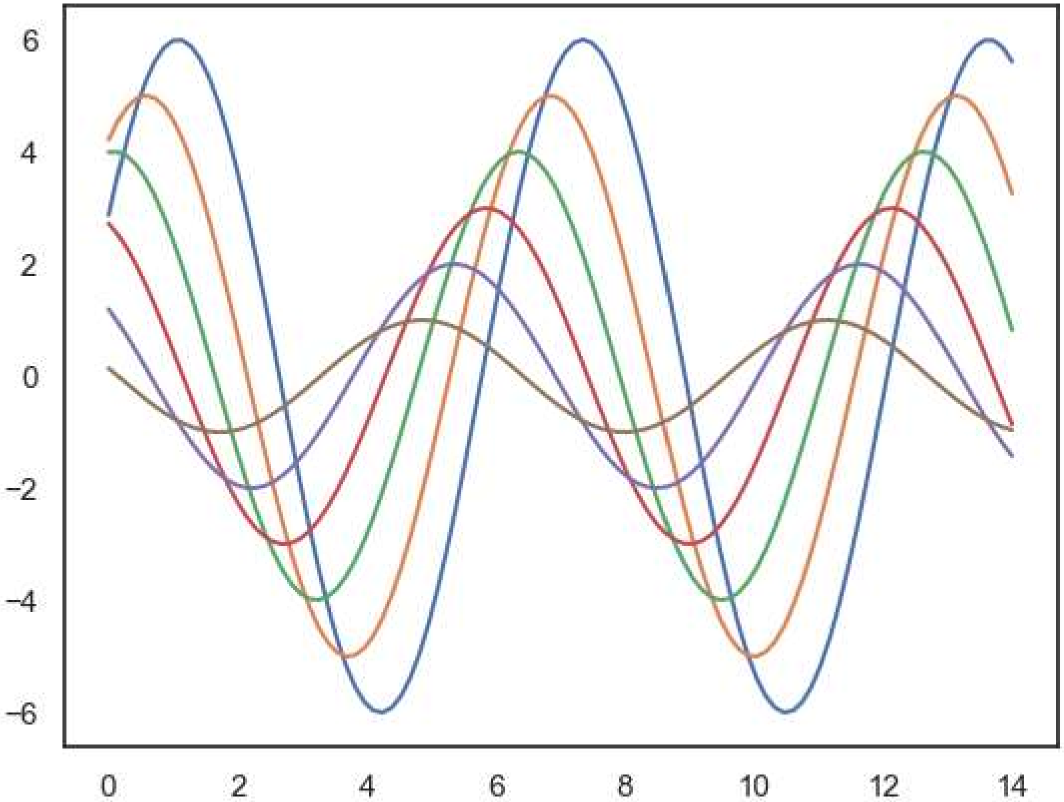
In [87]:
```
sns.set_style("whitegrid")
sinplot()
```



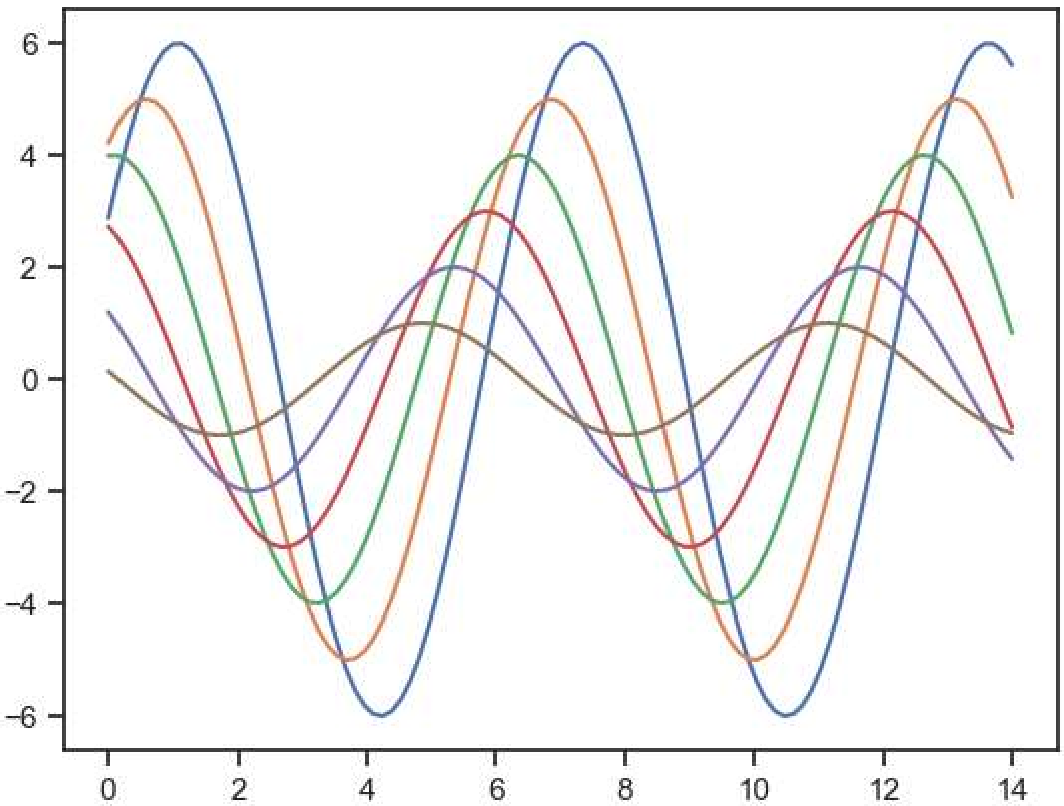
In [89]:
```
sns.set_style("dark")
sinplot()
```



In [90]:
```
sns.set_style("white")
sinplot()
```



In [91]:
```
sns.set_style("ticks")
sinplot()
```



In [ ]: